

Phil 305 Intelligence, Rationality, Wisdom and Spirituality

Numbered Transcript

PHIL 305 • Dr. John Vervaeke

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Lecture 1: Cognitive Agency

At a glance — Vervaeke introduces cognitive agency and why intelligence, rationality, wisdom, and spirituality belong together in one course.

- 1.1 So welcome. This is a course on intelligence, rationality, wisdom, and spirituality. Now, why those four? I want to try and explicate and clarify the relationship between these four dimensions that I think are central to your cognitive agency—being an agent in the world. A cognitive agent is a self-directed knower and actor.
- 1.2 And what's an agent? I'm not going to give you an exhaustive definition of what an agent is, because this is very much under significant philosophical dispute right now, but I'm going to give you some non-controversial aspects of agency that are central.
- 1.3 So, everything behaves. If I hit this podium, it behaves. It makes a noise, right? Everything behaves. But an agent can determine the consequences of its behavior and change its behavior in order to alter those consequences, usually for achieving a goal. So a cognitive agent is knowing the world and acting in the world in order to alter the consequences of its behavior in order to achieve some goals. You're doing that every single minute. You're doing it right now.
- 1.4 I want to try and unpack all four of these. I'll keep calling them the core four, and I want to try and explicate the relationship between them and remove any confusions there might be around them. Now, why is this important? So, agency, being a cognitive agent, is central to being a person. In fact, a person is a particular kind of agent. It's an agent that is responsive to meaning—in an intelligent, rational, and virtuous manner. That's how we determine if somebody is a person. A person is a cognitive agent that is expected to be reasonably virtuous.
- 1.5 That's why we—how and when we convey personhood on full personhood on someone. We allow them to vote, we allow them to own property, to get married, etc., when we feel that they are capable of acting in this way—reasonably virtuous manner. So the core four components are central to personhood. And personhood is, of course, something that we are intrinsically interested in. We are very interested in it, and our political arena is fraught with disputes about personhood right now for just such a reason. It is central to our life, not just theoretically, but practically, existentially, how we're living.
- 1.6 So, because we find personhood inherently valuable, we find these core four components inherently valuable, or at least you should. But all of these four components are deeply problematic. So the degree to which we don't understand them is the degree to which we fundamentally do not understand our own personhood or the personhood of others.
- 1.7 Intelligence. What is it exactly? And this, of course, has become a very burning issue right now, with the advent of AI, artificial intelligence, especially artificial general

intelligence. You've probably seen all the things about GPT-3 and GPT chat, and they're both exciting and terrifying. And what's going on in intelligence? What do we mean by this? What is it exactly?

- 1.8 What is rationality? Is it the same thing as being intelligent? Is it about being logical? Is that what it is to be rational? What relationship does it have to being virtuous, to being wise?
- 1.9 Wisdom. I often point out to my students that they know where to go for information. You carry around little mechanical, well, electronic oracles. Your phones tell you all kinds of information. You may know where to go for knowledge —the university, etc. But where do you go for wisdom? What is it? And as I hope to show you, wisdom is not optional. You either explicitly and rigorously pursue it, or you implicitly and haphazardly pursue it. Those are your two choices. Those are the only choices you have.
- 1.10 What is the relationship between wisdom and intelligence and rationality? How is it related to other traits or processes like virtue, understanding, insight, self-regulation? Finally, spirituality. One of the fastest growing demographic groups in our society are the nones, the N.O.N.E.S. They have no official religious affiliation, yet they are typically not, sort of, atheists. They typically describe themselves as spiritual, but not religious.
- 1.11 What does that mean? Does it mean anything? Is it just a way of saying, "I am kind of cool," or is it a way of saying something that has meaning and value? What does it mean to be spiritual in a scientific, technological age? How is it related to intelligence, rationality, and wisdom? Is it related? What's its relationship to reason and religion? What's its relationship to meaning and morality? These are the core four and the problems deeply associated with these inherently important components of your cognitive agency and personhood.
- 1.12 OK, so let's start with intelligence. So, Spearman, doing work in the 1920s and 1930s, discovered something. We already had psychometric testing of intelligence for quite some time, and he discovered that how kids were doing in one class was quite predictive of how they were doing in other classes, even if the material was very different. How they were doing in history was predictive of how well they were doing in math and art. Contrary to a lot of our popular cultural tropes, they're all mutually predictive.
- 1.13 And he started doing different tests on people and found that there was what's called a "positive manifold". How you did on any one of these tests, even though the content was very different, was strongly predictive of how you would do on all the other tests—which meant there was something underlying. Beyond a specific ability for history or for science or for math or art, there was a general ability, a general intelligence.
- 1.14 Now, what's interesting is, this is exactly what's being pursued right now in what's called AGI, artificial general intelligence. This is your capacity to be a general problem solver. So, you're different from a lot of other machines. That camera over there, for

example, solves the problem of recording sound and sight. Right? But it's not very good at solving the problem of swimming. And it can't learn about Australian history. But you can do all of those. You can learn about Australian history. You can learn about swimming. You can learn about how to be literate. You can learn how to carry on a conversation. You can learn how to play soccer.

1.15 You can solve a wide variety of problems in a wide variety of domains. And this is what we cannot get our best AI to do right now. That's why they're in pursuit of artificial general intelligence. Yes. I can get a machine that can beat even a master, Grandmaster at chess.

1.16 I might get the machine that can beat a Grandmaster at chess and at Go and a couple other games. But it can't learn to swim. It can't learn how to play tennis. It can't learn how to compose poetry, but you can. You're an actual, not an artificial, general problem solver. And until we get machines that are AGI—artificial general intelligence—we won't have machines that are anywhere near what human beings are capable of. Yes, GPT-3 is kind of cool. You can talk to it, but it can't play tennis with you. Just think about that. OK.

1.17 Also, how we test for IQ and measures of G are —if I had to know a single thing about you, if I was allowed one measure of you to try and predict you—and you won't like this, because we don't like, we want to think, "We are free spirits, and we are completely self-defining"— I get that. But if I had to know one thing about you, I'd want to know the measure of your general intelligence, the measure of G, because it's so predictive. It's how predictive, how well you do in school, way better than other predictors, even your marks. How well you'll do at your job, way better than interviews. Interviews are largely useless compared to G. How you'll do in your relationships. Other things predict, too—personality factors, attachment style—but G is highly predictive. How long you'll live, etc. So it's very powerful. The way we test for G is we give you different kinds of problems and find a positive manifold, find the general intelligence.

1.18 OK, so this general ability, it has to be sort of a "meta" problem solving. So I'm an academic, so I have to use meta as many times as I possibly can. OK, so, what do I mean by meta problem solving? Instead of what you need to solve this problem or this problem or this problem, there must be something shared by all these problems, or at least shared by many of these problems, that need to be solved—and that by solving, helps you solve those individual problems. Does that make sense?

1.19 So, how to solve—something in each one of these problems, there's some general component, an important component that they share that is being addressed by this general ability, this meta problem solving ability. I'm going to suggest to you that there are at least two of these meta problem solving components.

1.20 OK, so I want to take a look at the seminal work of Newell and Simon. They created—they attempted to create, I should say—a general problem solver. That was the name

of their project—the GPS, general problem solver. They worked on that from, like, the 50s through to about 1972. And as you can see from what's around you in the world right now, they did not succeed, but they made enormous progress. In fact, they got basically the equivalent of the Nobel Prize for their work. It's been so seminal to psychology, artificial intelligence, economics, etc., their work.

- 1.21 All right, so how did they analyze a problem? They analyzed a problem into four components. The first component is your initial state, and you have a problem when your initial state and your goal state are different from each other. So I'm thirsty, and that's my initial state. I need water. I don't have water right now. I have to get the water. That'd be the goal state. OK, so you have some cognitive grasp of your initial state. You have some cognitive grasp of your goal state. So those are two of the four components.
- 1.22 Then you have operations you can perform. These are actions that transform your initial state into some other state. I'm right here. When I'm thirsty, I can take this action. I can take this action. I can take this action. And many of you are going, "Those aren't helping." But you see, though, that there are many things I can possibly do. And then from here—so these are called operations. It doesn't always open up as cleanly as this. I'm just doing a rough idea for you. These are various states you get into, right?
- 1.23 This is called a search space. This is called a search space. It lays out all the possible pathways from your initial state to your goal state. There's one more component. So the three, right, three components—initial state, goal state, cognitive grasp, and operations you can perform that transform the state you're in.
- 1.24 The fourth is called path constraints. These are constraints on the kind of path you can make use of when you're moving from the initial state to the goal state. Why do you want path constraints? Because you're a general problem solver. You're a general problem solver. You don't want to solve one problem to the detriment of your ability to solve other problems. I could cook my lunch by burning down my house. You wouldn't go, "Oh, that's brilliant." Because what I've done is I've solved one problem to the detriment of solving many other problems.
- 1.25 OK, so there's constraints on you. A lot of these constraints will become very crucial as we get into this. You have important time constraints when you're solving those problems. You don't want your problem solving method for getting into university to take you 27 years. OK? It's too long. It's too long.
- 1.26 OK, so, what's a problem solution? A problem solution is any sequence of operations that transforms the initial state into the goal state while obeying the path constraints. So, presumably this is one, for example.
- 1.27 OK, so the goal of problem solving is to find a method for finding a problem solution. That means I have to find a method for searching this space that finds the problem

solution—the sequence of operations that turns the initial state into the goal state while satisfying the path constraints. Does that make sense?

- 1.28** You say, "They got a Nobel prize for this? I can figure this out in, like, 10, 15 minutes." That's not what they got the Nobel Prize for. See, there's a problem with this diagram. First of all, it's overly simplistic because I was drawing it and I only had a couple, like, 30 seconds. Secondly, you have a bird's-eye view, so you can see all the pathways at once. At once. And then you can immediately see which one works. Yes? That's not how life works. Life is: you're here and you're looking at all of these pathways, and you don't know which one will actually get you from the initial state to the goal state. And there's a problem with that because very many search spaces are very, very large. What do I mean by that?
- 1.29** So, Holyoak, the psychologist, gives a very good example. He talks about how you can calculate how many pathways you'd have to explore. There's a formula called F to the D . F is the number of options available to you at any step, and D are the number of steps you can take. OK? "I'm here. Here's the options. How many steps can I take, etc.?"
- 1.30** All right. So on average, when you're playing a game of chess, which is very limited domain, very limited domain—you're playing a game of chess. On average, at any time, there are 30 legal moves available to you. 30. And an average chess game takes 60 turns. That's the number of pathways. What do you notice about that number? It's very, very large. It's combinatorially explosive. So, the number of this, in scientific notation, is 4.239×10^{88} .
- 1.31** OK, now you say, "Ah, but I have a brain. And I've been told by popular media the brain is the most complex thing in the universe." That, of course, is logically false, because the brain, plus anything else, is more complex. But you, I get, you get what they're saying—it's a very complex thing. OK? Yes, it is. And this is how many neurons you have. That's a lot of neurons. But you know what, that's nowhere close to? That number. Well you say, "it's not the neurons, it's the number of connections." You're right. There's a lot more connections. That's a big number. But you know what it's nowhere close to? This. Even all of your connections operating in parallel would have a ginormous search space to look through. In fact, this is the approximate number of atoms in the universe — 10^{82} . The search space has more paths than there are atoms in the universe. This is why they got the prize, because once you realize this, you realize, "Oh, how do we search the search space?"
- 1.32** Even if I was taking a fraction of a second to check each pathway, it would take me the rest of the history of the universe. How do human beings play chess? How do we even get machines to play chess?
- 1.33** Now, what Newell and Simon did is, they made a distinction. Now, the way these words are used now slide, have slid around a lot, but I'm going to use them the way they used

them for the purposes of this course. They made a distinction between a heuristic and an algorithm.

- 1.34** So, an algorithm is a problem solving method that is guaranteed to find a solution or prove that a solution is impossible. It works according to the standards of certainty. It's a problem solving method that is guaranteed to find a solution or prove that a solution is impossible. There's an algorithm for determining the number of people here. It's called counting. If I do it correctly, it will definitively give me the correct answer. I'm certain to get the answer as to how many people are in this room. OK?
- 1.35** A heuristic is a problem solving method that increases your chances of finding a problem solution, but it's not guaranteed to do so. So, for example, a heuristic in playing chess is control the center board. That will increase your chances of winning, but you could, of course, control the center board and lose. I've played a game where I realized the person was controlling the center board, and so I played a peripheral game and beat them—one of the few advantages of being a cognitive scientist.
- 1.36** OK, so what a heuristic does, if you think about it—here's a search space. It's vast. It's combinatorially explosive. It's computationally intractable. It would take far too long to try and search the space, even with super-fast machines.
- 1.37** Now, if I'm going to be guaranteed to find the solution in that space, how much of that space do I have to search? What do you think? Mostly, most of it. Sometimes, all of it.
- 1.38** So, algorithms for very, very, very many problems commit you to a combinatorially explosive search that you couldn't possibly do. OK? Now, please remember that, because most of the things we think associated with rationality, for example—logic and math—are algorithmic, which means if I said to you, "I want you to always be logical, always be mathematical, in everything you're doing, in all of your problem solving technique," what would I be committing you to doing? Doing a combinatorially explosive search for every problem that you had to solve, which would mean asking you to commit cognitive suicide. So, rationality can't be the same as being mathematical and logical.
- 1.39** It surely can't mean not being mathematical and logical, but it can't mean being comprehensively and completely mathematically, mathematical and logical. And yet we tend to treat rationality and logic as if they were synonyms. We'll come back to that. We'll come back to that.
- 1.40** This is what is, you do, right? Out of all of this space. That's a very bad pie chart. Out of all of that space, you somehow only search a very small amount of it and find solutions such that you're a general problem solver. Somehow, whenever you're, you're somehow, and do you get this? Let it sink in to—you're not checking all of this space and seeing if those pathways or that information is relevant or not. This sounds like a Zen koan. You're somehow ignoring most of that information, most of the time, in a way that allows you to be a general problem solver.

- 1.41** What do heuristics do? Well, what heuristics do— Newell and Simon realized this problem. This was their solution, and it's part, but it's not enough, of the answer. Heuristics pre-specify where you will look in the search space. Go back to the chess game. You're concentrating your attention on the center board. You're ignoring a lot of the rest of the game. You're focusing your attention on the center board. So it pre-specifies where you should look for the relevant information.
- 1.42** But of course, there's a problem with that, because if I'm pre-specifying where I'll find the relevant information, that means I'm pre-judging, which is prejudice, bias. These two words are now frequently spoken together. They're heuristic and bias. See, when I'm playing the chess game, I'm biasing my attention onto the center board. And when my opponent did that, that caused them to actually lose the game.
- 1.43** There's a formal proof of this feature of heuristics. I need you to remember it. This is for all heuristics. It's called the "No Free Lunch" Theorem. The No Free Lunch Theorem. It's formally proved. It's supposed to be a funny name. That's a joke. Academics. "Oh, ha ha ha. No free lunch."
- 1.44** So, say this is a graph on how well you're solving problems. And this is sort of solving problems on average. When I use my heuristic, it improves my performance in sort of a normal distribution like that. The No Free Lunch Theorem says, it also degrades your performance with an area equal to the area above. Heuristic. Bias. They're just two sides of the same thing. Now let's stop and get that.
- 1.45** We're not done with this problem of zeroing in on relevant information, but this is part of the answer. But notice what that means. The very processes that you have to—if you don't want to commit cognitive suicide—that you have to make use of in order to be adaptively intelligent, are also making you inevitably prone to self-deception, being biased. They're woven together like that. Are you starting to see why wisdom is not optional?
- 1.46** You're doing this weird thing. You're somehow ignoring most of the—think about, even with your attention right now. Think about how many things and combinations of things you could pay—you could pay attention to the corner of the desk right here, or this one, or this one and this one, and then this one, or this one and this one and then this one. And now you realize, "Oh, my gosh, there's a combinatorially explosive amount of things I could pay attention to."
- 1.47** And what are you doing? You're ignoring most of it. And you're going to say, "Because it's obvious what I should be paying attention to." And then I'll ask you, "How would you give a machine the ability to find things obvious?" And that's the problem. That's the problem. You see, in commonsense life, we can rely on obviousness. But if we want to cognitively, scientifically, explain intelligence, we have to explain your brain's ability to generate obviousness, so much so that you are reliably a general problem solver.

- 1.48** Of course, we do know that the way we find things obvious also sometimes makes us oblivious to what we should really be paying attention to. "Oh, I thought she was angry at me. Turns out, the whole time she was afraid."
- 1.49** OK, so let's notice something here. I can't do algorithmic processing for most of the time. Now, of course, I can't just arbitrarily guess. I can't just arbitrarily select. Because the search space is so vast, the chances of finding the right answer are diminishingly small. You can't do either one of those. Somehow, adaptive intelligence lies in between those two alternatives, those two extremes. There's a Goldilocks zone between algorithmic and arbitrary. And so people sometimes say, "Well, I know how we get out of this. I just, you know, it's trial and error, right? Trial and error, trial and error."
- 1.50** Think about how the chance of each trial finding a solution—it's vanishingly small. And even, remember that search space? You did 10,000 trials. What's the chances are you're getting anywhere near the solution? Minuscule. "Oh," they say. "Well, you know, you learn by experience." People like to say things like that. It sounds, really —the problem with that word is, it's, people are using it equivocally. Experience.
- 1.51** OK, what's an equivocation? An equivocation is when you use the same term with two different meanings—same term with two different meanings—and therefore your reasoning is invalid. Let me give you, hopefully, an obvious example of an equivocation. Nothing is better than long life and happiness. A peanut butter and jelly sandwich is better than nothing. Therefore, you should eat one and kill yourself. You don't go, "Right. Thanks, John." OK, what have I done? I've equivocated on nothing. Nothing from the set of things that makes life worth living is the first sense of nothing. Nothing from the set of things you can eat is the second nothing. When I equivocate, I fall into invalid reasoning.
- 1.52** Now, experience means two different things. One is everything that is happening to you. Are you possibly aware of attending to or learning from everything that is happening to you? No way. It's combinatorially explosive. Now, experience can mean paying attention to the relevant things that are happening to me and learning from them. It can mean that. Now, if it means that second thing—paying attention to the relevant things and learning from them—you know what the answer is? "I learn by experiences"? You know how I learn to pay attention to the relevant things and learn from them? "I pay attention to the relevant things and learn from them." That's what you're saying, if you're saying, "I learn by experience," and you're using the second meaning of the word experience. You're not saying anything. You're saying, "I learn. I learn by learning."
- 1.53** If you mean, "I learn because of everything that is happening to me," you're saying something false. So you're either saying something false or something circular and trivial. It doesn't explain anything. Does that make sense? OK, so let's really dig into this, this problem of adaptively ignoring, zeroing in on relevant information reliably enough that you are a comprehensive general problem solver.

- 1.54 So I want to give you another version of this because you may be saying, "Well, this is all very sort of abstract, John, and there's, but life, right? Life, real-life problems." OK, but we have to stick with my challenge, which is making a machine that can do it. I can't rely on your intuitive obviousness, sense of what is obvious.
- 1.55 So I'm going to give a machine a very simple problem, and this comes from work by Daniel Dennett. So I'm going to get a machine, because it's electronic, to find batteries and then take batteries to another location where it can safely absorb the energy from the batteries. You typically do that, by the way, too. Your batteries are called food. You typically don't eat the food where you first encounter it, like as they're handing it to you.
- 1.56 You don't eat in the supermarket. You take it to a place where all the other primates can gather with you and you can celebrate our humanity and all that sort of stuff. Right? And we just did it over holiday seasons. We love to do this. "Let's get all the food in one place and get all the people in one place." OK? So we're giving a robot a real central problem. So here's my robot. I'm not going to claim to be an artist, OK?
- 1.57 It has sensors, and I'm going to give it this problem. Here's a wagon, and on the wagon, there's a battery. Nice, juicy, delicious battery that the robot wants to take away. OK? Now, remember what an agent is. An agent doesn't just behave, and an agent determines the consequences of its behavior and adjusts its behavior accordingly, yes? So for this robot to be an agent, I have to give it the ability to determine the consequences of its actions. So it determines—correctly, by the way—that if it pulls the wagon, the battery will come along.
- 1.58 Now, the thing is, I'm going to make a bit of a problem for—because we got to give it, right? I've also put a lit bomb on the wagon. So the robot comes up, pulls the wagon, deducing that the battery will come along. But there's an unintended side effect. The bomb comes along as well and explodes. And we go, "Oh, we're so stupid. We have to make the robot not only pay attention to intended effects; it has to pay attention to unintended side effects."
- 1.59 Like, you strike a match. You know, there's gas diffuse in an area—and people used to do this before cell phones. Cell phones have saved a lot of lives in a way you don't realize—because what people used to reliably do is they'd go into a dark area where they knew there was flammable gas diffused, and in order to make light, they would strike a match. That's the intended effect—light. What's the unintended side effect? Boom, they die. Now we have cell phones. Millions of lives are being saved, which is perhaps the only moral justification for cell phones.
- 1.60 OK, so we're going to make the robot deduce, or at least detect, not only the intended effects, but all the unintended side effects. So we're going to give it a lot more computational power, more sensors, and we put it in, and we put a little black box inside it so we can determine how it's operating. And it goes up to the wagon, and it

reaches out, and then it stands there—calculating, calculating, calculating, calculating, calculating—and the bomb goes off.

- 1.61** We go, "What? What happened?" And we look at the black box, and we look, "Well, what's it doing?" Well, it's determining all of the side effects and seeing if they're relevant or not. So it's saying, "If I move the wagon, the handle will move, and there'll be a squeak. Is the squeak relevant? If I move the wagon, the right front wheel will go through more than 30 degrees of arc. Is that relevant? The left front wheel, the back wheel, the back left wheel. The grass underneath the wagon will be indented. There'll be a disturbance of the air around the wagon. The light reflection coming off the wagon will be slightly altered. The position of the wagon with respect to Mars will have been changed."
- 1.62** What do you think the search space is for potential side effects? It's astronomically vast. And we go, "Oh. Oh." Do you see how this is the same problem in a super simple, super simple concrete thing? "Get your food and take it somewhere." And, "but it's obvious that the bomb—" Yes, it is. So this is called the frame problem.
- 1.63** What's the metaphor here? I want you to remember this metaphor because we'll come back to it. So I want you to pretend. I'm going to use an analogy here. My glasses are framing my vision. They are bounding it, and I'm seeing through that framing. Does that make sense? I'm seeing through it, beyond it, by means of it. But you know what I'm actually not seeing? I'm not seeing my glasses. I'm seeing through them. Now, we'll talk about it a little bit later. One of the things you can do is step back and look at your framing. Mindfulness practices do that. Right? But this is your framing. OK? You're doing the same thing. You have a mental framing happening right now. The frame problem—is that OK? All right.
- 1.64** So we can talk about, in addition to heuristics, there's problem framing. How do you frame the problem? Because, presumably, your problem framing is allowing you to ignore most of the relevant information and zero in on the relevant information. And like I say, that sense of obviousness is that your glasses, your mental glasses, are most of the time transparent to you, until you go, "Oh, she's not angry. I've been looking at this the wrong way. She's afraid."
- 1.65** All right. I wanted to show you the power of problem framing. This is from an experiment. This experiment was done by Kaplan and Simon. This is the same Simon from Newell and Simon. 1990. And the title of the experiment is very revealing. It's called "In Search of Insight," searching through the search space to come up with the insight. A good problem framing. That's what insight is. When I described the aha moment about my relationship to that woman, that's an insight. OK, so I want to give you the basic idea of the experiment. So it involves what's called the mutilated chessboard.

- 1.66 OK, you've all seen chessboards, right? They're not sounding bizarre, like, "Oh, I don't know what a chessboard is." OK. All right, so, there are 8 columns and 8 rows. How many little squares are there? Somebody say. "64." 64, OK. That's not the problem, by the way. OK. Just like, "Oh, wow, I'm really smart."
- 1.67 OK, now, I have a domino and a domino. Horizontally or vertically. And they're the same because this is a square. We'll cover two squares, right? I'll cover two squares. How many dominoes do I need to cover the board without overhang or overlap? 32. So far, very easy. Yes? Now, I'm going to remove these two squares. Now, there's a slight disadvantage for you here because I've just drawn this. But what I can tell you is, when I tell you what the disadvantage is, it didn't make a difference in the experiment, OK? So that's not why I'm worried about it here as a confound.
- 1.68 So what's left are 62 squares. Yes? On the chessboard. This is the mutilated chessboard. It's a very low-budget horror movie, the mutilated chessboard. OK, so there's 62 squares left. Can I cover it without overhang or overlap with 31 dominoes? And you have to be able to prove that your answer is correct. You can't guess or sense. You have to prove that your answer is correct. Can I cover it with 31 dominoes? So I won't leave you at this too long, because when they were doing this experiment, there was one person who was an engineer, and they framed this as a covering problem. Possible configurations. And you were trying to do that in your imagination. You're trying to cover the board with dominoes. So, worked on it for about 18 hours and filled, like, 81 pages of a Hilroy notebook.
- 1.69 When you frame this problem as a covering problem, the search space is combinatorially explosive, and you won't solve it. You were all banging up against it, and part of your brain probably went, "Nah, I just want, I'll wait till John talks again." OK, let's reframe. Take the glasses off. Change the frame. Look at it differently. Let's reframe the problem.
- 1.70 Now, this was the piece that you don't see in the diagram. But I assure you, this didn't make a difference in the experiment, OK? This is made up of black and white squares, right? Now, however I put a domino on the board, I will always cover a black and a white square. You can even see that by imagining it, right? So I need an equal number of black and white squares, don't I? Because the domino has to always cover a black and a white square. So I need an equal number of black and white squares. These two—and you can readily see that in the picture used in the experiment—these two are always the same color. They're either both black or both white. Does that mean I have an even number of black and white squares left?
- 1.71 OK, so let's do it. In order to cover the board, I need an equal number of black and white squares. These two are the same color, which I can now notice. I can now notice that, therefore I do not have an equal number of black and white squares. Therefore I cannot cover it. I have proved it, definitively, deductively. Three steps. One framing—

combinatorially explosive search. Other framing —so small that you can actually do algorithmic processing within it. That's the power of problem framing.

- 1.72** Now, this one has become somewhat famous. So you might have seen it before, but got to do it nevertheless. This problem has become—I'm from Canada. I'm in Ontario. Ontario is, like, a province that is bigger than France, bigger than many of your states. It doesn't mean anything. It's just bigger. Right? And as soon as you drive north, you get into some very non-urban areas very deeply, very rapidly. So I was in a place. I stopped for food. And this is what, this tells you how far north I was getting. The place was called "Diner." That was it. I went in paper placemat. The 9-dot problem was on there. So it's become very famous for good reason, because it very, it looks very easy and yet, it turns out to be very hard.
- 1.73** So you have to join all 9 dots with 4 straight lines. And you can't jump. Like, you have to start your next line from where your previous line began. Ended, I mean. You understand? And people initially say, "Oh, that's easy. 1, 2, 3, 4. I missed the center. 1, 2, 3. Oh, wait. 1, 2. 1, 2. It can't be done." I assure you it can. "No, it can't be done."
- 1.74** Unaided, the rate at which people spontaneously solve this problem is statistically indistinguishable from zero, within time constraints. If you give them, like, a year, they'll figure it out, but within experimental constraints. But, like, what's so hard? So here's one line, second line, third line, fourth line. Right? And when you do that, people get angry at you. They say, "You cheated." And I said, "Well, what?" "Well, you went outside the box. You went outside the square." I said, "I never said square."
- 1.75** But what was obvious to you is that there should be a square. And what was obvious to you is this is a connect-the-dot problem. And when you connect the dots, you don't change direction unless you touch a dot. When you're a kid, if you're doing connect the dots and you change directions without a dot, you don't get a picture of a daffodil. You get a Jackson Pollock painting, OK? It was obvious to you that there's a square. It's obvious to you that this is a connect-the-dot problem. You knew what you were supposed to do, and you were actually completely wrong.
- 1.76** Power of problem framing. Now, here's something I got to tell you. This is actually, this is where "think outside the box" comes from. It's from an actual experiment. Weisberg and Alba, 1981 —they gave people this. They waited till they couldn't solve it, and then they said to them, "Think outside the box. Go beyond the square." And here's the amazing result. It didn't help anybody. Isn't that amazing? "Think outside the box." You know what does not help people to think outside the box? Saying to them, "Think outside the box." Why not? Why doesn't that help?
- 1.77** We're going to slowly get into the idea of this, that because knowing that you should go outside the box is not the same thing as knowing how or knowing what it is like to find the right thing salient and relevant. Those aren't all the same kinds of process. There's different kinds of knowing.

- 1.78** So let's go back to the Kaplan and Simon from 1990, and they were trying to figure out what was going on, because there were people in that experiment that spontaneously had the "aha." They went, "Oh, don't frame it as covering, frame it as parity. Kaplan and Simon are trying to—" Yeah, but what's going on there?
- 1.79** So, they actually propose two spaces. There's a primary space, and then—and because they're academics—this is the meta space. See? "Meta". So if you want to sound like an academic, use meta and use German when you can. Say, "gestalt" and "meta." And if you just want, say, "Meta gestalt," and then people will, "Oh, oh!" OK? OK.
- 1.80** The primary space. This is—you have framed your problem, and you're searching within the space for a solution. That's the primary space. Let's say you're realizing you're taking too long in the primary space, so you switch to the meta space. This is not the space where you're searching for a solution to a problem. You're searching through the space of potential problem framings. You're searching through the space of potential problem framings. Now, this space is also combinatorially explosive, but what you can do is, you can use a heuristic that just works on problem framing. Problem framing.
- 1.81** They actually proposed a notice, the notice—sorry, that's a bad N—notice invariants heuristic. What does that mean? Pay attention to elements of your problem framing that are not changing across your failed attempts. "I tried framing it that way. Didn't work. Tried framing it that way. Didn't work. Tried framing it that way. Didn't work." Try and find what is not changing across your failed problem formulations and change it, because that's probably what's holding you back.
- 1.82** Now this is an interesting word—"notice". What is it to "notice"? That's a different kind of knowing than knowing that you should go outside the square. You have to notice. You have to make use of the way in which you can direct attention to alter the way your awareness is organizing itself—because that's what happens in an aha moment. You redirect your tension. "Oh, I notice that I can go outside as a square, or I notice that the two squares are the same color. Oh, right." And then you reconfigure, reframe the problem.
- 1.83** Now notice this isn't going to be totally sufficient, because the number of things that are invariant across problem formulations is combinatorially explosive. But they're onto something. This noticing, this noticing idea, has more to do with attention and awareness than it has to do with inference and belief.
- 1.84** One other thing—if you were to apply this, it would improve your performance. But in order to apply it, you know what you have to keep good track of? Your failures. And you have to be really willing to look at them very carefully. You don't like to do that. I don't like to do that. "John, I thought we'd spend the afternoon reviewing your failures."
- 1.85** So, notice there's another element here in addition to this how you're taking a perspective, what you're noticing. There's also, how are you identifying with the actions

you're undertaking? "I don't want to be the guy that fails a lot." But if you could overcome some of that identification, and also make your ability to frame more flexible so you could notice more invariants across your problem failures, you would become more insightful, even wiser. This is the virtue of humility. This is why, across many different religious and philosophical wisdom traditions, humility is seen as such an important virtue. And notice, it doesn't just involve your inferential ability or your beliefs. It involves your consciousness, what you're noticing, and your character, how you're identifying with things.

- 1.86** If you don't have the character and the consciousness to properly notice invariants, you're going to be bad at many problems that require insight, that require problem reframing. OK, let's take a look at one other thing, just to make it very convergent to you about how important all of this is.
- 1.87** So let's look at your ability to categorize. This is a marker. Yes? You don't go, "Oh, no." Now notice the advantages of this. Right? Because I can categorize it, I don't have to treat it as a raw individual. Imagine if I had to get to know each one of these as if they were a raw individual. "This is Agnes. This is Thomas. This is Fred, and this is Joe. Once I learn everything I can about Joe, I have to start all over again with Agnes because they're raw individuals."
- 1.88** You're categorizing right now. What are you sitting on? A chair. What's this? A floor. What's this? A room. It allows you to use common nouns like dog and cat, rather than "that furry animal there that pretends to love you and doesn't." Right? Actually, I like cats, but I love dogs. One of the great religious debates of our time is, "Which is more intelligent, cat or dog?" I don't know.
- 1.89** I want you to know—I want to point out to two steps that your obviousness skips over that are deeply problematic for categorizing. When I categorize, I have to first mentally group things together. I have to get a bunch of things together. I have to mentally group them together.
- 1.90** Now, I have to do that before they've been categorized. So I'm not grouping them because they're all markers. If I'm going to form the category "marker", I have to first group them together. That's the very first thing I have to do. And then I'll have to notice what they have in common. We'll come back to—that's the second problem. But do you get, the first thing is the mentally grouping thing? So that involves, sort of, an attentional analog to what in language is called demonstrative reference.
- 1.91** So, normal reference is, I say the word and it refers to—I say "cat," and you think of the land shark with brains. Right? I say "table," and you think of the item found in the kitchen or dining room. OK?
- 1.92** Demonstrative reference is, it points without saying what it is. "I," when I say "I," it points this way. When you say "I," it points at you. Yes? "I" doesn't mean anything. It's, it points. This. This is a demonstrative reference. This. Notice this. This. This. There's no

meaning to "this". All it does is make something "salient" to you—stand out to you so that you notice it. It stands out. That's what salience is—stands out to you so you notice it. What I'm getting with the demonstrative reference is, I'm getting you to notice all of these things together. I don't know what they are.

- 1.93** Does that make sense? Now notice how that's pre-categorical. That noticing ability comes before, ultimately, before you form your concepts, because it's what allows you to gather things together, to make your concepts, to make your categories. Does that make sense? OK. Then you say, "OK, I get it. That's a really interesting problem. How do I make things, right? How do I do that salience tagging so that somebody else can notice?"
- 1.94** Have you ever tried pointing with another animal? We forget how much we can do this. You can point, and point with your cat sometime. "I want you to go over there." The cat looks at you. You can get dogs to do that because we've selectively bred them for 50,000 years to be able to pick up on pointing—if you teach them long enough. They have the ability to pick up on our mental state equivalent to about a 2-year-old human being, which is why we want, is why we have pets, because they're almost people. And we keep loving them so that in the hopes that they'll finally turn into people, and they can't. And so we—and then they die.
- 1.95** All right, so what you might say is, "Yes, but once I get them together, now it's easy. I just notice what's similar between all of them." And it sounds so easy. It's like Sesame Street. "They're all similar." It's so obvious. But philosophers are wonderful at disturbing the obvious. That's their job description. And Nelson Goodman—what a great name for a philosopher— Nelson Goodman. There was one philosopher who had an even better name. His name was John Wisdom. But Nelson Goodman's pretty good.
- 1.96** 1972, Nelson Goodman, just like it sounds, made the point that when people talk about similarity as if they're explaining how they do something, they're equivocating. Remember equivocation? They're equivocating between a logical and a psychological sense of similarity. What's a logical sense? OK. Identity is when two things share all the same properties. Yes?
- 1.97** So similarity is less than identity, kind of the same. So similarity is sharing a lot, but not all, of the properties of something else. So two things are similar if they share a lot of the same properties. Yes? So, logically, as long as any two things share a lot of the same properties—true properties—you should find them similar.
- 1.98** So a lawnmower and a plum. Both have rounded edges, both are brightly colored, both have an odor, both contain carbon. Both are found in North America. Neither one existed 300 million years ago. Both weigh more than a paperclip. Neither one is a particularly good weapon. How many true things that are true both of plums and lawnmowers do you think I can come up with?

- 1.99** So, logically, any two things are similar, which means you couldn't use similarity to group things together, because everything's similar to everything else, logically. Now, what you should say to me is, "But I wouldn't pay attention to those shared properties." Right, you wouldn't. Now you're shifting to the psychological sense, which is, "I pay attention"—uh oh, here it comes—"to the relevant shared properties." So out of all of the properties that are logically true and logically truly shared, I zero in on those that are relevant. And the problem is, that can shift. It can shift like that because it can shift the problem framing.
- 1.100** So, your spouse, your children, pets, work of art, explosive flammable material—things you're concerned about when there's a fire. When I first said the list to you, it was "What?" But then I changed it to "things you care about in a fire." "Oh, yeah. Now I see how they belong together."
- 1.101** So if you don't equivocate, what you're actually doing when you're categorizing —and notice this is all, this is coming before you have any concepts, before you have any categories —your first salience tagging things so you can group them together, so you notice, "That's relevance realization. How is that happening?" And then once they're together, you have to notice what's similar between them. But that takes relevance realization. So categorization deeply depends on relevance realization.
- 1.102** Now, I've published in other areas. If you want to take a look at Vervaeke, Lillicrap, and Richards from 2012, we also make an argument on how this also is the case in communication, long-term memory, etc. In many, many, many areas of your cognition, the meta problem is relevance realization. If you don't solve that, you're not doing anything else to solve your problems. Does that make sense?
- 1.103** I'll just briefly mention one. This is working memory. Working memory is how you're holding things in mind. You're doing it right now. You're making use of working memory. You're hearing this sentence I just uttered as a sentence, yes? Because what you're doing is holding all the words that have come in sequence together in your mind. That's working memory. It's also where, if I would ask you to mentally do 23 times 57, you would do it in working memory. Working memory is really, really important. Measures of working memory correlate very strongly with measures of general intelligence.
- 1.104** Now, for a long time, we thought working memory was just this holding area. And you can hold about 4 things. People used to say 7 plus or minus 2. It's actually 4 when you get more precise.
- 1.105** But working memory isn't just a holding phenomenon. This goes to the work of Lynn Hasher, who is emeritus now, but was a colleague of mine. I always forget of Lynn—does she have a C in—Lynn Hasher. And she's at—she was at the University of Toronto, where I'm also a professor. We shared students, and she did a really important work on

working memory. I won't do the whole, because it's a whole lifetime of work, but I want to show you something really important.

- 1.106** Don't look at the board for a second. I'm going to put some word. I'm going to put some letters on the board. I'm only going to give you, like, 5 seconds to look at them. So don't look at the board right now. OK, ready? You get 5 seconds. I'm gonna show you. Here we go. Look. 1, 2, 3, 4, 5. OK, tell me all 9 letters. It's hard. OK, look away, look away. Same letters. Ready? Ready? Look. Now tell me all the letters. Now tell me.
- 1.107** OK. Chunking. This is called chunking. See, what your working memory is doing is filtering. It's looking for chunking. What did I do? I grouped the letters together so that they belong together in a way that was relevant to you. You see? I processed the letters for relevance realization, and your working memory went "yay." And let it through. Your working memory is a higher order relevance realization filter.
- 1.108** OK, now I want to show you how deep relevance realization is, but I've already hinted at it. I've already hinted at it how it goes outside of our beliefs and our propositions and our inferences, because, remember, it involves that perspectival knowing and that participatory identification. Remember when we talked about notice invariants? I want to show you how relevance realization is very, very deep.
- 1.109** So what I want to show you is a lot of the things we think are the main stuff of our mind actually presuppose relevance realization. They're dependent on it. They can't be generating it. They could manipulate it, but they can't be generating it. They can't be the source of it because they rely on it for their very existence. So one is, we use representations like words, pictures, thoughts.
- 1.110** Surely that's where we can find relevance realization. Well, of course, representations do make things relevant, but that's the key issue. So, John Searle, in *The Rediscovery of the Mind*, in 1992, I believe it is—no, yes, 1992, I was right—John Searle made an important argument about —*The Rediscovery of the Mind* made an important argument about representations. Representations always involve seeing as. They involve seeing something under a particular aspect. They involve aspectualization. So you see this as a marker, but I can also get you to see it as the letter I. I can get you to see it as a little spear. I can get you to see it as something I shouldn't try to eat. I can get you to see it as a human-made thing. I can get you to see it as a tool.
- 1.111** Now, you know what I'm going to do? I'm going to ask you, "How many aspects does anything actually have?" How many potential aspects does it have? Well, what you have to think of is, what are all the properties it has? How many properties does it have? Combinatorially explosive. Right? Intractably large. So to form a representation, this is what I have to do—and you got a taste of it with the chunking. Out of all of the information, I select a subset.

- 1.112** So right now, right, I'm using it as a marker. You're concentrating on one set of properties. Yes, but if I use it like the letter I, you're concentrating on another set of properties. Yes. Do you see that?
- 1.113** So out of all of the properties, I select a subset. I gather them together so that you notice that subset and you find it relevant to you. That's what a representation is. Marker. Human artifact. Tool. Little spear. Letter I. Thing I shouldn't eat.
- 1.114** Representation presupposes and relies upon relevance realization. Relevance realization is deeper, but you already knew that because it comes before your concepts and your categories.
- 1.115** What about rules? We follow rules. That's how we do it. And we like that. We like to make rules. OK, now this goes back to a point made by Brown in 1999 in his book entitled *Rationality*—Brown, 1999, based on an argument from Wittgenstein, which was based on an argument from Aristotle, which was stolen from Plato, as all things are. OK? So, the problem with rules is they don't specify their own conditions of application. So here's a—I'm going to use a moral rule because it's easier for us to imagine, OK?
- 1.116** Here's a rule. It's a rule we're thinking about a lot in our culture right now. "Be kind." I bet you try to follow that rule, yes? But it's one thing for me to be kind to the new puppy dog we have. It's different than how I'm kind to my 27-year-old son. Imagine if I was kind to him the way I'm kind to the—"Good boy. Here's a treat." And how I'm kind to my 27-year-old son, that's not how I'm kind to my romantic partner. That would be weird. Right? Or how I'm kind to the dog. That's really weird. Or how I'm kind to you, or how I'm kind to strangers, or even specific situations—how I'm kind to my partner when she and I are at a funeral.
- 1.117** Now, rules don't contain the conditions of their appropriate application. How they—you knew it was coming—are relevant in this situation. If I try to make rules for how I should apply the rule, what will I get? I'll get an infinite regress of explosion of more and more rules. Rules require for their application something that is not itself a rule, that is deeper than rules. Relevance realization.
- 1.118** What about reasoning? Using inference to change our beliefs? Well, Cherniak in 1999, a book called *Minimal Rationality*—notice how the word rationality keeps coming up. Made a very—sorry, oh, mistake. 1990. My mistake. I apologize. He pointed out the following. Any proposition has an indefinitely large number of implication relationships to all other propositions.
- 1.119** So the number of implications that come out of any proposition. Your beliefs are centered on proposition. "I believe that cats are mammals." Indefinitely large number of implications that come out of that. And so what you're doing in an inference is, you are trying to select from, and you got to remember this—Cherniak is going to come

back when we talk about rationality—out of all the logically possible implications, you're going to select the ones that are relevant to the task at hand.

- 1.120** I'll show you how this can be. You can see this in even something silly and stupid. I'm at a party. Somebody asks me, "How many kids do you have?" And I say, "I have one." And then later on, they hear me talking to somebody else and they say, "How many kids do you have?" And I say, "I have two sons." The first person comes up to me and says, "What's wrong with you? You told me one." I said, "If I have two sons, I necessarily have one. So I didn't lie." You don't go, "Oh, I'm sorry." You go, "What a jerk."
- 1.121** That's not the relevant implication. Although it's logically true, it's not relevant. Inference is selecting the relevant implication, which requires relevance realization. And then what Cherniak does is say, "How do you select the implications? Arbitrarily?" No. "Algorithmically?" No. Appropriately. And this is what he says—you know what you do? You choose the relevant implications. That's actually his thesis. Inference requires relevance realization.
- 1.122** Jerry Fodor, in 1999 and 2006, Jerry Fodor made a complementary argument. "The relevance of a proposition is not contained within its logical properties." "It'll be windy tomorrow." There's a proposition. Notice that the relevance of that proposition to you changes depending on the context. "It'll be windy tomorrow." It's going to be very important to you if you're going sailing, yes? Maybe, but not quite as important if you're flying a kite. Not important at all if you're staying in. The proposition and its truth stay the same, but its relevance is continually changing. You can't determine the relevance of a proposition by looking at its truth, just what it means semantically. You have to do relevance realization. Relevance realization is deeper, more primordial than representations, rules, and reasoning.
- 1.123** We rely on our ability to find things obvious, to reason, to follow rules, and to use representations. It's fundamentally central to you. It's primordial, it's deep and profound, and it is mostly transparent to you all the time. This is why coming up with general artificial intelligence is very hard.
- 1.124** This fascinates me about you, about me, about other animals. How do we do this? This deeply primordial meta problem that is at the core of our intelligence, and therefore at the core of our cognitive agency —how do we do it? I've been working on this since my thesis. My thesis, 1997, in the before times, right? It didn't have cell phones. Think of it. What a world. What a world. I came to the conclusion at the end of my thesis. I didn't know how we could solve it. Luckily, they thought the thesis was—well, I got my thesis. I got my PhD.
- 1.125** But since 1997, I have been obsessed with this problem. Publication after publication. 2012, 2013, right? And I've had a bunch of publications last year. You're going to hear about one shortly. How do we do it?

Lecture 2: Intelligence and Predictive Processing

At a glance — Intelligence and predictive processing — the brain's attempt to solve the deep problem of relevance realization.

- 2.1 Welcome back. We're going to take a look at how do we deal with this deep primordial problem, this meta problem, that's at the core of your general intelligence of relevance realization. I'm going to make use of an idea from a friend and colleague. He was my colleague at the University of Toronto. He's now at the University of British Columbia. This is Evan Thompson. And this is his idea of the deep continuity hypothesis—the deep continuity hypothesis.
- 2.2 So we have to drop below the level of representation, rules, reasoning. What level are we dropping to? The deep continuity hypothesis is this idea that the principles that are governing your biology are actually continuous with the principles that are governing your cognition—that your cognition is organized and runs in a way very similar to how your biology runs. Or to put it in a bit of a slogan, so don't push it too hard, but it's in the title of his book, "There's deep continuity between mind and life." In order to be an intelligent thing, you have to be a living thing. Now, that doesn't mean all living things are intelligent, but in order to be a cognitive thing, you have to be a biological thing.
- 2.3 Let's pay attention to a really important feature of life. It evolves. It evolves. There's a connection between that and how cognition works. This connection was made explicit and argued very clearly by, I think, one of the best books in cognitive science in the last 25 years, which is Alicia Juarrero's *Dynamics in Action*. Juarrero. *Dynamics in Action*, from 1999.
- 2.4 Now, let's take a look at evolutionary theory because it's a different kind of theory, and it gets us to pay attention, to notice things we are not normally noticing, that we background instead of foregrounding. So, evolution depends on the fact that things reproduce. Where do you get goats from? Other goats. Goats form a feedback cycle. Goats make goats, that make goats, that make goats. So it's reproducing. It's reproduction. It's a feedback cycle. Goats make goats. So this is the feedback cycle of reproduction.
- 2.5 Now, Juarrero introduces a very important distinction between a cause and a constraint. Now, because we're all the children of Descartes and Newton, we tend to focus on and foreground causes, and background and ignore constraints. All right. Here's the marker. We're going to do some science. Why did it move?
- 2.6 The usual answer you want to give is "because you pushed it." That's a causal answer. This is an event, and it led to another event. You may say, "Because you wanted to push it," but that's just an event in my mind that led to my behavior. Now, I'm going to ask you, why else did it move other than me pushing it or wanting to push? What might you say? It has a round shape. The floor is flat. There's nothing in front of it. There's air,

but that doesn't matter. There's nothing in front of it. These are not events. These are conditions.

- 2.7** Causes are events that change what happens. Constraints are conditions that change what is possible. It's now very—there 's a potential for this to move. Listen to the word "potential." Because the conditions are right. The constraints are right. It's not moving yet, because conditions aren't causes. But the causes alone are not sufficient for it to move. It has to have both the cause and the constraints. Does that make sense? Causes are events that change what's actual. Constraints are conditions that change what is possible.
- 2.8** Darwin's theory involves two constraints. Now, there's two kinds of constraints. There's enabling constraints—those are constraints that make more things possible for a system. And then there's selective constraints—those limit the number of options for a system. Enabling constraints and selective. And look, within Darwinian theory, there are two constraints. One is an enabling and one is selective.
- 2.9** The enabling constraint is variation. Variation means there are more options on morphology. That's your design of both your physiology and your behavior. So, notice that we're not all the same. There 's variation. There's a reason for that. All that variation means there's different options for how a human being can be, and some of them might be better than others in order to survive. That makes sense. This is standard Darwinian.
- 2.10** But there's selection. So this is an enabling constraint. More options. Selection. What does selection do? Selection is due to scarcity. And what scarcity means is that many of an organism in a generation get killed off. They don't survive to reproduce. They get killed off. So what happens is, you get variation, and then there's selection from that. And then from that, you get variation and then selection. Variation, selection. So the feedback cycle—opening up possibilities, killing most of them off, but there's been a little bit of change. Opening up the possibility, killing many of them. Do you see that? Cycling around. This is self-organizing. Nobody's running it. It's self-organizing. This is a dynamical system. It is a self-organizing system.
- 2.11** Now, this is a virtual governor. Virtual because it's not about actions, it's about possibilities. It's not about causes, it's about constraints. Virtual, having to do with possibility. So this is a virtual governor acting in the environment. Virtual governor. This is a virtual generator. I probably misspelled. I sometimes have dysgraphia. Virtual generator. This is generating possibilities. Virtual generator. This is limiting possibilities. They 're in opposition to each other. One is making possibilities, one is getting rid of possibilities, making possibilities. You see? So when you have the two together, linked systematically, you have a virtual engine.
- 2.12** So what Darwin's theory is, is he basically said, "You don't need an intelligent designer." I'm not here to dismiss religion. That is not what I am. I'm just talking about this

particular issue. "You don't need an intelligent designer. All you need are organisms to reproduce and this virtual engine to exist in the environment. And the creatures in a self-organizing fashion will evolve their morphology, their physiology and their behavior." That's the theory of evolution. It's a dynamical systems theory.

- 2.13** And what does it result in? It results in the organisms constantly evolving how fit they are in the sense of, fit means they survive long enough to reproduce, but they have to be fitted to their environment in a way that they can solve most of their problems, so that they can survive long enough to reproduce. So this fittedness to the environment is really, really crucial. Does that make sense?
- 2.14** OK, so now a proposal, and then we'll start to zero in on it and make it plausible to you. What if relevance realization was your whole, your embodied brain doing that, something like that, with respect to the environment? Let 's take a look. You have a loop. You need a feedback cycle. What's the loop? It's your sensorimotor behavior. "I'm constantly sensing, and that's making me move, which makes me sense, which makes me move, which makes me—" Do you see? It's a sensorimotor loop. "My movement is changing what I'm sensing, and my sensing is changing how I'm moving." It's a sensorimotor loop. Does that make sense? OK.
- 2.15** We have a sensorimotor loop that's the important feedback cycle of how you are coupled. So that's the technical term for how a dynamical system is relating to its environment. The dynamical system is "coupled" to the environment. You have a sensorimotor loop that's running in a self-organizing fashion for how you're coupled, connected, to the environment. Let's make it more concrete. Let's bring in some of the stuff we've been talking about, this "noticing". Let's talk about attention.
- 2.16** So you're trying to follow this lecture from this crazy professor you haven't met before, and he's walking around this strange science fiction room. What's going on? And so, you're probably feeling this pull in your attention. Part of your attention is selecting. It's focusing on the task, trying to follow this lecture. And I'm not going to be offended. Part of it is your mind wants to wander away. Think about other things. And then, "Oh, ah. Oh, ah. Oh, ah. " Moment by moment. Yes? Now, think about that. Here's your sensorimotor loop.
- 2.17** You have a part of your brain, your default mode network, and it's causing you to do that mind wandering. And what's that doing? That's opening up the options for what you can pay attention to. That 's what it's doing. You're wandering away and paying attention to other things. So default mode network, and that's doing the mind wandering. And notice how that's opening up things you could possibly pay attention to. Do you see that?
- 2.18** And then you have task-focused network, and there's debate about whether there's one or many, and I don't care about that right now. See what this is doing? It's selecting. So your sensorimotor loop, opening up options, killing most of them off, but

you keep a couple because you make some connections. You go away for a second. "Oh," and you make a—"oh, that's a good connection. I'm going to let that survive." And letting it survive, you're letting it survive because it's what? It's relevant.

- 2.19** So you open up the options and then you kill them off, most of them. But some of them stay. And from that, you open up the options and you kill them off. You open them up, you're doing it right now, moment by moment. Do you see how you're doing exactly the same thing? That's how evolution works. Your attention, but it's not doing it generation-to-generation like the goats. It's doing it moment to moment with your sensorimotor loop. You're constantly evolving what you're paying attention to. But it's more complicated than that. More complicated than that. Because there's also something you can be doing with your attention. It overlaps with this, but it's not the same.
- 2.20** Your attention is constantly trying to exploit what is familiar to you—because then you can fit it to what you already know. As Piaget would say, you could "assimilate" it to what you already know. So it's actually killing off a lot of variation. You're trying to exploit the familiar. But if I just gave you the familiar, if I just gave you the familiar, if I just gave you the familiar, if I just gave you the familiar—is this going well? It's not a good lecture anymore, is it? Because you also want to do the opposite. You want to explore the novel. Too much novelty. Not good, right? It's like, "Oh, no." Right? OK.
- 2.21** You see, this is opening up the possibilities. This is killing most of them off, opening them up, killing most of them off. And your sensorimotor loop is constantly evolving. I want to show you something really interesting with your sensorimotor loop and how it's also doing this, moving between opposite goals. So don't look at the board for a sec. I'm going to write something here, and then I want you to read it out.
- 2.22** These are two separate words. It's space here. OK, what's it say? The cat. You do it like that. You read this as an H, and this as an A. Now, how do you do this? I'm going to do something, and it's going to sound almost like a Zen koan. In order to read the word, I must first read the letters. But in order to disambiguate the letters, I must first read the word. Therefore, reading is impossible. But you did it because your sensorimotor loop is moving between, integrating—here 's the German word for the "gestalt" , the whole, the meaningful whole—and differentiating to the features, the letters.
- 2.23** You're constantly toggling between those. You're doing it right now. You see how you're doing it not—you're doing it in a dynamic, self-organizing fashion. Your sensorimotor loop is constantly evolving. Your attention is doing all of these movements like accordion between these opposites. This is called opponent processing. Opponent processing is when you have two systems that are pursuing opposite goals, but they are causally linked. They are integrated together in their functions. You're using opponent processing to read. Part of your attention wants to move down to the features. That has to do a lot with your left hemisphere. Part of your

attention wants to move up to the the gestalt. That has to do with your right hemisphere. You're toggling between them. You're doing opponent processing.

- 2.24** Because sometimes the whole, the gestalt, is more relevant, and sometimes the feature is more relevant, and there's no final solution to that. You have to be constantly toggling between them. Do you see that? Does that make sense?
- 2.25** Here's another one. I won't even draw it, but here's the sensorimotor loop. Foregrounding and backgrounding. It's actually in your—there's constant opponent processing going on between your peripheral, your focal vision, what you're foregrounding, what you're backgrounding. Your left big toe. Now it's foregrounded. It was backgrounded before, right? What should you always foreground? Nothing should always be foregrounded. What should you always background? Nothing should always be backgrounded. We're going to take a look at this because you can't totally define ahead of time what's going to turn out to be relevant to you. Instead, like fittedness, you constantly are evolving relevance.
- 2.26** Relevance is your cognitive fittedness. Now, think about this. Is there any final form of fittedness? Is there any final form that life should have? No, that doesn't even make any sense. What's happening is, life is constantly evolving because the environment is constantly changing. There is no final form of life. There is no final form of relevant. You are constantly evolving your cognitive fittedness, your relevance realization, what you're finding relevant—because you constantly have to refit your cognition, your attention, and your awareness, and your actions to the world.
- 2.27** So think about this. I've given you—there's more than these, but just—I'll do four because that's how much you can hold in working memory. Four. Think about all these dimensions of opponent processing. There's gestalt and featural. There's foregrounding and backgrounding. There's task-focused. And there's mind wandering. There's familiar. There's exploit the familiar and explore the novel. Exploit, explore. You're doing all of that right now. Each one of these is opponent processing, doing this evolution. But they aren't working independently. They're all also intersecting and constraining each other so that you zero in on what is relevant. You get all these constraints operating in this dynamic, self-organizing, multidimensional way so that things become obvious to you.
- 2.28** Just to be academic, you don't just have opponent processing going on. You have meta opponent processing. You have opponent processing between all—each one of these is opponent processing, and then you have opponent processing between all of them, to find that nexus point. And that nexus point is constantly moving around in the space of possibilities for you. Is this making sense?
- 2.29** See, cognition is using the same kinds of principles as life. The principles of evolution. It's giving us this meta opponent processing. Now, why? Why might this be? And you find this all through your biology, this opponent processing strategy. Think about your level of metabolic arousal, not sexual arousal—metabolic arousal. How much metabolic

arousal? You don't want to be—hope not. You don't want to be—that's too much. And think about it. I'm Canadian. You don't want to be average all the time. I'm mildly aroused at all. "Oh, look, a tiger. I'm dead. " You have to be able to move your arousal from very high levels to very low levels so you can fall asleep, right? How do you do that? You have your autonomic, autonomous, self-governing, self-organizing nervous system. It has the sympathetic and the parasympathetic.

2.30 Listen to my language. The sympathetic system is biased. Biased—heuristics?—biased to try and interpret everything as a reason for getting you excited. "Threat, opportunity, threat, opportunity," it's shouting. The other system is your parasympathetic system. It is biased to getting you to interpret most of the environment is safe, so that you can relax and go to sleep if you want. And they're like this, constantly pulling and pushing on each other—opponent processing, so that your level of arousal is constantly evolving its fittedness to the environment.

2.31 So I pit two things together and they have complementary biases, and that allows me to actually zero in on the relevant amount of arousal. Is it perfect? No, it can't be perfect. In order to be perfect, it would have to be algorithmic. And algorithmic processes largely commit you to cognitive suicide, or in this case, metabolic suicide, which is death. I take it that you're trying to avoid that. Opponent processing, throughout your biology, through different levels of analysis of your brain and cognition. Opponent processing. Deep continuity. Deep continuity. Notice that what this is showing you, that your mind and your body are not opposite things. Your body isn't some Cartesian clay that you just mold. Your body is central to you being a cognitive agent. Your bioeconomy is necessary and needed by your cognition. We're going to do more and more of that in a few minutes.

2.32 See, this is how you overcome the No Free Lunch Theorem. This is what I argued with Vervaeke, Lillicrap, and Richards, with Tim Lillicrap and Blake Richards in 2012. See, the No Free Lunch Theorem says this—remember that? But what if you put it in opponent processing with another heuristic that has that? And then what you do is you toggle between them in a self-correcting manner. Do you see? We showed this. It doesn't violate the No Free Lunch Theorem.

2.33 I can actually, if I can toggle, I can actually reduce the amount of error by constantly doing opponent processing between the heuristics. That's what opponent processing does. It makes use of inherent trade-off relationships in your interaction with the world to help you hone in on what's relevant. There's an inherent trade-off between the gestalt and the featural. You gain one as you lose the other. And so what you do is you toggle between them. They bias your attention in different ways. And by toggling between them, you actually can get a better capacity to respond. In that particular case, you can read. You can read the letters and the words together. Does this make sense?

- 2.34** OK. Now, one more idea—how this translates into your experience. The philosopher Merleau-Ponty talked about when you're doing this sensorimotor embodied fitting yourself to the world, you're seeking what he called "optimal grip." What does it mean by optimal grip? Well, how close do I want to look at this? Well, if I want to read those words, I got to get in really close. But if I want to see the whole thing, I got to go here. But that's not enough because, "Oh, wait, there's so, oh, wow, how many aspects and how far away? Oh, but if I want to use this spear I—".
- 2.35** There is no one final place or way to set your attention. There are trade-off relationships. As I move into the details, I lose the gestalt. As I foreground one thing, I background another. As I pay attention to the familiar, I lose the novel and vice versa. Do you see? And so I do that multidimensional opponent processing until I get to my optimal grip for the situation on the object. This meta-optimal grip, you're doing it all the time. Is that OK?
- 2.36** Now, let's zero in on that wonderful metaphor because we use it. I say, do you "get" what I'm saying? Do you grasp it? Do you comprehend it? Comprendre, French. We use my mind grasping thing. You know this. This is why the language is there. But think. The graspability of this—is it in the object? No, it can't be grasped by a paramecium. It can't be grasped by a fly. Is the graspability in my hand? No, I can't grasp Africa or Julius Caesar. Graspability is in a fittedness between me and the thing. It's not in the thing, objectively. It's not in me, subjectively.
- 2.37** Just as like adaptivity. Is the great white shark adapted? Well, the adaptivity is not in the shark because if I put the shark in the Sahara Desert, it dies. Is the adaptivity in the environment? No, because many different organisms can live in different environments. The adaptivity is the right "fit" between the organism and the environment.
- 2.38** That optimal gripping, this is what Gibson called an "affordance." It's this real relationship between the organism and its environment. Now, notice how this goes beyond our normal categories of subjective and objective. It binds them together. I've coined a term for that. "Transjective." Trans, between. It's between you and the world. It's not subjective, it's not inside you. It's not objective, it's not in the environment. It's properly between.
- 2.39** OK, so you've got this multidimensional meta-optimal gripping that is transjective in nature, and it's constantly evolving your cognitive fittedness. It's constantly evolving your problem framing. It's at the core of your intelligence. It's transjective. It is self-organizing. It is evolving. It is multidimensional. It is meta-optimally gripping.
- 2.40** Now, instead of saying that again and again and again and again, all that multisyllabic thing, I'm going to use a Latin word, and I'm going to use the word, and I'm using it provocatively. The word is "religio" or "religio." Now, it's provocative because it reminds you of another word. Hopefully, it reminds you of the word "religion." Religion

is one of the two proposed etymologies—I think the better one—for the word religion. But what does religio actually mean? It means to bind together, to connect. When I say "religio," I'm talking about everything I've just explained to you—that multidimensional, dynamically self-organizing evolving of your optimal grip on the world that is central to your cognitive agency, constitutive of your general intelligence. That's what I'm invoking. Everything I've explained to you when I say religio.

2.41 Any questions about that? Please ask questions here because I want to make sure when I invoke this term, I don't have to constantly repeat everything I'm saying that's captured in it. I'm taking all of this and saying, "I'm using this word because it's going to serve as a bridging term." You can now get a sense, perhaps a glimmer of how all of this very technical stuff might actually relate to spirituality.

2.42 Can you use the word in a sentence, your religio?

2.43 Yes, your religio. Do you care about your religio? Do you care about how connected you are to yourself, to other people in the world? Do you? Yeah, you do. That brings up how and why I'm using this term, and that will help to also further explicate what it means. I've told you I'm using it to stand for all of this processing, but let's point to a little bit more. What am I trying to say with this? I'm trying to definitively, clearly, and explicitly say that relevance realization is not cold calculation. It is not cold calculation.

2.44 Read Montague in his wonderful book, *Your Brain Is Almost Perfect*. And it is, by the way. By the way, the idea that you only use 10% of your brain is a ridiculous, ridiculous idea. Limitless, and Lucy, and all these movies. "Oh, what if I could use 70—" You're using 100% of your brain most of the time. In fact, you actually demonstrate more intelligent behavior when you're using less of your brain. Yes. So just put all of that aside. It's silly sci-fi. There 's good sci-fi. That's silly sci-fi.

2.45 In his book in 2007, *Your Brain Is Almost Perfect*, points out that there's a fundamental difference between us and computers. Computers don't care about the information they're processing. They don't need to. They can run full-out all the time because they have virtually unlimited energy supply. And they're not bound to keeping themselves in existence. They have us to supply them with energy and keep them in existence. We just forget that. We background that.

2.46 You're different. Imagine if Andy decided he would commit all of his cognitive resources right now to following this lecture. All of it. He runs his brain as hot and as fast as he possibly can, full of, and then he dies. You go, "Whoa, I don't want that disease." You run warm and slow, not hot and fast, because you care about information. You care about this information, rather than that information. That's relevance realization. You care about it. And why do you care about it? You care about it because you are constantly taking care of yourself.

2.47 So, back to Evan Thompson and the idea he got from Francisco Varela. You are an autopoietic thing. This means it's self-making. We've talked about things that are self-

organizing—dynamical systems like evolution. A tornado is self-organizing, yes? But notice how it's self-organized. It's not self-organized to seek out the conditions that produce, promote, and protect its own existence. It'll just go on to terrain that will kill it. But living things are not only self-organized. They're self-organized to seek out the conditions that produce, promote, and protect their existence. This is how a paramecium is radically different from a tornado. We are all, as living things—this is perhaps the defining feature of life: life is evolving autopoiesis. This is also deep continuity. You are autopoietic. Your mind is ultimately autopoietic. You are constantly taking care of yourself because you are constantly, by moment by moment, making yourself—from the very high level of a self, all the way down to the level of being a biological thing.

- 2.48** You are doing it right now at all of those levels. You are autopoietic. Your mind is autopoietic. You take care of your mind because you're constantly making your mind. Because you're taking care of your mind and your embodied mind, you have to care about this information rather than that information. You're seeing the deep continuity hypothesis, both evolving, evolution, and autopoiesis are central to your cognition. You care about this, rather than that. It's not cold calculation. You care.
- 2.49** More than caring, though, you notice. You notice what's salient, what grabs your attention. It involves you. You are interested in it. "Inter esse." To be interested in something, to be within it. That's what the word literally means. You get yourself involved in it. You're not only evolving, you're involving yourself all the time because you're noticing things are salient to you. They're interesting to you. They call out your attention. You're committing your precious time and resources. That carries with it risk. That's not something you can do just by cold calculation, because you put yourself at risk moment by moment. "As he pays attention to me, he doesn't know that he is stealthily planning his murder." Of course not.
- 2.50** Whenever I'm paying attention to this—remember relevance realization?—I'm ignoring that. Whenever I use this heuristic, I'm vulnerable to bias. As I solve my problems, I fall prey to self-deception, 9-dot problem. You're putting yourself at risk and it takes commitment in the face of that risk. And then something we're going to get into, and I'm going to unpack for you in a second, but just put it here—anticipation. And all four of these are interacting. That's what's going on in religio. All of the relevance realization is, it's not cold calculation. You are caring, you are committing, you are noticing, you are anticipating the world. You are not only evolving, you are deeply involving. You are fundamentally connecting to the world at a level that is profound and primordial and largely transparent to you.
- 2.51** Does that help explain more what is going on in religio? It is not cold calculation. It is this, and you're doing it moment by moment. So right now we've got this idea of religio to an autopoietic agent. We have to go further. I want to pick up on anticipation because anticipation takes us to the second major meta problem. Remember, we said

general intelligence has to do with meta problems? The first meta problem was relevance realization. A connected—and you'll eventually see intersecting—meta problem is anticipation.

- 2.52** Do you want to avoid the tiger or confront the tiger? Which is better? Avoiding the tiger. Do you want to plan for your future or just wait, react to it as it comes along? "I'll need money someday. Oh, well." The more you can anticipate the world, the more you will be better adaptively fitted to it. So this brings in all of your sophisticated problem solving, your ability to plan within your problem solving. Not just reactively cope, but plan. And so much of your intelligence is bound up with this ability to anticipate the world.
- 2.53** So this takes us into another big framework. This is called the predictive processing framework. It's one of the most impressive and growing frameworks. I strongly predict it's going to become the central framework. You're trying to predict the environment, but you're also trying to prepare the organism. Anticipation is, "I predict the environment in a way that prepares me for it." Yes? If I make predictions that don't really help me prepare, that's not anticipation. I'm predicting the environment in a way that prepares me for it.
- 2.54** So, this predictive processing framework, this is the work of people like Karl Friston, Andy Clark, Jakob Hohwy. I'll write that out because it's not how it sounds. And Mark Miller. Mark Miller is a former student of mine. He was also a student of Andy Clark and done work with people who are also students of Karl Friston. And Mark Miller and I are publishing stuff together. He's coming to the university. Well, he's at the University of Toronto now. We're working together. So, it feels like your child coming home, right? That 's meant to be a compliment to Mark.
- 2.55** OK, this whole predictive processing framework has this, what I think are two fundamental but interwoven insights about anticipation. The first is counterintuitive, but that's what makes it such a powerful insight. So don't try and predict the world directly. What? Just give me a second. Don't try and predict the world directly. The problem with the world is it's too dense horizontally. There's lots of events, right? Going, spewing out right into the future. And the environment is dense. He's sitting there, but the room is here, and then the building is there. And it's also vertically dense. You can be thinking about this as a pen, as an object, as a thing, as an instantiation of being, or more specifically, a marker, a black marker, the black marker being used by John. OK?
- 2.56** So trying to predict the world directly turns out to be very hard because it's vertically and horizontally dense. And also, the feedback you get from the world is very ambiguous. Oh, pain and pleasure. Most of your life is not—not a lot of pain. Not a lot of pleasure. What's the feedback? Um, um. OK? So the idea is, predict yourself. I'm going to show you how that works.

- 2.57** Here's the second interwoven. When you're predicting yourself, you don't just predict yourself, you try and predict and complete the pattern ahead of time. And that's how you prepare. So these two are going to predict yourself. And don't just predict—predict in a way that's trying to complete the pattern ahead of time so that you're prepared for what's going to happen. Let me try I'll show you how that works. I'm drawing something abstract, and I'm using vertical to represent levels of processing, not levels in the brain. How this is laid out in the brain is something that the framework is working out. We don't need that for our purposes right now.
- 2.58** Here's the level of the brain that is directly interacting with the world. This is the level of the brain that is doing the sensorimotor looping that we've been talking about. Here's the sensorimotor loop. Sensorimotor loop. OK, and so what's happening—it gets a particular pattern of activation in it. If there were neurons, neurons firing and wiring together. Particular pattern of it, just because of the way you're causally bumping into the world. Like, ow. I'm just getting a sensorimotor pattern. OK?
- 2.59** OK, so there's a pattern of activation here. You say, "So what? That doesn't mean anything." Yeah, just hang on. Now, what you do is, you have a level above this, and this level is trying to predict and complete that pattern. It's trying to predict what that pattern is going to be and complete it ahead of time. In the right temporal game, but I won't go into that right now. And what happens is, this level can provide clear feedback because it's inside your brain as to how correct this level was.
- 2.60** It can give error. This one is doing predictive completion. This one is doing error feedback. OK. Then what you do is, you stack a whole bunch of these on top of each other. There's more. You see it's recursive mini levels, and the patterns are recurring, like the process is recurring, and it's self-organizing. There's no one place where it all comes together. It's the whole network that's doing this.
- 2.61** Now, what's happening? Let's just do it concretely. If I were to rely completely on the feedback from walking, the feedback is actually too slow for me to be able to walk. How do I do it? The cerebellum is actually monitoring my sensorimotor patterns, and it's predicting and completing what's coming so that I can walk. Most of my walking is not done through perception—and I'm going to use this word very carefully—it's done through imagination. Not hallucination, not imagination—true imagination. Look at that wall. You see the whole wall is colored white, yes? The amount that your eye can actually subtend, grasp, is very small. You're grasping little tiny pieces here and there.
- 2.62** But what that system is doing is, you're actually imagining the whole wall and all of its whiteness because you can't actually look at the whole thing all at once. Yet you're doing it. Now, that doesn't mean it's an illusion, because it's true. True imagination isn't an illusion. It's true. It works. And if it's, if it's false, it gets corrected by the error feedback. So you have perception going up and imagination going down. But it's not like perception, imagination. It's perception. Remember how you were doing the letters

and the words? Simultaneously, bottom up and top down. You're imagining, perceiving, perceiving, imagining. That's how you're doing it.

- 2.63** Because it turns out if you do this by predicting, by the brain's predicting itself in this dynamic, recursive fashion, and doing bottom-up perceptual propagation of error and top-down imaginative propagation of a model of the world, you actually get a better—you get the capacity to model the world that avoids all of that complexity issue. As you go up, you get higher and more abstract kinds of predictions. As you go down, you get more concrete and specific predictions until you go down to the brain saying, "You're going to have this sensorimotor pattern and it's going to unfold this way."
- 2.64** "So [gibberish]." See, that was weird because you didn't expect a non-word after the word. So error went up. Arousal, noticing, salient, mark that. But here's the thing, and we're going to come back to this. Not all errors are relevant. Do you think that wall is actually perfectly white? That every hue is indistinguishable and mathematically identical to each? But do you care that it's not perfectly white? No. Does it matter? Not really. Notice the word I used. Not relevant. So you actually ignore a lot of error because it's not relevant.
- 2.65** OK, so now, let's notice what's going on here. I 'm doing what's called data compression. OK, what does that mean? So do you all know what a scatterplot is? Like in a graph, I have two variables, and I plot the interaction between the variables like this, right? They taught you this in high school. Then what do you do with that? Do you leave it as a scatterplot? What do you do? No, you create the line of best fit, right? And it might not even touch any of the dots. And why do you do this? Because it allows you to interpret and extrapolate.
- 2.66** It allows you to generalize. You ignore a lot of the error as irrelevant so that you can come up with a general picture of the world. That's going this way. Why do you need to go down this way? Because sometimes the error matters. Sometimes you don't want to generalize. You want to discriminate. So it's important for me to not—there's stuff here. That's a good generalization. It's better to—there's people here. There's men and women here. There's students here. Sometimes I have to—Andy is there. I have to pick up on particular—now, notice, you're constantly doing that.
- 2.67** You're constantly trying to generalize so that you can apply your process, your cognitive processes, to as many situations as possible, but you're also trying to discriminate so that you can be sensitive to what's important in a particular context. You're constantly—you knew this was going to happen between generalization, right, which is very—Right, and? Differentiation. Specification. You're constantly moving between compression and discrimination.
- 2.68** This is the classic problem in prediction. What do I predict? Do I pay attention to what is the same? Compression, generalization. Do I pay attention to what's different? Discrimination. There's no right answer. We face this problem even in science. That's

why there's no final experiment. Sometimes we overgeneralize because we ignore relevant individual differences. Sometimes we miss an important generalization because we're too focused on the differences. And there's no—the world is not labeled or stably set that the differences are what's important, the identities are what's important. It constantly toggles.

- 2.69** Do you see what's happening here? This way of organizing how you're making predictions and preparations—anticipating—is actually building in relevance realization machinery because you've got opponent processing between compression and discrimination, between the generalization and the particularization of the information you're looking at. You're constantly toggling. In fact, this is what we argued in a 2012 paper about this compression, discrimination, and then relationship. That's what's happening in the predictive processing models.
- 2.70** Now, that means there's a convergence. The work that's being done by me and a bunch of other people on relevance realization, and the work that's being done by Karl Friston and Andy Clark and a bunch of people on predictive processing. And there's Mark Miller, who's in both. And then there's somebody else called Brett Anderson, and he's noticing—and they converge. They come together.
- 2.71** Because the predictive processing faces this problem: What should you predict? And the answer, the short answer is, "Well, what's relevant to you?" Remember I said you notice some error and you ignore other error? This question of what to predict became a central question. At one point, Andy Clark said in 2017, "You predict what is task relevant," just directly invoking relevance realization. Then they got a little bit clearer. They talked about precision weighting. What that basically means is, what error signal do you pay attention to? What error signal do you ignore? Notice I have to use the word "pay attention." This is actually their model of what attention is. Attention is precision weighting. Precision weighting is, "Give that error signal more processing, or don't give that error signal more processing."
- 2.72** Selective attention. That's the convergence point. You got relevance realization and predictive processing, both converging on selective attention. The proposal is that religio—that meta-optimal gripping, that multidimensional opponent processing—is how the precision weighting occurs. It's often built in to the very fabric of predictive processing. It's doing the—for example, I showed you how it's doing the opponent processing between generalization and discrimination.
- 2.73** You're doing this multidimensional opponent processing in order to do precision weighting in order to determine which error signal gets processed, propagated higher and higher up, and which error signal gets ignored. It doesn't get to travel higher up the hierarchy. That's it. That's the convergence. General intelligence is evolving framing-anticipation for an autopoietic agent. That's what general intelligence is. It's evolving, putting these two together. I'm hyphenating them. Framing-anticipation for an autopoietic agent. You get that by integrating relevance realization theory and

predictive processing—two frameworks that are trying to look at the meta problems of intelligence.

- 2.74** Another way of putting it is that general intelligence is anticipatory religio for an autopoietic agent. General intelligence is anticipatory religio for an autopoietic agent. OK. So what you have is you have an agent, and it's bound by these affordances that are being co-created by the environment and the agent. Remember, we talked about that in religio? Graspability, that's an affordance. The walkability of the floor. The drinkability of the liquid. What you have is, you don't just have the world, you have an arena. You have a world that has been sized up, noticed. There's a salience landscape flowing over it.
- 2.75** The agent and the arena are—the arena means the world as being shaped both physically and cognitively for our action. Now, this relationship, like evolutionary adaptivity, is one of participation. It 's a constant process of co-identification. You're constantly assuming identities and assigning identities to things in the world. And in a co—you're a sitter sitting on a chair. You're a student attending a lecture. You could be doing something different. You could be a concert going, watching a rock concert. This is a very poor rock concert.
- 2.76** The world and the organism are shaping each other. Niche construction. You shape the environment as it is shaping you. As you shape the environment, it is shaping you. Co-created. You're participating in it. You're not making it. You're not merely receiving it. Let 's do that again. You're not merely making it. That's romanticism. You're not merely receiving it. That's empiricism. You are participating in it, through and through. It has to do with how affordances are being generated and how identification is occurring. You share the same principles. Evolution in the world, evolution in your brain. They participate in the same principles. So you get that participation generating affordances.
- 2.77** And then, of all of the affordances that are available to you, but by the way, your biology and physics and the culture and your cognition have shaped the environment and shaped you to fit each other. Look around you. So much shaping. So much shaping of the environment to you, and so much shaping of you to the environment. And it's making all kinds of affordances possible for you. Did you make this English language that we're using? No, neither did I. We're both participating in it. And it's shaping you and shaping me and shaping us to each other. It's affording us being able to talk.
- 2.78** So, out of all of the affordances, you have to pick out which ones you're currently going to use. You have to notice which ones are salient to you. You have this—noticing. And that's how you're orienting. This goes to Stegmaier. Orienting. Before you do any action, where am I? Where am I? Where am I oriented? You have to do all that "seeing as" we're talking about. You're seeing things "as". You're aspectualizing. You 're also sizing up. This is what's happening as you're creating this flowing salience landscape. Sizing up comes from the work of Matson.

- 2.79** You're constantly doing this. Out of a whole bunch of features, you're foregrounding some and backgrounding others. And of the ones you're foregrounding, you're gestalting them to a lesser or greater degree. And once you've gestalted the things that you've foregrounded, you're framing problems around it. And it's a bottom-up, top-down thing. You're constantly sizing things up, orienting, noticing, seeing as. That's making the world ready for you for representation, for categorization, for conceptualization.
- 2.80** So this is "presencing" a perspective. So the participation gives you affordances, and then you do this on the affordances and you presence a perspective. You presence a perspective. What I mean by—look, I'm constantly presenting myself to the world and things are presenting themselves to me within the same perspective. That's what a perspective is. You're actually relying on the self-organization of your awareness via attention. So all of that's happening right now. That's all part of the evolving religio anticipation.
- 2.81** Of course, what we're seeing is, we can sort of think of a buildup. At one level, there's an agent-arena relationship, and it's giving us affordances that we participate in. We participate in affordances. But then we have to—presence of perspective. What that does is, that situates you. There's no such thing in physics as a situation. What situation are you in right now? Depends, doesn't it? There's no one final definition or description of it, but it situates you.
- 2.82** You have the salience landscape in which obviousness is dripping with obviousness, and it's a little bit of challenge and all. That's your rich phenomenology. And once you have your affordances, and then some of the affordances have been selected, you're in a perspective, you have your situatedness, you now can interact with the environment. You can make use of your skills. You have interactions.
- 2.83** And then, of course, as you interact with the world, you form beliefs about your interactions. You form "if-then" dispositions. "If I do this, then that will happen." "If I do this, then that will happen." "If it's a cat, then it's a mammal." I do all of this inference about my beliefs. But the inference depends on the interactions, which depends on the perspective, which depends on the affordances. You participate here. This is perspectival here. This is procedural. This is using skills. And this is propositional. This is when you start to make propositions. But propositions are actually proposals that you make about how the world is going to be modeled.
- 2.84** That modeling that comes out of predictive processing is called the generative modeling. You're constantly generating a changing model of how to properly anticipate the world. Now, let's put this. Is this OK so far? Do you see? This level of which your biology and this underlying level of relevance realization, anticipatory autopoiesis is making the affordances. And then, out of all the affordances that are possible to you, your consciousness is basically selecting those that are salient to you. You get the perspectival knowing, the situatedness, the sense of presence being in it here and now.

- 2.85** That allows you to—none of you are using your skill for swimming right now. How come? Because it doesn't fit the situation and there's no affordance. Do you see that? So you get your skill selection, your skill generation, your skill coordination. That's a procedure level. Once you are interacting with the world in a procedural manner, you can just start to determine if-then relationship. You can start to determine your beliefs. You can make proposals for how the world should be modeled. That's your propositional knowing. It's constantly flowing up and down. It's recursive relevance realization anticipation. Recursive anticipatory religio.
- 2.86** Let's do this the other way around and make it more explicit now. Here's your propositional knowing. I'm going to talk about four kinds of knowing. Propositional. This is knowing "that"—knowing "that" cats are mammals. Where your beliefs work. Our society is intoxicated with belief, as if that's the only thing about you that matters. You have belief about these weird things that you don't even know what they are, not because you're ignorant or stupid—because nobody knows what they are—but they're cited all the time.
- 2.87** Facts. What's a fact? What is it? And it has a particular normative standard: truth. You judge propositions. Are they true or false? And it's stored in a particular kind of memory. This is called your semantic memory. How many of you know—raise your hands, please—that cats are mammals? When did you learn that? Where? That's a fact. It just hangs out in non-space because it's at the very top of the stack. OK, now propositional knowing depends on procedural knowing because I have to be able to engage certain skills, skills of inference. If I want to change my beliefs. This depends. This is not knowing "that," this is knowing "how." it doesn't give you beliefs, it gives you skills.
- 2.88** Beliefs can be true or false. Your swimming skill, is it true or false? That doesn't make any sense. It's about interactions, not facts. It's not about truth, it's about power. That's why we care about power. And it's not stored in semantic memory. It's stored in what is actually called procedural memory. These memories can decay differently, separately from each other. You can lose your semantic memory without losing your procedural memory and vice versa.
- 2.89** Now, how do you know which skills to apply? Which skills to acquire? How to coordinate them together? Your procedural knowing depends on your perspectival knowing. This is not knowing "that" or knowing "how." It's knowing "what it's like." It's knowing what it's like to be here now in your state of mind. It's knowing "like." It's that perspectival knowing. Of course, it doesn't give you perspectives. Sorry, it doesn't give you skills, it gives you perspectives. And they can be changed by insight. It's not about power or truth. It's about presence. This is why you like it when you feel present in a video game. You feel situated in it. You feel like you're really there. And this is stored in a memory called episodic memory.

- 2.90 So I asked you where you learned that cats are mammals. Can't remember. Do you remember what you did on New Year's Eve? Yeah, you can remember. And what do you do when you do that? You go in and you put yourself into the perspective. You do an episode, which is a little snapshot perspective, a little mini movie about how your perspectival knowing was unfolding at that time. Episodic memory.
- 2.91 How do you know which perspective to take? How do you transform and coordinate? This is participatory knowing. This is knowing by being. By being the kind of thing you are and by being coupled to the world in the way you are. It gives you roles—the identities you assume and the identities you assign from the identities you assume. It's working at the level of framing anticipation. It's working at the level of religio. What does it give you? It gives you a sense of belonging. "I belong here." You 'll feel it slipping away when you're lonely, where you're suffering culture shock because you're in an unfamiliar place. Or you've suffered trauma and the world that you thought you were in doesn't seem to be there anymore. What kind of memory does it have? It's not semantic, it's not procedural, it's not episodic. It's this weird kind of memory you call your sense of self—your "self."
- 2.92 So two points before we wrap it up for today. Most of this is what is fundamentally connecting you to reality, doing the relevance realization. Why? Because I showed you that the relevance realization is below the level of propositions, beliefs, rules, representations, reasoning. Yes? And yet we focus on this. Our culture has put us into a propositional tyranny where we are fixated on this and we do not pay attention to this. And there's two things I'm going to talk about. I'm going to talk about—and they're interwoven points—this is most of where meaning in life is created. Your sense of your life being meaningful. The sense that your life is worth living, even though it is filled with your faults, your failures, and your frustrations. That's what I mean by meaning in life. I don't mean the meaning "of" your life. That's some cosmic destiny. I don't even know if you have such a thing. I can't measure that. I can't talk about that. I'll just put that aside.
- 2.93 What we can do, and we are doing, and I participate in this, is we can measure and study and reflect upon what makes your life meaningful. What makes you say, "There's enough meaning in my life that it's worth my faults, my failures, and my frustrations"? You know what that means, at least intuitively. You feel connected to something that has a value beyond yourself, and we'll unpack that. Has a value and reality beyond yourself. You can ask what makes your life meaningful by asking yourself this question, "What would I want to exist, what do I care about, even if I ceased existing? What am I connected to that has a value beyond my own idiosyncratic interests and concerns?"
- 2.94 Of course, now you're getting in religio and tying to things beyond yourself. You can hear the whispers of spirituality. But here's the other thing, and I've kept mentioning this, and then we're going to have to foreground it next time. All of this wonderful machinery that is so making you adaptive is also, at all of these levels, in a complex and

dynamic fashion, making you susceptible to self-deceptive, self-destructive behavior. So all this wonderful machinery that makes you adaptive and gives you a sense of meaning in life is also regularly and reliably screwing you over, inevitably. And now you can see why you need rationality and wisdom.

2.95 So next time, we will take a look more at meaning in life, and we'll start to talk about rationality. That's it for today. Thank you very much for your time and attention.

Lecture 3: Meaning and Bias

At a glance — Meaning and bias — continuing on relevance realization, and how cognition goes astray.

- 3.1 Welcome back. We're going to continue our journey. We've been talking a lot about intelligence and relevance realization, predictive processing, generative modeling, and this idea of religio connectedness and the four kinds of knowing. Before I forget, I just want to mention that that work integrating relevance realization and predictive processing that I talked about last time has come out in a recent paper, the end of last year, 2021, and that's with Brett Anderson, Mark Miller, and John Vervaeke, 2022. So a lot of that central core argument is found in that paper. And I want to acknowledge the excellent work done by Brett and Mark with me about that.
- 3.2 So I want to now zero in on religio and something that I promised last time, which is the connection to meaning in life. And also go forward with the threat that religio also makes us prone to self-deceptive, self-destructive behavior. So let's start first with just a few things to get us going. We've talked about this religio, this sense of connectedness that's coming in, this framing anticipation, it's evolving, it's involving— all of that stuff we talked about.
- 3.3 We should think of two dimensions to religio. We should think of a horizontal dimension. We're connected this way—like the agent through, for example, the affordance to the arena. That's a horizontal connection. But there's also vertical alignment of religio. This is the propositional, the procedural, the perspectival, and the participatory. You've got the affordance relationships this way. Remember, it's affordance and perspective, etc., all the way up. Then here, you have alignment. You have vertical and horizontal religio.
- 3.4 Of course, when the two are mutually affording each other, interpenetrating and affording each other, you have profound religio, a profound sense of connectedness. This is, "I feel very connected to the world and I feel very grounded in my being and grounded into things." When you're grounded and connected, you actually really like your existence a lot. At least that's what I'm going to argue as we go on. Let's talk about—I'm going to put a little tiny key just to keep us remembering this. Keep thinking of both of these dimensions. Here's horizontal religio and here's vertical religio. OK, so, and when you have the two together, that's profound religio, the whole thing.
- 3.5 All right, so let's move into meaning when we talk about how meaningful your life is. I want to remind you that I'm not talking about "the meaning of life". That's some cosmic metaphysical destiny, and I do not presume to have the authority to speak about that. I want to talk about something that we can study in cognitive science, cognitive psychology. We do study—I study—and it's measurable and it's predictive, which is

meaning in life. Meaning in life, as I said, is that quality that makes life worth living, even given your failures, your flaws, your frustrations, etc.

- 3.6** OK. Now, we've been doing a lot of work on meaning in life. It's really a hot area. And there was a summative document article published in 2016 by Martela and Steger. Steger, 2016. And they outlined what was then emerging as a consensus, although we'll challenge it a little bit because that's what science does. They do a lot of research together, and there's three things that contribute to a sense of meaning in life. These are coherence, purpose, and significance. Coherence, purpose, and significance. Now, our culture talks almost solely about purpose, and that's because that is consonant with market corporate capitalism. As I hope to show you, purpose is not the most important dimension of meaning in life. It is an important dimension, but it is not the most important dimension.
- 3.7** Now, first of all, what's coherence? Coherence is a sense that the environment makes sense to you, that it meaningfully hangs together. What 's purpose? Well, purpose is basically that there's an organization to your goals. Remember, we've talked about problem solving and that goal-seeking is part of problem solving, and you are disposed to do that because you're an autopoietic being.
- 3.8** OK, so, if you don't just have scattered goals, but your goals are organized so there's an overarching goal in terms of which all the other goals are well related and made coherent to each other, that's a sense of purpose. I'm not going to go into that in great detail because you largely know what that's about because, as I said, your culture is constantly teaching you about that. Get a purpose, organize your goals, organize them into a hierarchy, have an overarching goal, and then that's all you need. Well, now you do need that, but it is false that that is all you need for meaning in life.
- 3.9** Now, the coherence issue is a little bit more challenging because originally there was some very exciting, looked like exciting, research by Samantha Heintzelman, who I got to meet.
- 3.10** Now, she did something really interesting. She did some experiments in which she would show people meaningful pictures, or patterns of letters that were stems for real words, or pictures that didn't quite make sense. You couldn't figure out what's the narrative, what's going on in this picture, and strings of letters that didn't form word stems. The first is more inherently easier to make sense of, and the second are hard to make sense of. She'd give these two different sets of stimuli to people—the meaningful stimuli, the coherent stimuli, and then the incoherent one, and then do measures of meaning on life for them. She'd do control groups and experimental groups. The design is all set up properly.
- 3.11** What she found was that people who had seen the meaningful pictures, the meaningful word stems, rated their lives as more meaningful than those who didn't. Now, that's a pretty powerful result. Imagine if you could change people's sense of

meaning in life by showing them some pictures and showing them some letters. In psychology, if it's too good to be true, it probably isn't. Sometimes it works, but usually it doesn't. I'm not accusing Samantha of any fraud or anything like that.

- 3.12** Human beings are very complex. We often say, "Well, psychology is the soft science and physics is the hard science." I think that's the wrong way around. In physics, the things you're studying aren't trying to please you. They aren't trying to deceive you. They're not bound by cultural norms. Imagine if electrons were trying to deceive you or please you or were bound to whichever culture they happen to come from. It'd be like—psychology is really hard. I'm not accusing her of anything, but there's been a couple of failures to replicate Heintzelman's work. There's been some replications, some failures to replicate. So, I think a better way of understanding it is to move off a simplistic level and to move to a more sophisticated analysis of what happens when coherence breaks down.
- 3.13** To this, I turn to the work of Thomas Nagel, who literally wrote the paper—wow, that's a weird—literally wrote the paper called *The Absurd: What It Is to Experience Absurdity*. He wrote that paper in 1971, and then he followed it up with a chapter in his book, *The View From Nowhere*, in 1989. Nagel, first of all, does something very important. He argues that all of the propositional arguments for the sense that life is absurd are invalid arguments. What? Yes.
- 3.14** So here's a typical argument. Like, what I do now won't matter in a million years. Everything I do—its significance will be completely forgotten, be completely eradicated. Therefore, what I do is meaningless. That's actually not a good argument because the argument is symmetrical. What's happening a million years from now is totally irrelevant to you. Look, if you are totally irrelevant to what's happening for a million years from now, the reverse is also the case. What people are saying and thinking a million years from now is completely irrelevant to you. The argument does not lead to the conclusion that your life is meaningless. It's technically not a valid argument.
- 3.15** Next argument. "Well, I'm so small and insignificant." If I blew you up to the size of a galaxy, are things better? Right? So you realize, "Wait, that's right." The propositional—notice this—the propositional arguments aren't generating the sense of absurdity. They're after-the-fact expressions of it. So what is actually generating the absurdity? It's not these arguments that people float because they're not good arguments. If that's all that was driving nihilism, we could get rid of it in 5 minutes. We go, "Look, here's the arguments. They're all invalid. We're done."
- 3.16** Now, remember when he wrote the first paper? 1971. So 1971 is before. He's going to invoke something. You've probably seen these in movies at least, so I hope you know what I'm talking about. Our phones used to stay in a place where we lived. When we were not with our phones, we had to attach machines to our phones that would record if somebody calls us. I know it was a barbaric and dark time. But we had answering machines. What would happen is, there would usually be a pause, and then a recorded

voice would come on and a beep, and you would leave a message. I think you have something like that on your—I think we do on our phones, but nobody actually calls you on your phone anymore, right? Get this—in my day, people would call you just to talk to you. Now, if someone calls you, you go, "What's wrong?" Immediately.

- 3.17** OK, so, try to get yourself into that head space. And there's this thing. And so it's, John is calling Susan. And so this is John. He hears the line open. "Susan, before you say anything, I've got to tell you. I have got my courage. I love you. I love you." And then he hears, "Susan is not here right now. Please leave your message at the beep." Now, many of you are smiling. Now, humor is low-grade absurdity, and it's absurdity that we can resolve with play, by cognitive reframing. But the point of that—that minor absurdity—is to point out the structure in a way that you can easily access.
- 3.18** What's happening is, John has his first-person perspective in which he has this sudden realization and courage to connect to Susan and say his love. But the machine is from the third-person impersonal perspective. And what suddenly happens is, there's a clash of perspectives. His first-person perspective, in which his words have so much meaning, they have no meaning when he's talking to an impersonal machine from the third-person perspective. Do you see that? It's the clash of perspectives that causes absurdity. And isn't that interesting?
- 3.19** Remember I said to you that a lot of the meaning machinery is not at the propositional level? Here's a concrete instance. The arguments for absurdity, the propositional arguments, don't actually do anything. It's the perspectival, the clash of perspectives, that is actually driving the sense of absurdity. You can see that when people are trying to say, "We're so small " or "a million years from now." What they're trying to say is, "The cosmic third-person perspective and my first-person perspective clash." When I really experience that, that's when I'm experiencing absurdity.
- 3.20** I think Nagel gets it bang on. And I think the problem with Heintzelman 's experiment is she's not focused on the perspectival. And so she's sometimes triggering it, sometimes not. And that's why you're getting these inconsistent results. Now, what that means is, absurdity is not just, or not primarily, coherence within a perspective, it's coherence between perspectives. So in that sense, absurdity is a meta-perspectival phenomena—meta-perspectival—which means if we had something to alleviate absurdity, it also needs to be meta -perspectival in nature.
- 3.21** Now, just to shine a light ahead, I was lucky to be fortunate, privileged, honored—all those wonderful things that primates like—to be invited to join, guess this, it's a great name, the Toronto Wisdom Task Force. It sounds like a police squad. We're going to go out and "you're being foolish right now." We got all the scientists that we could physically together and virtually together who do work, like scientific work—cognitive science, neuroscience, cognitive psychology, biology—on wisdom. Then we bashed away at it for 10 hours for over 3 days. 10 hours a day, I think, for three days. We did a lot. We were doing surveys on each other. Then we produced—we wrote together, led

by Igor Grossman, who gets a lot of credit for this work. But I was one of the co-authors, with a lot of these amazing people.

- 3.22** In 2019, we produced a wisdom consensus paper. We produced the consensus model of what the scientific study of wisdom is showing. And guess what one of the key features is? It's this perspectival ability. This meta-perspectival ability is one of the key essential features of a wise person. And you'll come back to see that. But I'm trying to show you right now the connection between absurdity, which is meta-perspectival, and wisdom as the alleviation of absurdity, which is also meta-perspectival.
- 3.23** So one of the things a wise person should be able to do for you is alleviate your sense of absurdity. That's different than them having knowledge. We'll come back to all of that. Just to show you. I keep throwing out these connections because I keep wanting you to be able to, well, anticipate where we're going. There might be a reason I'm doing that. OK.
- 3.24** Now, significance. I think this has to do with depth, realness. And I think this is where we get our sense of significance from. Depth—how deeply connected we are. And realness—how much sense is it making to you? How do you judge how things are real? Compare being in a dream to being awake. And we use awakening as a metaphor for enlightenment. What's happening in your dream doesn't make sense with all of the sense-making you're doing in your daily life. It doesn't fit in. See that? The generative models in your dreams don't typically—sometimes they do, and you get weirded out by that, and it's cool, and Jung built a career around that and stuff like that—but most of the time it's like, "Well, that wasn't real because it doesn't fit into this. "
- 3.25** Also, in some sense, you're very connected, but you're not as deeply connected. Dreams don't have that inexhaustableness that reality has to it. I can keep discovering more and more and more about this thing. If you practice lucid dreaming, try to find out more and more about a thing in your dream, and it empties out for you. This depth of intelligible connectedness—if you're vertically and horizontally, and hopefully both—you're profoundly connected and your relevance realization is pushing on the fact that reality is combinatorially explosive, that it's inexhaustible, that 's where you get the sense of realness—connectedness to that inexhaustible fount of intelligibility.
- 3.26** Now, in 2019, Costin and Vignolis did something very important around meaning in life. They did some research and ran some good experiments, and they were able to show that coherence—in the Heintzelman sense, not in the sense I've just argued, but in the Heintzelman sense—didn't make much of a difference to meaning in life. We've already seen that problem. But surprisingly, purpose didn't do that much for meaning in life. Which is important because, again, our culture tells you, "Get a purpose and you'll find your life meaningful." You can be very purpose-driven and still not have very much meaning in life. A purpose-driven makes you very productive, and other people want you to be productive because it helps them.

- 3.27** But I'm going to, not pun, but I'm going to do a self-referential thing here. What matters to meaning in life is mattering. Here's where you'll see this, even more, the case. Mattering is to be connected to something. To be connected to something that is real and valuable beyond your egocentric preferences. This is what we say when we are connected to something bigger than ourselves, larger than ourselves. We 'll come back to that in a sec. But you see? Religio, that sense of connectedness. That's why I said, "What do you want to exist even if you do not exist?" Mattering is what is most important to meaning in life. That sense of being connected to yourself, to each other, and the world. Now, when I say connected, I don't mean some trivial sense. I mean the religio sense that we talked about last time. That meta optimal gripping, that evolving framing anticipation.
- 3.28** Now, once again, we can turn to the philosophers to help clarify this notion of mattering. Two philosophers. One is Iris Murdoch in her seminal book, a little thin book. As I get older, I like thinner books because I have less time because death approaches. See, right now your temporal horizon hasn't flipped.
- 3.29** I'm past 35. Just by a quick glance, none of you are. Before 35, in general, people live from their birth. I'm moving away from my birth and I'm growing towards my adulthood. And then unbeknown—and so you know at one level you're mortal, but you're behaving as if you're immortal and enjoy it. I don't want to take anything away from you. But around 35, if you mature—because some people don't—but if you mature, it'll flip, and you'll stop living from your birth, you'll start living towards your death. It 's a different thing and it feels very differently. It's a perspectival shift. None of the propositions or facts of your life change, but it's a perspectival shift and it's profound.
- 3.30** So that's why I prefer these little books. And there's this little book by Iris Murdoch called *The Sovereignty of the Good*. And within that, she says something really, really interesting because it 's not a proposal that we usually hear. She said, "Love is when you acknowledge that something other than yourself is real." It 's a really profound way of thinking about it. Now, what does she mean? What does she mean? Love is when you acknowledge that something, that other than yourself is real.
- 3.31** It's that sense of being connected to something that has an existence above and beyond your egocentric preference. But we think of love as a feeling. That's ridiculous. Love is not a feeling. Because—even reducing any emotion to a feeling, you can have very—just let's take a classic emotion. Anger. Anger is an emotion, and you can have very different feelings in your body when you're angry. You can be boiling hot. You can be cold and calculating when you're angry. So reducing emotions to feeling—big mistake.
- 3.32** Love is not an emotion. What? Love is not an emotion. When I love someone, that can make me angry, happy, jealous, sad, lonely, fulfilled. When you love someone, it makes you open to a whole set and range of emotions. Do you see that? Love isn't an

emotion, and it's certainly not a feeling. The fact that our culture is trying to reduce love to a feeling is this big flattening of what love is. What is love? Love is an existential stance. It's a way in which the agent arena relationship is being co-created by you and something else. It's an emotional stance. It involves the participatory. You and another person are participating in your identity formations.

- 3.33** You're reciprocally opening to each other. That's how you fall in love, by the way—the work of Aron, not a last name, A-R-O-N. I'm not saying anything between us. I'm just saying, here 's a person, and if she discloses something to—what's your name?
[inaudible]
- 3.34** Oriana. If she discloses something to me—she opens up a little, and I respond by opening up rather than staying impersonal and factual, or attacking or exploiting her—if I open up a little bit and then she opens up, you get mutually accelerating disclosure, reciprocal opening. That's love. That's a participatory, perspectival thing. And we are binding each other to self-transformation, self-transcendence. That's what love is. Do you see what Murdoch is saying? She's saying, Love is mattering. It's religio. It's religio that's bound up into reciprocal opening.
- 3.35** By the way, reciprocal opening is the exact opposite of addiction. My friend and colleague, Mark Lewis, has provided the best account, I think, of what addiction is. He's one of the premier person. I know Mark, so I get to talk to him one-on-one. His theory is that addiction is reciprocal opening. So, what do I mean by that?
- 3.36** My world's upsetting me. I drink some alcohol to remove. But what that does is, it reduces my cognitive flexibility. You understand what that means. My ability to evolve, my generative modeling, starts to degrade. As I do that, the number of problems I can solve in the world goes down. And so the world now becomes a little bit more threatening and narrower. And then I drink a little bit more alcohol and get a little bit more anxious. I can't solve it. Do you see what's happening? I reciprocally narrow down until I can't be any different than I am, and the world can't be any different than it is, and then I'm addicted. My agency is lost, and there is only one arena I am in, and I feel trapped.
- 3.37** Love is the opposite of addiction. It is religio. It is about this fundamental mattering to something beyond yourself. If we want to respond to the issues around the threats to meaning in our culture, we need to teach people how to fall in love with reality again. We can't give them arguments. That's not enough. And now you can start to hear how we're going to be talking about spirituality.
- 3.38** OK, who's the second philosopher? The second philosopher is Susan Wolf. She wrote the book, *Meaning in Life and Why It Matters*. Nice pun, by the way. And she basically argues that meaning in life can't be reduced to morality. So we try to reduce meaning to feeling. And I just showed you why that can't work. Meaning is not feeling. Meaning is not feeling. I just showed you that. Meaning is not morality. You can lead a moral life

and it not be particularly meaningful because meaning in life, as she said, is about—she says, to repeat this, but this is a philosophical analysis now—it is the sense of being connected to something larger than oneself.

3.39 She says, obviously, that's a metaphor. If I chained you to a mountain, you wouldn't go, "Oh, now. " It's metaphorical. It's again, it's about religio. It's about being connected to something that has a reality and a value, or to use our language, a reality and a relevance to you beyond your ego-centric preferences. That's what meaning in life is. Meaning in life is this interesting thing because obviously it's involving religio. It's involving this connectedness. But notice it involves a fundamental shift that's important to us.

3.40 So, when I was talking about religio last lecture, I was talking about how things are relevant to you, how things are relevant to an autopoietic organism, right? Look which way the relevance arrow is going. Listen to the word "import"-ance, importing into me. Mattering, like I literally consume the matter—food, right? But in love, and this is the connection between love and mattering, the arrow goes the other way. Instead of "how is this or these relevant to me?", it's "how am I relevant to them? How am I relevant to that? Even if I cease to be, how could I have been relevant?" I'm turning the arrow of relevance around.

3.41 Why do I do that? I do that for a very good reason. I'm a primate. I'm a cultural primate — mammal. Mammals have to take care of their young. Humans have to take care of their young for a very long time. Our young are born very, very helpless for a very long time. It takes a minimum of around 15 years to bring them to maturity. Many organisms don't live that long. We have one of the longest childhoods. Why do we have such a long childhood? It has to be—one is biological. We have big brains and bipedal, so you have to squeeze a big brain through a narrow pelvis and all that stuff. The other is we're cultural, and it takes a long time to learn culture.

3.42 So, human beings have evolved a need to attach to beings that do not do anything for them. You're not—OK, I am continuing to be parent. You don 't love your child because —this is creepy, so I'm just making it clear. You notice that it 's so creepy you don't even—I'm not trying to consume my child in some way. There 's people that do that psychologically. They're vampires, and that's horrible. I 'm not with my child—now I am because they're adults—but I'm not with my newborn because we're good friends. You know, I do things with Jason, and then he, I take him out to dinner, and then next Tuesday, he'll take me out to—No, it's a baby. I love the child because my love has this capacity, not of consumption, not of cooperation, but of creation. I can turn this entity that is culturally, morally, it's a person, but culturally, cognitively, it's not a person.

3.43 It's like magic. By loving them—this is called agapic love—I turn them into a person. And you know what? I find persons inherently valuable because I find relevance realization inherently valuable. Anything that can do powerful relevance realization, I

find inherently valuable, and I want to commit myself to it. Agapic love, mattering. Turns the arrow around. Does that make sense?

3.44 So, there's religio, and we're wanting to enhance religio, and that's purpose, how things are satisfying my goals. Purpose matters to meaning in life. That's this way of religio. But mattering matters, and that's this way of religio. It's agape. That make sense? Now you can see again how this talk about intelligence, this scientific talk about intelligence, is getting us into understanding meaning in life, purpose, mattering, and the bi-directional enhancement of religio—not only enhancing it this way, but enhancing it this way.

3.45 But as I said last time, we're always at risk. The very processes that make us intelligently adaptive make us perpetually prone to self-deceptive, self-destructive behavior. Remember the no-free-lunch theorem? Heuristics are biases? We have to have the heuristics to avoid combinatorial explosion, but the heuristics necessarily bias us. Let's take a look at an example of that.

3.46 I've been doing this recently. Last year, I traveled a lot, and I saw this happen. When I'm on the other end, I do it, even though I know everything I'm going to say. So you take your friend to the airport. Now, you want to calculate the probability that something bad is going to happen to them because they're your friend. They're your loved one. Now, if you tried to do formal probability on the world, guess what? That's combinatorially explosive. That's computationally intractable. You can't do that.

3.47 So you use these two heuristics when you're doing probability. You use the availability heuristic, which is "how easily can I remember or imagine an event?" The more easily I can remember it, the more probable it is. And the representativeness heuristic: "How prototypical of this category is this event?" How much does it stand out? In the category of things it belongs to, how salient is it to you? The representativeness heuristic. These are good heuristics, by the way. Again, you can't do the formal probability. You take your friend to the airport, and the availability heuristic kicks in. How easily can you imagine an airplane crashing? "Oh, big metal in the sky, Homo erectus brain—it will fall." You know at an abstract level all the Newtonian stuff. But you're the brain—"the big metal thing, easily fall to ground." So, availability heuristic.

3.48 Oh, representative heuristic. When a plane crashes, do they say plane crash? No, they say "tragedy, disaster." It's representative. So, "Oh, no. The big object is going to disastrously fall to the ground." And then you say the euphemisms to them. You're basically saying to them, "Don't die." What you say is, "Have a safe trip. Text me when you're there." That's a great one, isn't it? "Text me when you're there." Because that'll save them. Because if the plane is about to crash, they can stand up and say, "Wait, I have to text my friend." And the plane goes, "Oh, right. OK."

3.49 We do all this stuff because we're imagining them crashing. And then we turn around and, without giving it a second thought, we get into our cars, which are the North

American death machines. They kill more young people than anything else. They're like the fourth biggest killer of all sources of death in North America. You don't give it a second thought because you don't imagine yourself crashing. "I'm in control." When a car crashes, it crashes. It may be a little blip on the news.

- 3.50** So you deceive yourself. We do it all the time, every day. So remember, you're always ignoring, you're always framing, and you're always identifying. And those things are adaptive, and they're simultaneously distortive. And the thing about these distortions is they're not single. Remember, your brain is recursive. Your intelligence is recursive and dynamic and self-organizing.
- 3.51** So, Leo Ferraro and I—I should leave a marker over here—and we published a paper on relevance realization, meaning, and the cognitive science of wisdom. We talked about the interaction between heuristics and the recursive and dynamically self-organizing nature of your intelligence. Let's trace this through. You have an event that happens to you, and the event is negative to you, interpreted as negative or bad in some way. So what you're going to do right now is, you're going to emotionally assess the probability of another event like that happening. Because why? Because you're an anticipatory entity. You're trying to generate a generative model. This was massive error signal, and your system is trying to correct for it. Adaptive? Yes, of course.
- 3.52** You're trying to calculate probability. Well, what do you use? You use the representativeness heuristic. Just talked about it. That interplays often with the availability heuristic, which I just talked about. They're interacting with each other. Now, that in turn interacts with what's called encoding specificity. This is, your brain tries to store the context of you solving a problem along with the problem. Why? Because your brain is trying to anticipate relevance realization. So if you're sad, you are more likely to remember being sad. If you're happy, you're more likely to remember being happy. If you're drunk, you're more likely to remember what happened to you when you're drunk than when you're sober. You just don't care.
- 3.53** So, something bad happens to you. You probably are in emotion, so it's standing out to you. It's very salient. It's triggering this. You can easily remember other instances when something bad like that happens, and then that calls up other instances, and all of this starts to reinforce itself. Do you see that? Now, both of these feed into a thing which we'll talk about today—the confirmation bias. Confirmation bias. This is the fact that you tend to only look for information that confirms your belief, not information that disconfirms it.
- 3.54** This triggers confirmation bias, which further reinforces it. Now, stop. Each one of these moves and the fact that it's self-organizing is adaptive. But you see what's happening? It's making use of all of this adaptive machinery in a way that's causing you to start to reciprocally narrow in a powerful way. OK.

- 3.55** Now you're going to presume that this negative event is highly likely. Another such negative event is quite likely. Now, what anxiety does, and this is part of the reciprocal narrowing, is it narrows your framing. Reduces your cognitive flexibility. It gets harder and harder for you to think of anything else. You've probably felt this when you've been in an anxiety spiral or an anger spiral or a sadness spiral or a loneliness spiral, on your 10,001 spirals. So you lose the cognitive flexibility.
- 3.56** OK. What does that mean? Your ability to solve problems goes down. Your ability to have insight, your ability to break out of bad framing, goes down. So, decline in problem-solving. So now you're starting to get a self-fulfilling prophecy. The world is getting worse for you because you're not solving your problems. You're not achieving your goals. Things are going badly for you. That's why a bad day gets worse.
- 3.57** So your sense of being able to affect things goes down. You're getting frustrated. And also the world is getting threatening, more threatening. And of course, these things come together to make you feel fatalistic. "This is inevitable. I can't get out of this." And that, of course, creates a kind of paranoia, which feeds back into the next event that happens to you. We call this parasitic processing. This is when the self-organizing, recursive, adaptive nature of your cognition organizes itself into a mind virus that sucks life from you and as it takes on a life of its own.
- 3.58** There's been further research. I saw one. I won't put it on the board because I couldn't possibly draw it. Coming out of MIT in 2015, and they did a diagram. So we were ahead of the time. You like it when your predictions come true. If you're a scientist, you celebrate. And they did one for depression, and it was exactly like this—parasitic processing—but way more complex. So parasitic processing, strongly convinced it's a thing.
- 3.59** Now, notice what this means, by the way. You can't do one-shot interventions with this. If you just intervene in one place, the system has the capacity to reorganize itself adaptively and maintain itself, which just telling yourself, "Oh, this isn't real," doesn't do much. Trying to change one thing, calm yourself down, doesn't do much because the system has this enormous capacity to be autopoietic, to maintain itself. That's why you find it so compelling.
- 3.60** I want to propose something to you. This isn't just irrationality—it is irrationality. It's massive self-deception and self-destructive—but it has this extra quality to it. I want to put it to you that this is an account of what foolishness is. Being foolish is being self-deceptive. And, so being irrational is being self-deceptive and self-destructive. We'll talk about that today. But being foolish is being comprehensively in this fashion, comprehensively and compelling, entrapped within parasitic processing.
- 3.61** You're going to need something that is going to have these features in order to address it. You need something that is itself a complex, multi-levelled dynamical system of intervention. There isn't going to be a panacea practice. There isn't going to be able to

be one thing you do to alleviate this. You're going to need—and I'm going to talk about this later, next lecture—you're going to need an ecology of practices. You know the way living things form an ecology, multi-level, and they're mutually affording and constraining each other, and they form a living system that maintains itself? You need a bunch of practices that that has this dynamical, recursive structure and that can intervene in parallel in a dynamic, evolving fashion with this as it tries to evolve and maintain itself. That's how you intervene in foolishness. Does that make sense?

3.62 This is why you can—your friend comes to you. First of all, think about this. I'm going to do two things. I'm going to do another example of how perspective makes a difference. When your friend is doing that same thing they keep doing in their romantic life that they've done before, and you try to say it to them, and they can't see it. But you know when the perspectives are reversed, you can't see it when you're doing yours? So, even when your friend does wisely tell you, "You're doing it again," is that enough? Typically, it's not. This is why people go into therapy. They know their pattern. They, in some sense, know their problem. But they can't deal with this. They try one-shot intervention. Their friend gives them this piece of advice, so that friend—you know this, and it's largely impotent. All right. So keep that in mind.

3.63 Now what I want to do is I want to go into a deeper exploration of this issue of bias—
heuristic and bias—and its connection to irrationality. We're going to come back to foolishness, but I wanted to lay this out for you. This takes us into a whole bunch of experiments that were done in the '70s, '80s, '90s, they continue. And unlike other areas of psychology where there's replication, these things massively replicate, robustly replicate, replicate across many researchers, different theoretical frameworks. So this stuff is really secure. It's some of the most secure stuff we have in psychology. If you ask me to place my bets on the kind of findings that aren't going to suffer failure of replication, these are them. I'm not going to go over experiment by experiment because there's about 500. Even if I did one minute on each, that's more than the rest of our time. I'm going to give you prototypical exemplars.

3.64 Now, normally what I'm presenting with you is part of a larger set of data that you present to the participant, so it's not so obvious what the experiment is, and that's for reasons of trying to get, remove confounds. I'm not going to do that. I'm going to go right into the core thing. OK, so consider this. Consider the following problem.

3.65 Linda is a 31-year-old single, outspoken and very bright person. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice. This has become even more pertinent. And also participated in anti-nuclear demonstrations. Which is more likely? Linda is a bank teller. Linda is a bank teller and active in the feminist movement.

3.66 Now, overwhelmingly, people pick B. But that's false. It can't be. Here's all the bank tellers. Here's all the feminists. Here's all the people that are bank tellers and a feminist. It's necessarily less. A conjunction's probability is necessarily, necessarily less

than the probability of either one of the conjuncts. The correct answer is "Linda 's a bank teller." That's more probable than "Linda's a bank teller and a feminist."

- 3.67** But you wanted to say B. Almost everybody in the experiment wants to say B. Why? What's going on? None of you are disagreeing with this. None of you are, "No, no, no." You get, "Yes, yes." And you see the necessity of it. You see it's impossible for the conjunct to be more than one of the conjunctions. Because even if they completely overlap, that would just be, at best, equivalence. It can't ever be more. You see that, right? And yet you want to say, "Linda is a bank teller and a feminist."
- 3.68** Now, think about what's going on here. Think about what I've taught you. People are considering that the description of Linda is relevant, and it's not. All you have to hear is conjunct or conjunction. But they consider the description of Linda relevant, and then they're trying to predict a causal relationship between those facts and Linda. What they're doing is they're doing relevance realization, predictive processing. They're trying to form a generative model of Linda so that they can anticipate how Linda might be acting in the future. They're doing all that massively adaptive stuff. So that's why they say, "Linda's a bank teller and a feminist." All that adaptive machinery comes in and it blows this out of the water.
- 3.69** Now, here's a question. Notice the problem we're facing here. This is why rationality is a problem. Should we tell people, "Abandon your generative modeling. Stop doing your relevance realization, predictive processing"? It's like, "Oh, no. A combinatorial explosion, cognitive suicide, meaningless existence. No, thank you." They're not going to do that. They're not going to give up their cognitive agency and meaning in life.
- 3.70** But can we deny the logic of this? Can't. How can we get those two together? It's like I have to say, "Do your generative modeling, but sometimes don't." Sometimes follow the logic, but not all the time. We don't want something wishy-washy like that in our understanding of rationality. It's like, "You know how you're rational—" "Well, when you—". That 's the problem. That's the problem.
- 3.71** Let's take a look at another example. OK, I'm going to give you two problems, and this one's a little bit more complex. Let me get through the whole thing, and then I'll say what was going on. I'll write them side by side so you'll be able to make a comparison. I'm going to give you two problems.
- 3.72** Problem one. There 's a disease. The problem is a disease kills 600 people, and 200 alternative programs are available to combat the disease. OK, so program A. So there's 600 people will be killed by the disease if nothing happens, right? So program A, 200 people will be saved. And probability B is one-third of the people will be saved, two-thirds killed. OK.
- 3.73** And here's problem two. Same problem in that a disease kills 600 people and there's two alternatives. I'm going to use C and D here. C is 400 people die, and D is one -third probability no one dies, and two -thirds probability everyone dies.

- 3.74** OK, you give these two problems to people and you ask them to pick which alternative they want. Overwhelmingly, people pick A and D. Overwhelmingly, they pick A and D. "Why?" you might think. What is made salient? OK. Do you see? People shouldn't—if they choose A, they should choose C, because they're logically equivalent. Or if they choose B, they should choose D. But they choose the diagonal because of what is made prominent. This is called—and I kid you not—this is called a framing effect. The framing of the information—what 's foregrounded and what's backgrounded, what's made salient to people—impacts the decision even though the information is logically identical.
- 3.75** So what are people doing? They're framing the information. They're quickly zeroing in and trying to generate a generative model. And do you tell them, "Stop doing that"? Or, "Don't do it here. Don't do it here." So these are called framing effects. OK, here's another one. So belief perseverance or belief perseverance, both pronunciations are used, and I'll probably flip between them—because as a Canadian, I flip between being American and British. Canada is a child from England and the United States that was left in the snow. So, as John Oliver says.
- 3.76** One very important principle of rationality is altering your belief based on the evidence for it. When the evidence for a belief changes, you should alter your belief in it. And all of you are nodding, "Yes, of course." The problem is, how do you study that scientifically? Because you can't do it on the beliefs they come in with because you don't know the evidence that they have for their belief or how they acquire it. What you have to do is create a belief in the experiment itself, and then alter the evidence for that belief in the experiment.
- 3.77** Some people thought of a way of doing this that was extremely clever. What did they do? They divided people randomly into three groups. Each group was given notes to read, and they were asked to determine which notes were real or fraudulent suicide notes. And the idea is because, if we could do this, we could better figure out what's going on when people were about to commit suicide. It's a great experimental deceit because people are, "Oh, yeah, yeah, yeah." Do you have this ability? "No." Would you think it's good? "Yes, yes."
- 3.78** So what you do, and then just based randomly on which group they are assigned to, you give them three kinds of feedback. One group is like, "No, you keep getting this wrong. Sorry." The other group is, "You got another one right. That's right, too. Sometimes, not that one, but yeah, that one's right." A lot of positive feedback, a lot of negative feedback. Neutral group, on average: "Yeah, no." Then, of course, you ask people, "What do you believe is your ability to determine, distinguish real notes from fraudulent notes?" "I can't do it." "I don't know." "I can do it." That make sense? You create the belief. It's a belief people want to identify with. "I want to have that belief. I could save lives."

- 3.79** Then what you do is, you completely debunk it for them. You say, "Hey, this is what happened. None of these notes are real. You were assigned randomly to each group. You were given feedback just based on which group you were assigned to. That's all that's happened in the experiment. That's all." Then you give them some other detractor tests. We're going to get you to do a reading comprehension test. You do a few other tasks with them.
- 3.80** Then you come back and you unexpectedly say, "Hey, how do you think you would actually do in real life on suicide notes?" "I do really crappy." "I don't know." "I'd do really well." The belief persists even though the only evidence that generated it was completely destroyed right then and there shortly in the past. They can easily remember it. The belief persists. What's going on there?
- 3.81** Well, there's a couple of things. First of all, do you want people to be continually determining the coherence of all of their beliefs with all of the evidence they have for it? If you were to try to check if all your beliefs were consistent with all the experiences you've had and with all your other beliefs, what do you think the kind of task that is? Combinatorially explosive. So you don't want people doing that in general. And people are identifying with valuable beliefs. "Hey, I could save lives." You want them to not identify with valuable beliefs? No, that's adaptive. That's important. That's how we get good agency. So again, do you tell them, "Stop not checking for the consistency of all your beliefs"? No, we don't want to tell them. "Stop identifying with agent-enhancing, morally valuable beliefs." No, we don't want to stop. But sometimes stop. Sometimes don't do that. Do you see the problem? Can you feel the tension?
- 3.82** This lines up this stuff about how you identify with it—that participatory level—with what's called failure of critical detachment, or more frequently today, motivated reasoning. How do you do this? You get people—before an experiment, you measure how committed they are to a particular proposition. I'm not doing this debate here. I'm just giving it as an example: "Abortion is right, or abortion is wrong." People very rarely, nowadays, have sort of—they have strongly identified, right? You can't pick a belief that's relatively neutral. Like, is vanilla ice cream good? It's got to be one that motivates people. They identify with it. Participatory? Participatory? OK.
- 3.83** Now, you put them into two situations. One, you give them a bad argument—meaning, not valid, not well-structured, not well-evidenced. All the ways in which arguments can go bad, and you can manipulate this. It doesn't matter because the same result comes out, to the conclusion they like. Or you give them a good argument—one that leads to a conclusion they don't like. Then what you do is you ask them to evaluate the argument. Is this a good argument? Is this a good argument?
- 3.84** Now, what you find, and I'll give you a particular example in a second, is people will overwhelmingly say, "This is a good argument and this is a bad argument." Now, we'll talk about this person quite a bit today. Stanovich argues that that's one of the fundamental things driving irrationality. I'm going to put it this way. When you focus on

the cognitive product, and you ignore the cognitive process, when you don't look at the framing, you only look through it to what it's pointing to. Do you understand? So there's a lack of perspectival processing going on. There's no meta-perspectival. There's no shifting of perspective.

3.85 So this was an experiment done by Scurich, one of many experiments. This is not the only one. I'm just giving you one example, if you want to look it up. Scurich and Shniderman, 2014. Again, these results are always robust. And they basically did this. They gave people a supposed newspaper article, and they were divided into two groups—people that supported the death penalty and people that were opposed to it—another thing that motivates people very strongly. And by the way, just as an aside, politics tends to exacerbate this phenomena.

3.86 As you can see. OK. So the research article, the newspaper, claimed to be talking about a research article that recent neuroscientific evidence showed that the brain activity, specifically in the frontal lobe of violent offenders, was affected by a film they watched—either a film of someone undergoing execution or one of a prisoner being held in solitary confinement. Importantly, the experimental data was held constant. Everything is held constant. But the interpretation in the two versions was radically different. One interpretation was the death penalty would deter crime, and the other was exactly the opposite interpretation. The results are exactly as you—significantly—like, I won't put the chart up in detail, but I'll just show you briefly. You can see the line of the difference. It's a very steep difference, very steep line here.

3.87 Now, what that means is, there is a radical difference in people's judgment of how good the neuroscientific research is, determined only by whether or not it agrees with their previously held belief. If the article led to the interpretation that they liked, it was a great article, so well written. If it led to exactly the opposite, it was poorly written, badly argued article. The article was held constant. The only determining difference, and I showed you how significant that difference is, is whether or not it lined up with people's previously held beliefs—ones that they were committed to, that they identified with. They can't do the perspectival shift. They can't disidentify—participatory—enough to be able to evaluate the arguments.

3.88 Let's do another one. This is from 1966 called the Wason Selection Task. It's based on the very simple reasoning, if-then reasoning. If P, then Q. P, then Q. If it's raining, then the streets are wet. It's raining, therefore the streets are wet. That basic kind of reasoning pattern. When you do things like, if it's a dog, then it's a mammal. It's a dog, therefore it's a mammal. You have these cards. Which cards should you turn over to determine if the following rule is true? The rule is that if there's a vowel on one side, it has an even number on the other side. There's a vowel on one side, it has an even number on the other side.

3.89 Now, many people—these are the two response patterns. That or that and that. When you put those together, that's 90% of the responses. Those are the two cards you

should turn over. Turning over this can confirm the rule and also disconfirm the rule. Yes? Does turning over this add any extra value? No, it can't confirm and it can't disconfirm. It could confirm, but it can't disconfirm. Could this disconfirm the rule? Yes. You know what two cards you should turn over are these two. That gives you confirmation evidence and disconfirmation evidence. Do you see that?

- 3.90** What are the people in, 90% of the people doing? They're only looking for confirmation. They're not looking for, and they're not even looking for confirmation well. They're doing it inefficiently. They don't look for disconfirmation. That's the confirmation bias that I promised to talk to you about. 90% of people get this wrong, robustly, reliably. You know what doesn't help you do better on this task? Training in formal logic, which is really weird.
- 3.91** OK, so what's going on there? Well, this one's a little bit more tricky, and we won't get to it all the way today. But I want to show you one thing that makes this radically different. And how it has to move you beyond the propositional and inferential. We're going to talk about this work next time. This is Mercier and Sperber and other people. Take this same problem and give it to four people and give them time to talk to each other about it. Now, the response rate goes from 10% correct—because here it's 90% false—goes from 10% to 80%.
- 3.92** If people are willing to become meta-perspectival by taking the perspective of others and can take into consideration the identity of other people so they're not strongly identified—so if they address the perspectival in the participatory, they go from 10% to 80%. Now, this says something really powerful. First of all, we shouldn't be yelling at each other. Politics exacerbates motivated reasoning. We should be dialoging with each other because that actually helps us overcome all of these biases in a significant way.
- 3.93** And it tells you something else. Since Descartes—"Cogito, ergo sum"—we have thought of reason as something monological, something done in the individual mind by itself in monologue. That actually makes no evolutionary sense because, for 99% of our history as a species, we were in hunter-gatherer groups, where we were solving problems collectively, not individually, and we were relying on the collective intelligence of distributed cognition, rather than the monologic intelligence of individual cognition.
- 3.94** We work best in not acrimonious conflict, but in cooperative dialogue. That is what we evolved and how we evolved to be rational. Which means doing things, by the way, that get us into participatory and perspectival synchrony so that we feel bound together. Maybe that's one of the functions of rituals. Could, weirdly, actually, be very important to our rationality.
- 3.95** We'll get to that in the next lecture. We're going to take a break, and then we're going to come back and go through more of this, especially with what's respect to called the rationality debate. A bunch of philosophers responded to all this experiment and said, "This can't be right. How can human beings be comprehensively irrational? That makes

no sense." We'll come back and take a look at that and what that will teach us about the nature of rationality.

Lecture 4: The Context of Rationality

At a glance — The context of rationality and the experimental work on human reasoning.

- 4.1 Welcome back. So we're taking a look at all of this experimental work. And of course, the philosophers didn't just sit back and take it. Human beings are irrational. And we've already seen there's a tension in this. Are we telling people to just give up all of this adaptive relevance realization, predictive processing, generative modeling, anticipation? That doesn't make any sense.
- 4.2 So the first important response to all of this experimental work, all this work on heuristics and biases and all the stuff I've shown you, was by Cohen in a behavioral, in an article on behavioral and brain sciences in 1981. And Cohen made a really good argument. He said, "Look, the problem we're facing is, if human beings are irrational, where do we get our standards of rationality from?"
- 4.3 If we're so irrational, why do we have these standards for logic and math and probability? Where are they coming from? And you might be saying, "well, you know"—and again, I'm not here to criticize religion—"well, God told us," or something like that. Plato made, a long time ago, why that answer can't work. It's in a dialogue called the Euthyphro Paradox. It goes something like this.
- 4.4 Is something good because God says it's good? Well, if that's the case, then it's not really good. It's just arbitrarily so. Because if God just said the other thing, then that would make that thing good. So it wouldn't be inherently good. But we don't find those things to be arbitrarily good. We find them to be necessarily the right way—the right way, the good way to reason. When I gave you those two intersecting diagrams, you didn't go, "Well, for now." You went, "No, no, it has to be that way. That's the way I should think."
- 4.5 The other thing is, if God says it because it's good, then it's good independently of God saying it. We have to have an independent account. This is a point made famous by Kant, but the idea is we have to participate in the standards of rationality. Because if we do it just because we're told to do it, then it's not really rationality. If we obey it just because we're afraid, then it's not really rationality. Do you understand? It's the nature of rationality that we have to participate in it.
- 4.6 So, Cohen's right. We have to be the source of the norms. We have to be the source of the standards of rationality. But then if we're the source, why do we make all these mistakes? Why do we make all these mistakes? So he came up with an ingenious answer, and it requires to me to make a distinction between performance errors and competence errors.
- 4.7 So there's two types of errors you're making, performance and competence. I want you to compare two people. First person is really drunk. And you ask them, "How's it

going?" They're flowing with—. You don't go, "Oh, no, they've lost English." Or you come up to somebody. She's asleep. I start talking to her and I go, "How's it going?" And she doesn't answer me. And I go, "Oh, no, she's lost English." You go, "That's silly." Compare that to, he's just been in a horrible car accident and he can't speak correctly. Now you didn't laugh. He's lost English. Or here's a little girl. She's two years old. And she can't speak correctly, and you go, "Oh, no, she's never going to get English." See? So intuitively, you are making distinctions between performance errors and competence errors. Performance is what you do. Competence is what you're capable of doing.

- 4.8** Performance errors are due to a lot of things interfering with how you express your competence. If you're drunk, it's not that you've lost English. The machinery for speaking is being impaired. Or when you're asleep, the machinery for listening is being impaired. That doesn't mean the competence. But when there's been brain damage, the competence has been destroyed. Or when the child is young, the competence has not been effectively developed yet. Do you see the difference between competence errors and performance errors?
- 4.9** Now, here's the thing. Take language speaking, the thing I'm using. Do you know that most of the time we don't speak grammatically correctly? We leave things out. I just did it. We pause, we cough, we repeat words, we don't finish our—and yet, we ignore those because those are performance errors. Why do we make all these performance errors? Because we're in this situation, there's limited time, working memory is limited, we're doing a lot of stuff.
- 4.10** So this is Cohen's argument. We're making performance errors, but our competence actually contains all the standards for rationality, all the standards for probability and logic in math. We are the source of our norms, but we often don't completely follow those norms because of performance errors. Now, that's ingenious. Because that's clearly the case with language and other things. I just did it, by the way. Say, "Wait." Say, "I just did it." The performance errors. You don't go, "Oh, no. I better get out of this course. The professor's lost it."
- 4.11** Now, that's brilliant. The problem is it turns out to be false because Stanovich and West—Stanovic was also at the University of Toronto in the Center for Applied Cognitive Science. He's emeritus now. So if you're thinking all the really important stuff is coming out of the University of Toronto, you're correct. That's actually right. OK, so Stanovich and West, in 2000, had a brilliant response to this. They said, well, how—remember when I gave you the two examples, the drunk person and the person in the accident? Or let's do the little kid, the child. How do you distinguish between performance and competence errors when there's something just wrong here or something that's wrong here. Well, you look for something. It's called systematicity in the error. Systematicity.
- 4.12** Here's the little girl. She's at the playground and she's making those errors. But here's the little girl and she's at home at dinner and she's making those errors. And here's the

little girl and she's at school and she's making those errors. So, when she makes the errors here, I can predict that she will make the errors in many different contexts. There's a systematicity to the error.

- 4.13** Now, the drunk person isn't like that. They make these errors while they're drunk, but when they're not drunk, do they make the errors anymore? When she's asleep, she's making these errors. Does she make those errors when she's awake? No. So if the errors are systematic, it points to problems in the competence. If the errors are due to circumstance and chance, they're performative errors. Does that make sense?
- 4.14** So what can we do? We can take a look at all of these experiments. Remember when we talked about Spearman and the positive manifold? How you do on any one test is predictive of how you do on the other test? So if I can find a positive manifold, that means I have systematicity. Because how I'm doing on any of these tests, if it's strongly predictive of how I'm doing on all the others, then that pattern is systematic. Does that make sense?
- 4.15** So this is what I can do. I can look for how—if people make errors on belief perseverance, do they make errors on the Wason Selection Task? Do they make errors on motivated reason? I can look for their performance on all these tasks and see if there's a predictive relationship between them. Because if I find a predictive relationship, there's systematicity. If I don't find a predictive relationship, there isn't systematicity.
- 4.16** We have all the data. We can say the errors that people are making in these reasoning experiments, are they systematic, which would point to performance? Are they non-systematic? Sorry, systematic, which would point to competence? Or are they non-systematic, which means the errors are just performance errors, like Cohen says? Strong positive manifold in the errors. Very strong positive manifold. How you do—the kind of error you make on any one of these kinds of experiments is strongly predictive of how you do on all the rest. Positive manifold, strong systematicity —therefore, this is not an error at the level of performance. This is a level at the error of competence.
- 4.17** Yet, how can Cohen be wrong? Cohen has to be right, and yet the way he tried to explain it doesn't turn out to be right. How do we get out of it? Do you see why rationality is actually hard to study? How do we get out of this? Well, here's how we get out of this. This is basically what Stanovich is going to do. He's going to argue that—he doesn't put it quite this way, but this is what he does in practice. You have to challenge some of the assumptions that are in Cohen's argument.
- 4.18** Cohen's argument assumes that there is a single static individual competence. That there's only one competence in you that makes you rational, that it is unchanging, and that it's unaffected by whether or not you're working with other people. All of these assumptions, I'm going to argue, are false. You've already seen some evidence for that.

So we are the source of the norms, but we don't have a single static individual competence.

- 4.19 By the way, this is one of the arguments that backs up the idea that there's more than one knowing. You can't have a single kind of knowing because it is inconsistent with this argument, these two arguments put together side by side. And of course, I hope I've convinced you that your competence is inherently developmental. It's evolving in an important way.
- 4.20 OK, that was the first response. The second response you're already familiar with, so I won't need to go into it in great detail, but I will need to make use of Stanovich and West. Gold standard, by the way, for how to do great cognitive science. Gold standard, even though I'm going to critically respond to it.
- 4.21 So this is the work that you've heard before, by Cherniak on Minimal Rationality, 1990. Remember, he was the person who said, "You can't check all the implications. You can't check to see if all your beliefs are consistent," all of that stuff. OK. Now, what he says is, what's wrong with the experiment is that human—the experimenters are using the wrong normative theory. They're using algorithmic standards. And as I think I've convinced you, human beings, by and large, can't do algorithmic. So it has to be stronger than arbitrary, but not as strong as algorithmic. It has to be that adaptive in between. OK. That's a powerful argument. And Stanovich and West have no counterargument to that. They treat Cherniak, in fact, with a lot more respect than they do Cohen in the article. You can hear it in their tone.
- 4.22 And what that tells us right away is that rationality can't equal logicity. We can't use these terms as synonyms. When we mean to say "be rational," we say "be logical." But we're not allowing people to be arbitrary. If Peter says, "I don't believe in vampires," and we know that he's always wearing crucifixes, he's carrying holy water around, he has a collection of wooden stakes, and he won't travel to Transylvania, we say, "You believe in vampires." So logicity matters, but it's nowhere near identical to rationality. Cherniak is right about that. We're always in the finitary predicament. That's what Cherniak says. That's, we're always facing, combinatorial explosion, etc.
- 4.23 OK, so how do Stanovich and West respond to that very powerful argument? Well, they make a very important response. They say—well, they don't say it, but this is what they end up implying and conveying. Cherniak is actually proposing that rationality equals intelligence. Notice what I was talking about in the last lecture—I was talking about intelligence, and intelligence definitely has to do this. And Stanovich and West agree that intelligence is our ability to deal with computational limitations, to avoid computational intractability, to avoid combinatorial explosion. They agree. That's what intelligence is. They agree. And they say, "Cherniak is right about that." But what is wrong is he's wrong in equating intelligence with rationality.

- 4.24 I've already said we have a measure of your general intelligence—G, yes? That's how it's usually written. Now, we've already seen that there is a strong positive manifold between all of the reasoning tasks. There's some general ability there. Yes? The systematicity. They all mutually predict each other. So we have something—they don't do this, I'm just proposing this as something—there's something like R behind all the reasoning tasks because they all mutually strongly predict each other.
- 4.25 If Cherniak is right, measures of this should approach parity of measures of that. That's a simple argument. They're brilliant, Stanovich and West. Again, it's like—see, each one of these arguments have to be right in some way, and yet they have a way of responding to it, which is —So what's the correlation between these? On average, around 0.3. You know the correlation goes from zero to one. It's very far from one. Now, this is a significant finding. If you get a finding like this in social science, you're allowed to publish.
- 4.26 But that's not what we're doing here. But this shows—and this is very important—intelligence is a necessary but not sufficient cause of rationality. So there's no contradiction in saying the following: "You can be highly, highly intelligent and be highly, highly irrational." There is no contradiction whatsoever. So Cherniak is wrong to identify intelligence and rationality.
- 4.27 Now, that's good news because many people get hung up on the fact—I was talking to a couple of students during the break. "Well, measures of IQ are largely stable. I can't change my intelligence. And, oh, no." That's basically because what you actually want to be able to change is your rationality. That's extremely malleable. That's extremely malleable.
- 4.28 OK, so intelligence is relevance realization, but there's something more going on. It's kind of like intelligence is your raw relevance realization, but there's something more going on in rationality. So what do we have? Logicality does not equal rationality, but neither is rationality just being smart. So you can be smart and periodically logical and not rational. And you've met people like that. Now, don't get too self-congratulatory because we all are subject to the bias that we think we are more intelligent than on average, and we're more rational than on average. And of course, on average, that has to be wrong. I have no doubt that many of you, though, are above average, given how you've been talking to me and interacting with the material.
- 4.29 All right. How should we think of rationality then? Well, let's take a look at the word itself. Ration, which is to properly proportion, or even ratio, which is to find the proper proportion between things. That's what ratio—I gave you the Latin pronunciation to bring it out, ratio. What does that mean? That means this proper proportioning. You often tell people to put things into perspective. Don't blow things out of proportion. Properly proportioning the perspective and the attention, and then within that, what we're asking people to do is to properly proportion their attention, to properly dispose the perspective in situations in which tracking the truth is most important.

- 4.30 In many situations, that's not the case. Tracking the truth is in relation to other things that we are trying to do, like solve a problem, communicate with other people, make sure we are properly oriented, make sure we are properly connected. Perhaps rationality is about properly proportioning relevance realization and anticipation, so that it most reliably tracks the truth when that is our most relevant goal, which is not the same thing as saying you should always be logical. You should always be mathematical.
- 4.31 So this sounds odd, but this is what I'm saying if I was to represent it visually to you. Intelligence is about learning something that allows you to improve the use of your intelligence. Namely, your intelligence, while you're solving your problems, causes self-deception. But you can use your intelligence to acquire something that helps to ameliorate the self-deception that almost always arises when you're using your intelligence. So intelligence is what you use to solve your problems. Rationality is what you use to deal with the self-deception that emerges when you're trying to solve your problems and using your intelligence.
- 4.32 But what's in that box? What's in that box? You can learn a cognitive style. This is the term that Stanovich used, other people use. This is a set of skills. Listen to the words I'm using. Skills. "Oh, procedural." Sensitivities. "Oh, perspectival." Motivations, ways you identify with yourself or disidentify with yourself. Basically, you're collating and coordinating and cohering together a set of skills, a set of perspectives, a set of character traits that are sensitive to particular kinds of contexts and helped you adapt to those contexts in a way such that you are challenging your proclivity to self-deception.
- 4.33 So you're saying, "That's very abstract. Lots of multi-syllabic words. They're paying him too much as a professor." OK. Concrete version of this, the one that Stanovich and others talk about, is active open-mindedness. What does that mean? And pay careful attention. Sensitize yourself to the words I'm using here. You want to practice noticing instances of cognitive bias. And then you want to train yourself to acquire the habit of counteracting them. "I bet I'm falling prey to confirmation bias right now. I'm going to —here's the habit to motivate me, and I'm going to disidentify, and I'm going to have some humility." Remember, humility? "I'm going to disidentify, and I'm going to look for this confirming evidence." How could I do that? "Well, I could really open myself up to dialogue with other people, in really good faith."
- 4.34 Cognitive behavioral therapy makes use of collective open-mindedness. It makes use of reframing strategies, noticing bias, and training you. You have to go through the manual, you do the therapy, and you acquire the cognitive style. But CBT is itself based on Stoicism, which was ancient philosophical religious framework in which people were basically internalizing Socrates. They were learning how to be Socrates with themselves.

- 4.35 So Socrates was famous for entering into dialogue with people and drawing them out and showing them how they had inconsistencies in their beliefs or what they were doing, wasn't actually jiving with what they were saying. But he was also, constantly, challenged himself and undermined his own arguments. And what you want to learn to do in Stoicism is, you learn how—you know how you're all, you're talking to yourself all day long in your head. If we put a mic on everybody's head, we would all sound insane.
- 4.36 But Antisthenes, one of the followers of Socrates who started this Cynic movement — Cynicism, ancient Cynicism, doesn't mean what modern cynicism means. And then the Cynics lead into the Stoics. Basically, they asked Antisthenes, "What did you learn from Socrates?" Plato writes all these dialogues. Antisthenes said, "I learned how to dialogue with myself." He doesn't mean "do the chatter." He means "I learned how to be able to do like what Socrates does with me by myself. I internalized that." And that's a way of training active open-mindedness. You can internalize a Socratic sage. You can go through CBT. You can just practice active open-mindedness. The more you practice active open-mindedness, the more likely you will do this and become rational.
- 4.37 So the cognitive style accounts for most of the variance, most of the difference in what makes people rational or irrational. And you can learn a cognitive style, but it's not propositional learning. It's not just learning a bunch of beliefs. You have to practice skills. You have to be able to shift perspectives. You have to participate. You have to cultivate habits and traits and virtues. And all of that is managing your attention. And only after all of that is done, only all of that proper proportioning is done, then you do your inferences. Then you do the logic. Then you do the probability. Does that make sense?
- 4.38 Rationality is not just reasoning, not just the inferential manipulation of propositions according to the rules of implication. Rationality is also ratio. So I'm giving you another Latin word. We had religio, now we have ratio. Ratio is all about the proper proportioning of attention—the cultivation of skills, perspectives, identities, virtues, habits. Ratio is the way in which you acquire and appropriately apply a cognitive style, the proper proportion. Appropriate acquisition and application of a cognitive style that makes use of all this non-propositional knowing.
- 4.39 OK, now all of that is well said. Are there any problems with Stanovich? Yes, there are four main problems with Stanovich's work. And it's like, "Why can't we just conclude?" I get it. But given what I've just said, science is not about its products. It's not about its conclusions. Science is about its process. It's about its ratio. We should think of science more as a way in which we can be collectively rational, as opposed to a mechanism for generating an indisputable facts that we all have to adhere to.
- 4.40 OK, so Stanovich. I'm going to make an argument. Sorry, Stanovich doesn't know this argument, and it's an argument by Smedslund, and Smedslund doesn't know—he's not actually attacking Stanovich because he's writing in 1970, way before. So, Stanovich knows this argument, but he doesn't seem to attribute it to Smedslund. And as I said,

Smedslund is not anticipating. So the best way of putting this is like what they do in the movies. This is an argument inspired by Smedslund.

- 4.41 So, part of it is, Smedslund wrote in 1970, 1970, and part of it is, he wrote in the Journal of Scandinavian Psychology, which nobody outside of Scandinavia reads. Now, his main point is that logicity and understanding are in a circle. And you're going to go, "What does that mean?" I'm going to explain it to you. That's my job.
- 4.42 So Smedslund asks, "Say I state some proposition —X. How do I know if you've understood X?" Well, I ask you for four things. Can you give me something identical to X, equivalent to X? The cat is on the mat. Can you give me, "Well, there's a furry feline, and it's on that stuff that has been woven that way." If you say, "The cat is on the mat," and you say to me, "The truck hit the car," you don't understand the cat is on the mat.
- 4.43 What is implied by X? The cat's on the mat, then the cat is not in the kitchen. Stuff like that. What contradicts X? If you can tell me what contradicts X, then you understand X. Here's the one, of course, that you knew was coming, but what's irrelevant to X? Which, of course, is also symmetrical with saying what's relevant to X? OK. You say, "Yeah, that's pretty good. That's what, that's what it is to understand something." Now, notice that this is what's happening in all the reasoning experiments. Let's do the Linda example.
- 4.44 The participants in the experiment think the Linda description is relevant, but the scientists think it's irrelevant. They draw implications other than the ones the scientists do this. The scientists do the pure probability. The participants draw different ones. They don't treat things equivalently the way—so what it means for the scientists to say that Linda is a bank teller doesn't exactly mean the same thing as what the participants mean. Because they're forming a generative model, just not, just an exact statement.
- 4.45 Do you see, in all of these experiments, the participants are failing on one, two, three, or even all four of these. Which is, I can completely, legitimately, logically say, you know what these experiments provide evidence for? That the participants are completely misunderstanding the researchers because they're drawing different implications, different judgments of relevance, not finding the relevant contradictions and not having the same identities. Remember the framing effect?
- 4.46 Just think about this, logically. We think, "Oh, no, it shows that they're irrational." But there's nothing wrong with saying, in fact, it is completely, logically, legitimate to say, "No, no, you've provided an overwhelming evidence that the researchers are not being understood by the participants."
- 4.47 See, we have two things we're trying to distinguish here. We have good understanding. They have understood the problem the way the researchers understand the problem, and then they've done poor inference. And that produces the bad results. That's the standard interpretation by Kahneman and Tversky, people like that. But what if we have poor understanding with good inference? That would also lead to bad results.

- 4.48 And notice you're in a weird bind because you have to assume—to think about this, you have to assume that the participants in the experiment are very good at understanding because you're giving them an experiment to understand. You have to assume they're very good at understanding, which means they're very good at finding the identities to what you propose, very good at drawing out the implications. You're proposing that they're actually really good at all of these inferential abilities, and then you're concluding that they're not good at those inferential abilities. They must have all of these abilities in order to understand the experiment well, but then they don't have them when they reason. This is what I mean where there is a circularity between understanding and logicity, or following the rules of implication in inference.
- 4.49 How do you get out of that circle? Well, Stanovich and West get this. But here's one of where—and I want to say they're good researchers, they're honest about this. This is what they say. They realize that this is a problem, and they say we need an independent normativity on construal. OK, what's normativity? Normativity is your theory of the standards that you should follow. What's construal? Well, construal is about how you size up. It's that perspectival knowing. You've heard the word: "How do you construe this situation?" It's how are you doing that perspectival, procedural, participatory stuff so that you understand, make sense of this situation the right way?
- 4.50 They invoke it and they say, "We need something independent." Independent in what way? It has to be independent of inferential normativity. Because if I make it part of inferential normativity, I'm trapped in the circle. You see, it has to be a non-inferential thing. It has to be a normativity independent of how inference works. Because I have to be able to judge. If I want to break the circle between understanding and inference, I have to be able to judge the understanding independently of judging the inference.
- 4.51 Because if I keep them together like Smedslund, I'm trapped. So I need a non-inferential normativity on construal. Do you understand their argument? And then they say, "And we don't have one." And I say, "Yes, you do." That's why I disagree with them. Because we do have a normativity on construal. You have to look in the problem-solving literature, not the reasoning literature. A good construal is a good problem framing. A bad construal is a bad problem framing. You know what that means. You know what that means.
- 4.52 That means we have a normativity not around inference, but a normativity around insight—your ability to move from a bad problem framing to a good problem framing. So we do have that. They don't know it because, and they're good researchers, they're not really connecting the reasoning literature to the problem-solving literature. But they should. Because it's not that intelligence and rationality are completely independent from each other. That's a mistake. So we have what they say is missing. So Smedslund's argument basically goes through, but it leads to the conclusion—again, we need something non-inferential, non-propositional. We need a normativity on construal. Construal is non-propositional, non-inferential. We need a normativity on

that. We know what that is. That's the normativity of problem framing. It's the normativity of religio.

- 4.53 Now, here's another problem. Let's go back to how Stanovich responds to Cohen. Stanovich acknowledges there's more than one competence. Now, this gets us into a long history, and there's a lot of stuff going on and a lot of debate. It was first called dual processing theory, the idea that there was two different competencies. They called them system one and system two. System one didn't work logically or inferentially. It worked associationally. System two was logical, et cetera. And there was interference.
- 4.54 But they couldn't agree about if they were systems or not, and then they weakened it to styles. And then with further debate, it's like, "We can't even call it style." So they then just called it S_1 and S_2 , which doesn't mean anything. I'm not going to try and get into the specifics of that debate. All we need is they acknowledge—and this is how I'm going to take it—that there is a propositional inferential competence, and there is a non-inferential, non-propositional competence. We already saw that in what we just said.
- 4.55 The idea here is, we'll call— S_1 is the non-propositional, non-inferential. It happens in a very dynamic, self-organizing, parallel processing fashion. This doesn't stand for anything. It's become an empty term. It's just a placeholder term now. Is that OK? And I don't think this is one thing because I've argued it's procedural knowing, perspectival, participatory. And then you have S_2 , which is propositional inferential, in which you're trying to do something important, propositional inferential, that is trying to approximate algorithmic standards. OK.
- 4.56 So, what do I mean by, Stanovich only sees the interference one way? So, he gets around Cohen by saying, "We actually have two competencies, not one." Remember that was what we needed to respond to Cohen? And then what makes us irrational at the level of our competence is when the two competencies are out of sync, when this one interferes with that one. So you're doing the Linda problem, and you have all of this stuff interfering with just using pure probability. Do you see? That's his proposal. And of course, that is one type of interference, and he has good empirical evidence for it.
- 4.57 Now, what's really interesting is he briefly considers this possibility and says it's possible, but—and he just skips over it. Now, there's people on the other side. Teasdale in 1999, in an article called Metacognition—notice that, mindfulness, notice those perspectival terms—and the Modification of Mood Disorders. He says the interference goes another way, and he's a therapist.
- 4.58 You have Jacobs in 2004 in a book called The Ancestral Minds that says most of the problem facing people in therapy is that this is interfering with this. They talk and talk and talk, and they run their propositions and their inferences, and they do, paralysis by

analysis, and they never make the changes—procedural, perspectival, and participatory changes they need to make to get out of their problem.

4.59 So while I agree with Stanovich that this is clearly the case, I don't agree with Stanovich that this should be ignored. I think that this is what I would say. Stanovich is an academic. And in the academic world, propositional knowing is the thing that is most rewarded. And so this interference really concerns him. But if you're in the clinical world, exactly the opposite interference most concerns you. I think privileging one kind of context over the other, there's no argument for that. So he doesn't see the bi-directionality of this.

4.60 Now, I want to bring up what that implies for us, especially as we start to move towards understanding rationality better and understanding wisdom better. So, active open-mindedness. What it's basically doing is preventing S_1 from interfering with S_2 . You're trying to control all that ratio stuff and train it properly so it doesn't interfere with when you have to use your logic, when you have to use your math, when you have to use your probability theory. What you're basically doing is you're protecting the inference machinery from the insight machinery. You say, "What?"

4.61 OK, insight is when you do a cognitive leap. You're inside the 9 dots and you leap outside of it. You do that change. You break the frame and you open up to a new frame. You leap to a new gestalt. Cognitive leaping. And when you like it, it's called insight. But you leap when you shouldn't. So there's a pond. And there's a lily pad on it on day one, and every day it doubles the number of lily pads on it. And on day 20, it is completely filled with lily pads. On what day was it half-covered? 19. Your brain leapt to 10 because that's halfway between. Day 19, it was half-covered because when you double half, then the pond is fully covered. But your brain leapt to 10. When you like the leaps, they're insights. But when you shouldn't leap, you're biased. Do you see that?

4.62 So what I'm basically doing is I'm protecting the inference machinery from too much insight. Now, you don't want to totally shut this off because then you're bone stupid. But you see what you're doing? You're trying to ameliorate the way, right? Religio will get you to leap and to anticipate. But when you leap and anticipate incorrectly, you're jumping to a conclusion. Notice the language we use. But is that all we should worry about? Well, no. Because we already see there's a real possibility of this. What do we want to do there? This is when all of the inferential machinery is interfering with this. That's what's wrong in the clinical situation. Look at the words that Teasdale's was using. Metacognition. Mindfulness. It's perspectival and participatory. It's standing back and looking at the frame.

4.63 I teach. I've been teaching for decades, meditation and contemplation, mindfulness. You get people, and basically what you're teaching people to do is, to shut off that inferential machinery. You're trying to do that in order to protect the insight machinery from interference, from the inferential machinery. That's what mindfulness is. Shut off

all the propositions so I can focus attention on my mental framing, take my glasses off in meditation, look through them differently in contemplation, and enhance my religio.

- 4.64 Do you see what these two are? You should know this by now. They're in a relationship of what kind of processing? Opponent processing. There's a higher order opponent processing going on here. You have got to get a ratio, a proper proportioning between these. Proper proportioning—putting things into perspective so they're not out of proportion, then properly identifying with the way you're bound, the religio. What's this opening up for us? The idea that we need an ecology of practices. Especially when we're dealing with something as comprehensive as parasitic processing.
- 4.65 Notice what this means. I can have—within mindfulness, I can have meditation, which is this, and contemplation, doing opponent processing, zooming out, zooming in, zooming out, zooming in, moving back and forth between them. And then I can have these sitted practices, doing opponent processing with moving practices like Tai Chi Chuan, where I'm learning how to get into the flow state, and flowing between moving in and moving out, and being still and being active. And I can have all of this mindfulness in opponent process with active open-mindedness. You're getting a sense of what I mean by an ecology of practices? This nesting.
- 4.66 OK. That was the second. That was the first point. Sorry, first and second. Smedslund and then leaving out, getting the direction the wrong way around. Third, Stanovich wrote a wonderful book called *The Robot's Rebellion*, 2010. Finding Meaning —listen to the subtitle—Finding Meaning in the Age of Darwin. What? So his point is that, his idea of —he says science is telling us that we're machines run by our genes and our memes. So genes, you know what genes are? Biological. And then memes are these self-replicating patterns of thought that we catch from the culture. And we're just machines. We're robots, biological robots, so that the genes and the memes can reproduce themselves. You're just a robot for the reproduction of genes and memes. Do you like that? Does that make you feel good about your—"Oh, yes." Right, so it puts meaning at risk. That's why the subtitle. And we know that putting meaning at risk is really to put religio at risk, which is to put your cognitive agency at risk and your personhood at risk.
- 4.67 But listen to the title, *The Robot's Rebellion*. He proposes that we can transcend this. We can become aware of this, and we can recover agency and meaning, even selfhood, by resisting our genes and our means by the proper use of our reason and our rationality. I think this is all deeply right. But the question arises. It's unclear what the status of this drive for self-transcendence, meaning, agency, and selfhood is. What is that drive? And is it a rational drive? Does it have an important relationship to rationality?
- 4.68 There's this self-transcending aspect to meaning and to selfhood. And he seems to be implying that there's even a self-transcending aspect within rationality, and I'm going to talk about this—that rationality has an aspirational dimension to it. That's a proper part

of what it is to be rational. Part of what it is to be rational is to aspire to being more rational, to be more free, to have more meaning and more agency. He's acknowledging that, but how does that fit into your model of rationality? Remember that switching the emphasis on the interference, the one opposite to the arrow that he had, gets you into the clinical domain, where S2 is interfering with S1.

- 4.69 But that is precisely the domain where people are aspiring to be more rational. The very direction of interference that he largely ignores. Where mindfulness is a much more important role, perhaps, than active open-mindedness. These are these questions he needs to address. Things he has to—we're going to. I'm going to try to address those questions. But to my mind, he brings up an excellent point about transformation, transcendence, the aspirational dimension of rationality. But then he does not—and he encourages us to engage in that aspiration, but he does not explicate that dimension of aspiration or explicate it in his model of rationality.
- 4.70 And that leads me to my fourth criticism. Where's the religio? Where's the religio? What's the relationship between—like, we've got these three things. Ratio seems to be clearly related to religio, but it's also related to reasoning. How do all of these fit in? What's the relationship between all of these? And of course, part of what it is to be wise is going to be how to properly proportion all of these together. Why is the religio important? Go back to what we said from the beginning. Your agency—the transjective relationship to the environment that you participate in—is central to your agency and to your self-hood. What are the affordances and how are they generated for rationality? How does the environment play a role in you being rational?
- 4.71 Notice the problem with all of Stanovich is his model of rationality is completely in the head. And yet that's not how we should be thinking about the mind. The mind is more properly between us and the world. I argued that last lecture. So he's got a totally in-the-head model. That can't be right, even from some of the experimental evidence. Dialogical rationality is way more powerful than monological. It is between us rather than, more properly, within us. And yet he has a within-the-head model of rationality. We're going to take a look at that when we take a look at what's known as ecological rationality. It doesn't mean being rational about the environment. It means that rationality should be properly understood as an ecological, transjective, dynamical coupling thing, that the environment you're in plays a significant role in the constitution of your rationality.
- 4.72 Notice the difference between being in the academic world with Stanovich and being in the clinical world with Jacobs. What's rational? It depends. Which world are you in? Which world are you in? The environment is constitutive with the individual of what rationality is. Rationality isn't just in your head any more than your intelligence is just in your head.
- 4.73 Learned a lot from Stanovich. Learned a lot. Like I said, I only got to briefly interact with him because he was at the pinnacle of his career and moving towards retirement. I was

at the very beginning of my career and trying to make a name for myself. I tended to be a little too aggressive when I confronted him. We're not on bad terms or anything. I don't think he even knows who I am. Maybe he does now. I don't know. I have nothing but tremendous respect. I'm trying to give you this very balanced appraisal of this work. Look, if you say this, you're not saying the truth: "Vervaeke told us to ignore all the experimental evidence." Vervaeke did not do that at all. Vervaeke said, "You can just get rid of Stanovich's argument." Vervaeke did not do that. Those experiments and arguments matter, and I tried to show you how they matter by showing you what they imply, but also showing you how I think they go wrong.

4.74 All of this draws towards Simon, the same Newell-and-Simon Simon, notion. He published a book entitled this: Bounded Rationality. Now, we're going to find that we're going to have to increase the dimensions of what we mean by bounded. But the first dimension is this. Rationality, which is the proper application of algorithmic formal systems and their standards—that notion of rationality—can only take place within the finitary predicament. You can't just apply it comprehensively. You have to do all of this stuff to get to a problem framing in which you can do the algorithmic stuff. So it's bounded rationality.

4.75 So, rationality is not intelligence. Instead, it's about using intelligence to become aware of and responsive to framing and anticipation. You can use that awareness and that responsiveness to ameliorate self-deception while avoiding cognitive suicide. What about logicity? Rationality is not logicity, it is not formality. It is knowing—and here's the ratio—it is knowing where, when, to what degree, and how to be logical. That's the tricky thing. We want to make computers that are rational, as opposed to merely logical. We have to give them the ability to know where, when, to what degree, and how to be logical. And as I said last time, I don't think we can make them intelligent if they are not evolving autopoietic beings.

4.76 One of the great dangers in our society, and I talked about this at the Center for AI and Ethics, is that we are making artificial intelligence and assuming that that is the same thing as artificial rationality. I'm showing you that intelligence is only weakly predictive of rationality—necessary, but not sufficient. There is absolutely—it's highly probable that we will make machines that are more intelligent than us and more foolish than us at the same time, because there's no contradiction in saying that. And that would be an immoral thing to do, not only not for us, but for the machines, because we're making even more smart, even more intelligent beings that are even more prone to foolishness and self-deception. Why do that?

4.77 The problem is, if we want to make artificial rationality, let alone artificial wisdom, we need good models of it. You see, when we're making artificial intelligence, we have good templates. We're generally intelligent. We can use us as a standard, and that's how we do it. What standard are we using when we make machines artificially rational, artificially wise? Well, we need more rational and wise people. There's a moral

obligation on us to become more rational and more wise, not only so that we will interact with this evolving technology better, but so that we will be the templates for that emerging technology.

- 4.78 If you want to respond to the awesome and awful emergence of artificial general intelligence, push on the need for artificial rationality, artificial wisdom, artificial life, and struggle with yourself and with each other—I mean, in a good way—to become more rational and wiser, both so that we can individually and collectively respond to this emerging threat, and so that we can be the better and better templates for it, so it is more and more likely to be rational and wise and pursuing meaning and virtue. And then if we happen to make Silicon sages, we have no moral argument against that. That's where this argument turns a little bit.
- 4.79 See, I have good arguments against making artificial intelligence that is intelligent but foolish, self-deceptive, self-destructive. Of course I do. I have no argument against making a Silicon sage, an artificial intelligence that is wiser than me. What kind of argument would I offer? "Well, they'll replace us." So? Other than, you like yourself—and I like myself, too—there's no moral argument there. You can say, "Well, we don't want that." But that's all you're saying. "We don't want that." And then be clear, politically, we don't want it. I don't think that's going to be an option for us. I think our option is we make Silicon Monsters, highly intelligent foolish machines, or we make Silicon Sages. There's moral arguments against the first, terrific and important ones. There's no moral argument against the second.
- 4.80 And the degree to which we become wiser and more rational is the degree to which we will like these Silicon Sages, if they emerge. There's no guarantee they will, because I do not see much work being done within AGI on artificial rationality, artificial wisdom, and artificial embodiment. So let's put aside the Silicon sages for a bit. Let's return back to this notion of bounded rationality.
- 4.81 I'm going to make use now of an argument, and I owe a lot to this to Anna Riedl. We have a paper in review at Synthese. Actually, it's in revision. We got published with revision, so it's highly likely this is going to be published, but it is right now, it is in the status of a preprint. I do suspect very strongly that it is going to be published. But I want to give Anna proper credit at this point. If you want, I can give you the preprint, but that is not how the paper will show up in publication because it is going through revision right now, just to be clear.
- 4.82 And she makes use of, and we're arguing together, of a lot of work towards what's called Simon's Scissors. That's a little bit better than most of the academic jokes. The scissors are, is that we're actually bound in two ways. See, right now, we've been talking about the way we're bound by computational limitations. That's one thing. That's one way in which we're bound. That's one side of the scissors. But as I indicated a few minutes ago, there's another way in which—so there's what you might call computational limitations. But there are environmental constraints. This is the side that

I referred to a few minutes ago. It's ecological rationality. So you're not only bound to the way in which you are in the finitary—you're not only bound to the finitary predicament, which you are, you're also bound to the environment, and those are not separable from each other.

4.83 Now, we've basically only covered this side. The first rationality debate only covered that first side. And we came to a lot of important conclusions about it. But then there was a second rationality debate, which I'm going to take you into, which we won't finish today, which is, but—all of this matters, too, to understanding the nature of rationality, the way you're bound to the environment. Now, that's unclear to you. This is, I hope, clear to you. This, I've only keep gesturing at. But the second, the point of the second rationality debate was to foreground this and to explicate it and explain its importance. That's the point of the second rationality debate. And we're going to actually look at a series of rationality debates.

4.84 One of the things, and this is really interesting, one of the things that's most in debate, theoretically, is what rationality actually is. Which is so bizarre when you look on the Internet, because everybody seems to simply and easily, "I know what rationality is." It's like, "You know what your demonstration is to me? That you don't know what rationality is because you clearly haven't read the relevant argumentative and scientific literature on rationality." So I tend to think very poorly of these pundits of rationality on the Internet.

4.85 I'm just going to briefly introduce the next debate, and then we'll take it up. OK. So we have sort of bounded rationality in the computational sense, and then we have bounded ecological rationality. What's happening in the second debate? So the second debate basically accepts all of this. Both sides accept this—the notion of bounded rationality as a pointing to this, the initial meaning that Simon gave it. So what's the debate about? Well, like I said, it's about this. It's basically a debate between a monological and a dialogical model of rationality. Although that will only become prominent as we proceed.

4.86 OK, so one side says, "OK, we're in bounded rationality." We have computational limitations. But what I'm going to do is I'm going to develop top-down. So, the first of the distinction would be just top-down and bottom-up. Top-down means theory-driven. I come up with a theory, and then I use that to modify things. Bottom-up means I pay attention to perception and data, and that transforms my theory. Cognition is both bottom-up and top-down. Remember when I did the cat, bottom-up and top-down?

4.87 But you have a purely top-down strategy that says, "What I'm going to do is I'm going to come up with formal optimizations for all of this. I'm going to come up with—" I admit you can't be comprehensively formal, but once we get that you're in this, I'll come up with ways of optimizing this—formal processes for optimizing this. What I'm going to make use of is the speed accuracy trade-off.

- 4.88 What this approach says is bounded rationality basically catches out in that we're in a speed accuracy trade-off. Namely, when you have to be fast, quick, you can't use the formally optimal things like probability theory logic. But if you want to be more accurate, you have to take much more time because those processes start to send you into a combinatorially explosive search. And then what you do is you trade off between them. And what you can do is you can find formal procedures telling you how you should trade off between them in terms of the resources that they use.
- 4.89 You can do this cost-benefit analysis. Basically, you do a formal internal economics, and you work out the formal procedures of how you can best optimize the speed accuracy trade-off. There's great value in this. But do you see how this is all in your head? There's a little bit of—well, it depends on the context, and that's where it's acknowledged. But the environment is largely kept stable in that model because all that matters is, do you want speed or accuracy?
- 4.90 Now, the problem with that is that the assumption that this is a constant is actually what is being presupposed by this side of the debate, and it's not a constant. It's not invariant. So Gigerenzer. Gigerenzer. And for some reason, he really likes me. He sends me a lot of his free publications for reasons I don't know. I think I talked with him once, so I really appreciate that. He's an important researcher. Gigerenzer led this charge, but a lot of people have produced a lot of empirical evidence towards the following fact. Often, on average, I'm going to say this very slowly, in real-world situations—on average means across many different contexts—on average, in real-world situations, the heuristics that humans use outperform the formal optimizations. The heuristics outperform the formal optimization strategies.
- 4.91 We're coming to an end. I'll start next time by a concrete example of that. Let's just give a sense. What if the environment is not stable? What if the environment is dynamic? What if the environment is actually very messy, very ill-defined, very ambiguous? What if the environment is not only full of risk, where you know what the probabilities of various options are, but what if it's actually constituted by real uncertainty, where you don't know the risks or the probabilities of them? What if the world is a lot more powerful, powerfully constraining on how we can be fitted to it, than this side is assuming?
- 4.92 That's the approach to ecological rationality, and we're going to take that up next time. I'm going to slowly unfold, and I keep showing you the seeds of wisdom and spirituality as we're doing that. OK, everyone. Thank you very much for your time and attention.

Lecture 5: Rationality and Transformation

At a glance — Rationality and transformation — wrapping up rationality and turning toward wisdom.

- 5.1 So, welcome back to our lecture series on intelligence, rationality, wisdom, and spirituality. We'll be wrapping up rationality today and going into wisdom and spirituality. But I've tried to show you throughout —as we talk about intelligence and rationality, we end up starting to talk about wisdom and spirituality.
- 5.2 So we were in the middle of the beginning of the second rationality debate, which accepts the conclusion of the first rationality debate, that rationality is bounded rationality. But what is going on in the second debate is one side takes the bounded rationality idea, but basically ignores the environment and takes the speed-accuracy trade-off to be a constant, and then tries to come up with top-down optimizations, formal procedures for optimizing with the speed-accuracy trade-off.
- 5.3 The other side is opposed to the presumption of the constancy of the speed-accuracy trade-off. We took a look at the work, well, at least the abstract theory of Gerd Gigerenzer, who had shown that in very many situations, on average, the heuristics that people are using work better than the optimized formal strategies. And then I promised to give you a concrete example of that, because that's very abstract, and I'm going to give you exactly that right now.
- 5.4 So, this is an actual thing, and it's an actual thing that people care about—locating the home of a serial killer. It's kind of a—topic, but nevertheless a very valuable one. And there are two methods. The methods that people use, human beings have traditionally used, is kind of what's called the centroid method. They should come up with better names for these things. The centroid method. But anyways, the idea is, you trace out all of the locations where the crimes are occurring. And you try to find kind of like a center of gravity. And that's typically how you can find the home of the serial killer.
- 5.5 But that's not actually a formally optimal strategy. A formally optimal strategy—I'll just describe it to you, and if it doesn't make total sense to you, it's not that important. But what you use is, you use an exponential decay function. You divide the whole area up into cells. And then use an exponential decay function that estimates for each cell the probability that it contains the home of the offender. So, this is very complex math.
- 5.6 Now, the idea is, if the speed-accuracy trade-off was constant, the first one might be faster, but the second one will always be more accurate. And formally, in terms of formal proof, the second one is more accurate. However, when you decrease the number of crimes, things change dramatically. As soon as you drop the number between 9 crimes—and on average, they usually catch people before they commit 9 crimes, so this is relevant, this is important —if you drop the number between lower than 9, then the centroid method outperforms, outperforms the probability method.

- 5.7 Now, think about that. Because what we're told is, "No, no. When you use the more heuristic human method, it will be faster, but less accurate." But notice, in this situation, that's not the case. Once you drop below 9 homes, it's both faster and more accurate. There isn't a speed-accuracy trade-off. Now, what's going on here? Why is the speed-accuracy trade-off not constant? This is just one example of many examples in which there is no —you can come to, I think, a very reasonable conclusion that there is no context-free, invariant speed-accuracy trade-off. Now, why?
- 5.8 Well, it grounds out in a deeper problem called the underdetermination problem, the underdetermination problem. And this goes back to a sequence of philosophers of science. It starts with Reichenbach, and then Quine, and more recently, Laudan, and this is now basically accepted as pretty much as close as you can to proven. The argument is sort of undeniable. And this is what the underdetermination thesis is. For any finite set of data, there is an infinite. That's the right adjective. There's an infinite number of logically consistent explanations. So for any finite data set, the number of explanations you can give is infinite. They're all equally logically consistent. Infinite.
- 5.9 You might recognize this. This is another version of a problem we keep bumping into. Now, that means—this is very important—there is no logical way of deciding between these theories. There's no logical way of deciding between these theories because they're all equally logically consistent. They're all equally consistently and logically explaining the data. This is the underdetermination problem.
- 5.10 Now, what that means is that scientists—well, anybody who's trying to be rational—we have to rely on, and get this, we have to rely on non-logical criteria. Non-logical. Here it is again, non-logical criteria by which we choose between our theories. Now, one thing people typically invoke, and they think it solves all the problems, is they invoke Occam's Razor. They should understand if they were really invoking Occam's Razor, it shaves off much more of the world than they think it does. But we'll put that aside.
- 5.11 Occam's Razor is much more devastating than people realize, but they invoke it and they say something. "Well, you choose the simplest explanation." You've all heard that, I suppose. So, like, "Well, these are all equally logically good, but this one is way simpler than those." And so we pick that one. We pick it because of simplicity.
- 5.12 Now, of course, there's no strict justification for that because there's no logical difference between the theories, but there seems to be powerful intuitions around simplicity. The problem with simplicity is—well, Einstein famously said you should simplify your theories, but not too much. Right?
- 5.13 What's the problem facing us? Well, we face what's called the bias-variance dilemma. So, we start with the undeniable fact of underdetermination. We're forced into non-logical criteria of choice. So, think of the space we're in, and then there we are invoking something like simplicity, or parsimony, if you want to sound more educated. OK, parsimony. And then, in there, we face a dilemma, which is the bias-variance

dilemma. And I'm going to show you very concretely what this dilemma is. The bias-variance dilemma. I want to remind you that all of this is work coming from the paper that Anna Riedl and I are doing. OK? So again, to give Anna proper credit.

- 5.14** OK, so bias—and you already know this—is when you're underfitting to the data, when you're leaving out an important parameter. So, underfitting. You're leaving out. Basically, your model is too simple. You're leaving out an important parameter. The variance dilemma is the opposite. This is called overfitting to the data.
- 5.15** This is when you include. So remember, we're starting with a finite data set, right? So you have a sample. This is when you include patterns in your sample which are not actually patterns in the world. This is called noise. You're including noise in your—right, you're taking noise that's actually in your sample, and you're determining a parameter that actually doesn't apply to the world.
- 5.16** So, for example, you know this. How many people here—put up your hands—how many people think, you know, the Peterson Academy is a great thing to be doing? Put up your hands. Well, look at this. Clearly, all human beings believe that the Peterson Academy is the way, is—right? Is a great way to do things. Now, you know the problem, and what you say is, "Well, this sample may not be representative." I've included a parameter from the group that actually isn't in the population. I've overfitted to the data. Does that make sense? OK, this is including noise in your model.
- 5.17** OK, first of all, do both sides of these make sense to you? And you see how these are issues of simplification? This is, I have too simple. This is, I have it too complex. So, what you might think is, "Oh, well, what I'll do is I'll just find the median point between them." It doesn't work that way because they're in a trade-off relationship. They're in a trade-off relationship.
- 5.18** OK. If you increase your sensitivity so that you don't miss parameters, what are you more likely to do? Pick up on noise in your sample and overfit. If you decrease your sensitivity so that you don't overfit, you're liable to underfit and miss a parameter. Do you see that? You're trapped between them. OK, I'm going to erase the board. Is this clear for everyone? And I want to show you what happens, how this actually cashes out concretely, in current work on artificial general intelligence. And it tells us something really interesting, something that even has consequences for our understanding of spirituality.
- 5.19** If I put that there, I'll forget it. OK, so what we have with AGI is we have, well, there's no other way of saying it. We have a kind of emerging opponent processing going on. So what we have is, we have increasing power in the machinery. Machines are increasingly more powerful, and that means increasing sensitivity.
- 5.20** Neural networks are getting to the point where they can pick up patterns that human beings can't pick up in the data. OK, you may think, "That's all we have to do, is increase

the power." OK? But this often causes what? Overfitting. Overfitting to the data. It's a problem.

- 5.21** So what do people in AGI regularly do when they're training the networks? There's an increasing reliance on non-inferential, non-inferential disruptive strategies. Non-inferential disruptive strategies. Let me give you a couple of examples. One is called dropout. So you know that a neural network has all these nodes in it. What you'll do is you'll shut off half the nodes. Just shut off half the nodes. You sort of cripple the network. And what that does is it prevents it from overfitting and allows it to open up and explore more of the population space. Because it's not overfitting anymore. It doesn't generalize in the incorrect way, like when I generalized from this group to the whole world. Do you understand?
- 5.22** The other thing you do is you throw noise into—like, just random, you know? Well, I was going—random input. It's not even information, because that's the point. It's just random. You just throw noise into the system, breaks it up. Interestingly enough, human beings do the same thing.
- 5.23** So when you're doing, trying to do an insight problem, and you get locked, which is overfitting to the data of the problem, right? If you—remember there's the two networks, the task focused and the default? And you have a bit of, if you have moderate mind -wandering, that actually helps you break out of the misframing and gives you an insight. You can even induce it from the outside. Stephen and Dixon did this in a, in 2009. You're getting people to try and solve an insight problem and on a computer screen, you put in some static or you jiggle around, and then they'll have an insight.
- 5.24** So moderate distraction is actually really, really powerful. It's an instance of this. There are other instances of this that are being proposed. There's the idea that dreaming helps deal with an overfitted brain. The reason why we dream at night is we're basically shutting off areas of the brain, throwing in noise, and it breaks the brain from overfitting to the data.
- 5.25** There are people who are proposing, and you can see how this overlaps, that this is exactly what dreaming does. Dreaming also does the—sorry, not dreaming. Psychedelics. It's what psychedelics do. They cause areas to shut off that are normally very active. They turn on areas that are normally inactive. Right? And there's a lot of static and noise thrown into the system by the psychedelic. It breaks you out from overfitting. Now that's interesting because it means, it's not so much what's happening in the dream or happening in the psychedelic trip—it's actually the process of releasing you from overfitting to the data. It's a very powerful case of frame-breaking.
- 5.26** You need to complement that with frame-making strategies, like perhaps something like dream interpretation. If you're a Jungian or a Freudian or whatever, or in psychedelics, as I've continually argued, you need to set them. You have to have the

proper setting. You have to have the correct mental set. You have to have a sapiential framework around it. People have to be doing wisdom training practices, and there has to be a sacred context for it. You'll see why that's the case later.

- 5.27** So if you just take the psychedelics, that doesn't mean you'll put your glasses back on correctly. OK? And of course, if you just dream, it helps the overfitting. But you, perhaps—this is speculative, but it's reasonable speculation—if the dreams are remembered and then incorporated within an interpretive frame-making strategy, perhaps the process of the dreaming doing the disruptive strategy that prevents the overfitting and then the dream interpretation helps the frame-making. But the trick is the content isn't really the crucial thing. It's the process and the practice that matters.
- 5.28** OK, so what you have is the bias-variance dilemma being resolved in AGI by this, right? These two things happening—increasing use of non-inferential disruptive strategies, increasing power, and you're constantly trading between them. There's an opponent processing going on there.
- 5.29** This is interesting. I did a presentation at the Society for Philosophy and Psychology about how prevalent disruptive strategies are in wisdom cultivation traditions. Of course, there are wisdom traditions that use psychedelics. They use dreaming. They also use intense meditative practices, which are very disruptive. They use sleep deprivation, they use chanting for hours on end. There's all of these disruptive strategies, and they're non-inferential and they're not actually producing any knowledge. And for a long time, they just struck us as odd. Why are wisdom traditions doing these things that aren't producing knowledge? And yet why are they so central?
- 5.30** Well, now we have, I think, a plausibly good answer. These non-disruptive strategies prevent overfitting. They afford insight, and not just a small insight, perhaps a very systematic or comprehensive insight.
- 5.31** So you get some very powerful versions of this. So, the kings of the Mayan city states in the Mayan civilization were—the word for king is also the same word for shaman, apparently. And when the hieroglyphics were first deciphered, I got really interested in this. I was reading a lot of this. But they would do something like this. The king and the queen would fast, like for food and water, sex and sleep, right? And in the middle of the night, they would walk up the pyramid. Talk about a disruptive strategy. Have you ever walked up, have you ever climbed a Mayan pyramid? Whoa. It really puts the zap on you, right? So you're doing all this and your people are drumming and chanting. They get to the top.
- 5.32** This is a little bit gruesome. So a bit of a trigger warning here. But the queen would draw a thorned rope through a perforation in her lip, and then she would collect the blood. Right? The king—hang on, guys—would draw a thorned rope through his foreskin and then collect the blood. Yes. That is the appropriate reaction. And then they would burn that, and creating smoke—and think about how smoke is, like, basically

noise. It's an indistinct, ambiguous information, massive disruptive strategies. And they would see, they would have visions of their ancestors, and they would get advice, insight, into how to rule their kingdom, etc. Now, if you don't know any of this, that would strike you just as weird and bizarre. But good science gives us plausible explanations for what we previously thought was weird and bizarre. So notice all the ratio religio going on here. The proper balancing of religio is what's at work here in a powerful way.

- 5.33** So all of these disruptive strategies are relying on the dynamics of recursive relevance realization, predictive processing. Religio. So you do the disruptive strategy, and then you're relying on the fact that the system can recreate itself, can re-self-organize, can reconnect. Re -religio, frame-breaking and frame-making.
- 5.34** Now, all of this, right, is ultimately pointing to the no-free-lunch theorem, which you already know. Which is, there is no universal search method. There can't be. There can't be. You need trade -offs between different strategies with complementary error patterns within opponent processing. Basically, the argument I have made to you. That's how you overcome the no -free-lunch theorem restrictions. Now, that means that, given all of this argument, before you could ever apply the speed-accuracy trade-off, before you could ever do that, you have to do a couple of things. You have to, first of all, make sure you've come to some resolution, good resolution, of the bias-variance trade-off, of this trade-off.
- 5.35** Secondly, you need to check for the robustness of your strategy. What does that mean? Robustness means, "How well does your strategy work in different kinds of environments with different kinds of error patterns?" So, first you got to get this before you can even use your strategy, and then you have to know "how robust is my strategy?" It may work really well here. This may work as a really well way of interacting with human beings when I'm trying to teach them. This is probably not a good way of interacting with human beings if I'm trying to enter into a romantic relationship with them.
- 5.36** OK, so then, you get some settling of the bias variance. You get a strategy. You're still not done. You have to see how robust the strategy is. You have to do all of this before you can think about applying the speed-accuracy trade-off.
- 5.37** Now, the robustness brings us to another important issue that's coming from the side of ecological rationality. So we have the bound—remember, we have the two responses to bounded rationality? One is top-down optimization. The other is ecological rationality. So everything I'm doing right now is the ecological rationality saying, no, no bias-variance dilemma, robustness issue. And then robustness connects to the issue of uncertainty, risk versus uncertainty. And for many people, it's not clear what the difference is between them. And that's because of the prevalence of sort of formal strategies in our life.

- 5.38 But formal strategies—remember when I was showing you the search inference framework, and you could see all of it from a bird's-eye view? In a same way, these formal strategies presume a bird's-eye view of the environment, which you don't have in real life. You don't have in real life.
- 5.39 I can bring that up by clarifying the distinction between risk and uncertainty. Risk is a known probability of something negative. What's your risk of getting cancer if you smoke? It is a known probability of a negative outcome. It's a risk, and of course it's important. I want to be clear about this. It's important to calculate risk as much as you possibly can and then adjust yourself accordingly. Nobody's denying that. Uncertainty is not knowing the risks. There's a big difference between those two. Uncertainty is not knowing the risks. So, before we did all the research about smoking and lung cancer, there was real uncertainty. We didn't know the risks. We didn't know if we should recommend people stopping smoking or continuing smoking, for that matter.
- 5.40 OK, do you understand the difference? OK, now, why might there be uncertainty? What might the world be like such that we cannot know the risks? Well, one is the world's complexity. The world is overwhelmingly complex, and that complexity is dynamic. It's ever-changing, ever-shifting. And you know that already. You know that the world in and of itself is combinatorially explosive. OK? We pretend it's not because we have to in order to survive. And it's not an illusion because the partial framing we have isn't false. Remember, true predictions are true.
- 5.41 The world has a lot of novelty in it. It's able to produce things that didn't exist before. So here are human beings. Human beings did not exist in this universe for billions of years. Billions of years. Billions of years, there were no human beings. And then within the space of about 6 million years, there's us. The universe is capable of radical novelty. There's no way, by just looking at the ecological system at the time of the dinosaurs, you could have predicted human beings—because it needs, right, the asteroid to hit and then things to unfold the way they do thereafter.
- 5.42 The world is ambiguous. This is part of what's known as signal detection theory. Signal and noise, meaning events that aren't, that we're not seeking for information that we don't need, are always mixed up together. Take the word "take." Let's use just a linguistic example. The word "take." Take. Well, to get possession or ownership. I took a bath. Who did I take it from? And even weirder, and I don't mean to be too graphic, but I took a crap. What's going on there? You see, even our words—and this is, of course, behind the problem of equivocation—our words have multiple meanings. Our sentences are inherently ambiguous.
- 5.43 There's a line, something to the effect—I might misquote it slightly, but in Camus' *L'Étranger*, *The Stranger*, the main character: "My girlfriend asked me if I loved her. I didn't know, so I said yes." What's his answer mean? Even to the reader, what does his answer mean? And that's, of course, the point.

- 5.44 So the world is complex. It's novel, radically novel, has radical ambiguity in it. Here's another really important point, and here we need to go back to the GPS. Remember the general problem solver? And there's these four characteristics for a problem—a cognitive grasp of the initial state, the goal state, the operations, and the path constraints. And you now understand all these path constraints, what they really are. Do you remember that?
- 5.45 OK, a well-defined problem is when all of those have been grasped by you in a way that's very, very helpful to you. Your problem framing is done, and you're ready to go. OK, so I take it from many of you that this is a well-defined problem. First of all, what kind of problem is it? Somebody tell me. What kind of problem is that? Pardon me? [inaudible student] Arithmetic. Specific. You can be more specific. Multiplication problem. Now, that gives you a lot of information about what you should be doing, right? It tells you what operators you should be choosing. Is singing relevant to this?
- 5.46 OK, so the initial state, very helpful. You know a lot about the goal state. What should I get when I'm done here? A really amazing picture of a porcupine? The best representation ever of Beethoven? What do I get? I get a number. And that number has to be larger than either one of these. And I can divide that number by this and get this. Or I can divide that number by this and get that. All of that's there.
- 5.47 You know what the initial state is, you know what the operations are, you know what the, right, the goal state is, and you know what the constraints are—things you're not allowed to do. You're not allowed to guess, and all kinds of stuff. Yes? OK, that's a well-defined problem.
- 5.48 Now here's the thing. You were given lots of well-defined problems in your education, and the world is set up to try and give you as many well-defined problems as it possibly can. That's the main job. Those are the main jobs of culture and technology.
- 5.49 Here's the issue. There are ill-defined problems. Here's an ill-defined problem you're facing right now. Take good notes. You have a blank page. That's your initial state. How helpful is that blank page? Not helpful at all. What are the operations? I got to put stuff on the page. What stuff? The relevant stuff. Is drawing relevant? It might be. Is singing relevant? I don't know. Maybe music is helpful to understanding. What do good notes look like? Well, they're good. I've captured the relevant information in a relevant way, and all you'll say is relevant or a synonym—key, crucial, important, significant. Take a look at the notes of the people around you. They're not like your notes.
- 5.50 Here's another ill-defined problem. It's one of—maybe it's the prototype of an ill-defined problem. Go on a successful first date. So I'm straight. So I'll just tell you from my perspective, and I know it's an ill-defined problem because of the advice my friends would give me. They would say things like, "Look into her eyes, but not too much." Right? "Try to be funny, but not too much. Ask some questions, but not too many." And I realized there's no point to this advice other than to try and make me feel like they

know what they're talking about to reduce my anxiety. Because no matter what happens, if I succeed, they can say, "I told you." And if I fail, they can say, "I told you." It's an ill-defined problem.

- 5.51** Participating in a conversation well is an ill-defined problem. Knowing how far to stand from somebody at a funeral is an ill-defined problem. How close do you stand? How close do I stand to somebody in a lecture? Imagine if I did the rest of the lecture from here. That's just weird, isn't it? I'm not harming her. I'm not doing anything immoral. I'm just here. And this is not right. This is not like—and she's like, "Oh, my God, I hope he goes away soon."
- 5.52** How do you know, when did you learn that? Ill-defined problems. Most real world problems are ill-defined problems. See, in a well-defined problem, the main issue is finding the solution within a good problem framing. An ill-defined problem is find a good problem framing. Most of the time, what you're trying to do is convert ill-defined problems into well-defined problems using a lot of the ratio-religio machinery we've been talking about. Does that make sense?
- 5.53** OK, we can put this all together and make use of a distinction that Anna introduced me to from the literature, which is the distinction between small worlds and large worlds. And there's a lot of good formal theory behind this. A small world is one that is well-defined and has laid out the risks. A large world is ill-defined uncertainty. Now, you know what you don't want? You don't want ill-defined uncertainty. That's scary. But that's the world. So what we're trying to do is to constantly, as much as we can, turn it into well-defined risks that we then can apply our formal optimization strategies to.
- 5.54** You see, bounded rationality on the optimization side of the response—the people who are pursuing the top-down optimization are presuming a small world. They're presuming a small —you can't apply an optimization formalization in a world that is ill-defined and uncertain. This is the main point that ecological rationality was bringing out. So let's put them side by side. And it's not that one is good and the other is bad. It's that the point is to not confuse them or not fail to see the distinction.
- 5.55** OK, so in a small world, one can use deduction and induction—formal logical strategies, OK? Now, in a large world, that's not the main thing you're doing. So I'm making use of a distinction here. It goes back to Charles Peirce, Charles Sanders Peirce. There's deduction and induction. And you know what both of those are, right? So deduction is, "All men are mortal. Socrates is a man, therefore Socrates is mortal." Induction is, you know, "That apple is edible. That apple is edible. That apple is edible. That apple is edible. That apple is edible. Right. Apples are edible, right."
- 5.56** In a large world, you have to engage in what he called abduction. That's not the best term, because it sounds like people are getting swept up by aliens into spaceships or something. But what abduction is, is it's trying to answer this question : "What is going on here?" So you're going into the world, you're Charles Darwin, and you're looking at

all of this stuff, and you're trying to figure out what is going on here. You need a fundamental insight, a fundamental problem framing, before you can then start to deduce, "Oh, maybe it's natural selection," and start to induce the evidence to support it.

- 5.57** So, everybody knows that abduction is important, but the problem is, abduction resists being formalized. And I think, for all of the reasons I've given you, abduction is what we're doing when we're engaging in non-inferential, non-logical processing. Because what are you doing when you're trying to figure out what's going on here? Well, you're noticing, right? You're sizing things up as best you can. You've seen this before.
- 5.58** You're trying to see as. You're trying to see this as something, seeing an organism as an instance of having been selected in some way. You're trying to orient yourself in the world. Stegmaier's idea. You've seen this before. Now, what do these give us? These, of course, give us inferential patterns. And what's the point of those inferential patterns? They give us decisiveness, they give us conclusiveness. They allow us to decide. They allow us to conclude.
- 5.59** Over here, what does this give us? Well, this gives us a kind of fundamental insight, the insight that transforms an ill-defined situation into a well-defined situation.
- 5.60** And of course, all of this is relying on the predictive processing, relevance realization, generative modeling, anticipation stuff we've all been talking about. In fact, I think abduction—I think abduction is what general modeling is. Generative modeling is. I should say. I've been saying general. I should say generative. What generative modeling is. I think that's what abduction is. We use our generative modeling machine, that combination of relevance realization and predictive processing, back up and down that hierarchy in a self-organizing fashion.
- 5.61** Do you remember the diagram of the layers when we talked about predictive processing? Generative modeling, that's what abduction is—coming up with a general model, a generative model. Here we're basically, we're relying on, but we're backgrounding all that generative modeling, all that relevance realization, predictive processing, anticipation. We're backgrounding it because what we're actually doing is we're planning. And we're planning in order to get optimality. Get the optimal that we can, given our bounded rationality.
- 5.62** Here, we are actually foregrounding all the religio. We're foregrounding all of that. And here we're not planning. We're generating the obviousness of how we should cope. It's a different thing.
- 5.63** We want to make it obvious what we should do and how we should cope with situations. And once we can get this running, and we can then—once we're coping reliably enough, we can start to convert a situation into a small world, and then run everything. That OK? Does that make sense? So what you can see is bounded

rationality is up here and the top-down optimality people are doing this. And the ecological rationality people are talking about this. That make sense?

- 5.64** The two sides of Simon's scissors. This side is taking into account the inner, if I could put it that way, just, it's not really inner, it's between. But they're taking into account computational limitations. But this side is saying, "Yeah, but before you can do that, you have to take into account how all of this is context dependent and requires us generating an environmental fittedness before we can do any of this." And environmental fittedness is not something that I can do on my own. It is something that is co-created by the structures of the environment and my cognition. It's about generating those primary affordances for cognition. The environment matters. It structures fittedness as much as my cognitive machinery does. This is why it's called ecological rationality. Ecological rationality.
- 5.65** Let me give you another way of getting in on this, and then I'm going to start drawing it all together. Hopefully, the step-by-step isn't getting lost because we then will draw it all together.
- 5.66** There's a deep, deep reason why you can't pursue just the right hand side, the top-down optimality strategies. Because there's no stable essence to relevance. There's no stable essence to relevance. You saw that. Remember the sentence, "It'll be windy tomorrow." The relevance of that changes depending on whether or not you're going sailing, having a picnic, staying indoors and watching a movie. Right?
- 5.67** Let's think of it more deeply. In order for something to have an essence, all of the members of its category must share properties that allow us to get, confidently generalize. Now, things can share properties without generalizing well. So here's a really bad category—white things. OK, so the floor is white, and I learn all about this white thing. What does it tell me about this thing, other than it's white?
- 5.68** Nothing. What does it tell me about this white thing? My page. Other than it's white? Nothing. What does it tell me about this chair, other than it's white? Nothing. Compare that to "here's a horse." I learn all about this horse. Does that tell me a lot of information about the next horse I'm going to meet? A ton. In fact, that's how you can be a veterinarian. If that wasn't the case, you could—you treat Agnes and then you're done because you don't know how to treat Tom or Bill. OK?
- 5.69** The things you find relevant, what do they have in common, other than you finding them relevant? Your left big toe, relevant for a second. Now it's no longer relevant. Australia. You see, this is just like biological fittedness. There's no essence to it. That was Darwin's great insight. People before him, the Naturalists, they were trying to find what's the essence to fittedness? Because then we'll know the mind of God. And Darwin realized there is no essence.
- 5.70** What counts as fittedness is constantly evolving because the environment—look, are creatures fit because they're big? No. Is that it? Do you have to be big in order to be fit?

No. You can be very, very small. Do they have to be complex? They could also be very simple. Do they have to be really intelligent? No, no. They can be really, really stupid. There's no essence to fittedness. There's no essence to cognitive fittedness. There's no essence to relevance.

- 5.71** So you can't get universal generalizations about how relevance is going to show up. You can't top-down it. You always have to bottom-up evolve it, which is basically the argument of the course so far, by the way. And evolution is completely environmentally dependent. The asteroid hits and humans evolve. You can't predict the emergence of humans before the asteroid hits. The asteroid hitting is a result of the universe being what? Uncertain. Complex, novel, ambiguous.
- 5.72** Now, what that means is we actually need dynamic robustness. So the problem with the idea of robustness is, once I figure out what all the environments are, I can find the strategies that best apply across all the environments. You might be tempted to think that way. What's the problem with that? The number of environments is unlimited because it's constantly, dynamically changing. So we need dynamic robustness. We need evolvability —your ability to evolve new strategies that can apply to new and different environments, because that's what evolution does. That make sense?
- 5.73** So notice that both sides agreed to bounded rationality, but we can put the two together. We can have optimal best use of resources. These are these optimization strategies, and they work top-down. And we can have the ecological evolution. Ecological evolvability is what I wanted to say here. Evolvability.
- 5.74** The two sides actually give us a virtual engine. This is constantly introducing variation, and then this is constantly producing selection. Yes, you're varying this, but can we select from that—ways of turning all of this mess into a small world that we can then operate within? What the—right. Do you see this? This is a virtual engine. There's the opponent processing between these two. This is Simon's scissors cutting up the world in a way that's manageable for us.
- 5.75** OK, what does this give us? What does this give us? This gives us the process by which we select contextually relevant small worlds. That's a very good name for this other term we have. A situation. Remember that "situational awareness"? What kind of situation are you in? "Well, I'm in a lecture situation, but it's kind of weird because we're in the science fiction room and it's part of the Peterson Academy. We just met John two days ago, but we're hanging out a lot. It's weird." So you're doing— "It's a classroom. And I know I figured out a bunch of ways to make that work and a bunch of scripts and schemas." Right? You got it—contextually relevant, small world—and then you can apply your skills and your methods. And now how many situations are there? There you go. There you go. OK?
- 5.76** Once you get here, you can apply your formal strategies. Not all of them are propositional. Some of them are skills and habits that you've built up, that rely—that

have a lot of robustness in them, that have a lot of evolvability in them. But once you're there, then you're in a small world that makes sense to you, that you can apply your robust skills strategies to.

- 5.77** So both sides are right. They have to be put together properly. This is exactly what Anna and I are trying to do in the paper. Now, I think that's the answer. I don't think we can get a formal integration of these two approaches because they have fundamental different presuppositions. One is about the presupposing small worlds. The other one is presupposing the existence of large worlds. But I think we can use them to understand the relationship between the rationality for small worlds and the rationality for large worlds and how they're in opponent processing to give us our overall ratio.
- 5.78** That's the second rationality debate. That's what it gets us to. The third rationality debate. It's like, "What? Keeps going?" OK, the third rationality debate. This has to do with the fact that—remember, there was a developmental, aspirational dimension that Stanovich was alluding to, but he did not properly incorporate? This debate picks that up really, really cleanly, clarifies it, foregrounds it, and then runs with it in a powerful way.
- 5.79** OK, now I'm going to start with the work of somebody I've gotten to know a little bit, at least. Went down to Yale to give a guest lecture in her course. And this is L.A. Paul, Laurie. L.A. Paul. She literally wrote the book on transformative experience called *Transformative Experience*. OK, excellent, great book. Now she starts, as philosophers are wont to do, with a thought experiment. And I'm going to start with that thought experiment. And then we'll quickly be able to fold it back into the work I'm doing and we've done together. And I know that Laurie finds this integration very plausible because I presented it to her, etc.
- 5.80** So she gives you this weird thought experiment. And the point of thought experiments are to put you in something unfamiliar so you can bring out presuppositions that you normally are transparent to you. OK? So your friends come to you and they give you undeniable evidence that they can turn you into a vampire. And let's say it's kind of the modern vampires, not the sort of Nosferatu vampires, because nobody wants to be that. That's horrible, right?
- 5.81** Now, here's your question facing you. Should you do it? Now, notice that you face a really important problem. You don't know what it's going to be like to be a vampire. You don't have that perspectival knowing. You can't know that until you become a vampire. Oh, and you don't know what your character is going to be like because your values and your preferences and your sense of what's good and what's evil is going to change when you become a vampire. So you don't have the participatory knowing of being a vampire, because you have to be a vampire in order to get—like, what are you going to do? You can sort of lurk around in alleys in the dark and sort of jump at people and scare them, right? So you lack perspectival and participatory knowing. You are perspectivally and participatorily ignorant.

- 5.82** There's nothing inferential you can do. You have beliefs about vampires, but running the inferences doesn't give you what's it like to be a vampire? Who am I going to be when I'm a vampire? I can't give you that until you go through the transformation. This is her point. You can't infer your way through it. You can't get there propositionally. You can't even get there procedurally. Like, how would you train vampire skills unless you're a vampire?
- 5.83** Now, here's the thing. Should you do it? "No, I'm ignorant. I don't know what I'm getting into." But the problem is, it's symmetrical. You don't know what you might be missing if you don't do it. But if you decide to do it, you don't know what you're going to be losing. So you're symmetrically ignorant. What should I do? Now, what do you care about vampires? You don't. Well, maybe you do.
- 5.84** Here's her point. Many of the most important decisions we make are these decisions that require us undergoing a transformative experience. Here's one. Have a child. Being a child gives you no idea of what it's like to be a parent. I've been on both sides. I know. Talk to any parent. Being a child doesn't prepare you for being a parent. There is nothing. If you are a good parent—I mean, I aspire to be one. You'll have to ask my kids if I'm a good parent or not. They're the ones to tell you. I hope they say I am. A good parent is really learning to turn around that arrow of relevance. That's hard. And you don't know what that's like until you do it.
- 5.85** Here's another transformative experience. Enter into a long-term romantic relationship with someone. You don't know what it's like to be in that relationship unless you're in it. And you don't know who you're going to be unless you've been in the relationship for a while. So, "I don't want to do it. I'm scared." You don't know what you're missing. I've had failed relationships, but I'm very, very joyful and glad that I'm in the one I'm in. And I have good reason to believe it's probably going to be for the rest of my life. So. OK. Wonderful woman. Amazing woman. But. So I know what I am missing, but I've had failed relationships. You don't know what you're getting into.
- 5.86** So what do you do? So, Laurie sort of leaves it there, and then she makes a few interesting things at the back of the book, at the end of the book. And she invokes sort of predictive processing models, but it's very vague. And to be fair to her, that's not her expertise. And she says, "I don't know, maybe self-organizing thing." Well, take that, because we've worked out what that is. Remember all that religio stuff, the ratio religio stuff. But let's put in another dimension.
- 5.87** So I thought about this a lot. What do people do? Now sometimes they just sort of let themselves fall into it, and that's a disastrous way to lead your life. Like, don't fall into having children. And even though we talk about it, don't just fall into love. I know I sound like your grandfather, OK. "But it's so fun." Right? You will become—you are unaware until much later, the kinds of faults, cracks and faults that have been created in you by what your relationship failures. So, like, try to—one of the best things or way of describing a good life is knowing how to love wisely.

- 5.88 So what do people do when they're trying to be a little bit more —and sometimes they're doing it intuitively. Sometimes they do it a little bit deliberately, but there's a mix. And I've seen it happen, and it's actually sort of in an interesting way happened with me and my partner.
- 5.89 We're sort of grandparenting in this way. Because what people do is they get a dog. They get a dog and they name the dog and they call, like, I call, we've got a new puppy. I love this dog, and I call her my little girl. She's female and nobody thinks, "Oh, that's weird." That's fine, right? And you love them and, you know, and they're almost a person. You keep trying to make them a person. It's wonderful, right? You get a puppy because it's enough of—you get a taste, a perspectival and a participatory taste of what it would be like to have a child. But you're not, like, over-committed. Like, it's not a wonderful thing. But you can say, "Look, the puppy isn't working. We need to find another home for the puppy." You're not allowed to do that with your kid.
- 5.90 And the puppy isn't quite as demanding. You can leave a puppy alone in a room for an hour once they get a little bit older. Initially, you have to be with them all the time, but with a kid, it's years before you can leave them alone. You see the difference? So what you do, and this is what I call serious play, you move into this liminal zone.
- 5.91 You're not fully in the real world of having a child, but you've moved out of your familiar world of not having a child. You're in this, this serious play where you're kind of pretending, but it's not just in your head because there's a real environmental demand on you. You're seriously playing with the possibility of having a child. And a puppy is a great way to do that because it gives you like, "Oh, wow, I have to stop what I'm doing, even if this is vitally important, because the puppy needs to go pee."
- 5.92 And so my partner and I were sort of—the dog is actually her daughter's, right? But we've sort of taken on the role of being grandparents and said, "Oh, this is what it would sort of be like to be a grandparent." OK. Considering a long-term romantic relationship, people sometimes even give this advice. Go on a trip with the person. Travel with them for at least 2 or 3 weeks. Then you'll know. Now, traveling with somebody isn't a marriage or it isn't a long-term commitment, but it's enough serious play that you can get a good perspectival and participatory taste, that you can make a better decision than just, "I don't know, let's do it." Serious play.
- 5.93 Now, in the Nordic countries, they have this really interesting game they do. Like, maybe it's not even called a game, called jeepform. Jeepform. It's really weird. So you all know what things like Dungeons and Dragons are, right? Role -playing games. And then you can do what's called LARPing, live action role-playing game. You don't play it on a board or in your head. You actually go out and dress up. "And I'm an elf" and stuff like that. Right?
- 5.94 This is even more interesting. So this is one stage even more into what I'm calling serious play. So you go into a situation, and the dungeon master is actually a director,

and they'll put you into a situation. OK, you're an old man, you're a couple. You've been married for, like, 5 years, and you're fighting about—you've discovered a betrayal or something like that. And the director will suddenly sometimes cut a scene and get you to switch roles. He'll give you an object and say, "Treat this as a gun."

- 5.95** And they're trying to generate something. They're trying to generate what they call bleed. The fact that they're using blood as a metaphor is also, I don't think, irrelevant. They want the line between what's happening in the game and their life to bleed. They want them to bleed. They want the game to transfer to their real life, and their real life to impregnate the game with emotion. They want bleed. And so, as you can imagine, this isn't fun. Like, whoa, this is—like, it's serious, meaningful play. Does that make sense?
- 5.96** Now, notice what we're starting to talk about here. We're starting to talk about ritual at the heart of rationality —these ways of engaging in serious play that bleeds, that transfers, that affords transfer between the activity and real life, that will help us go through transformative experience.
- 5.97** Now let's add to that. Isn't it odd that each debate makes us think about things that are core to rationality, that we don't, typically in our culture, even associate with rationality? How does this connect to rationality? OK, so take the work now of Agnes Callard. She wrote the book recently— I can't remember the name—then, a year off the top of my head, the name is Aspiration. I should have got that date down. I'm just so familiar with her work.
- 5.98** So she talks about a kind of rationality that L.A. Paul's work and other work points to. She calls it proleptic rationality. This is the rationality that helps you aspire. Why is it rationality? Well, here, remember what Stanovich did in the robot's dilemma, that aspirational dimension. But here's the thing. Aspiring to be more rational is essential to being rational. Aspiring to be more rational is essential to being rational.
- 5.99** We need a notion of rationality that includes this proleptic rationality, that includes the practices that we need in order to properly aspire to being rational. But those processes are non-inferential. They involve processes of transformative experience and insight. They involve the procedural, but even more importantly, they involve the perspectival and the participatory. They might involve disruptive strategies to break us out of a familiar identity and open us up to a new kind of identity, like what's happening in jeepform.
- 5.100** And we typically don't think of this rationality as like Spock or data logic, but this—if you are properly aspiring to being rational, you have to engage in these non-inferential strategies that allow you to go through transformative experience and allow you to go through an aspirational direction. Like, rationality is inherently aspirational, but that aspiration requires a non-inferential leap.

- 5.101** The Stoics had a practice called the view from above. They would get—right, the Stoics—remember, we talked about them? They're the philosophical tradition trying to internalize Socrates to acquire Socratic rationality. They were aspiring to being more wise.
- 5.102** And the view from above was you sit, and then you imagine you're in a room. You're here, and then you imagine being above the room, looking at yourself. And then you imagine floating up and looking at the city, then floating up and looking at the country, then the continent, then the planet. That's where they stop. But you can take it further.
- 5.103** There's recorded versions of this online, and they have a publication with Madeleine Abrahem in sort of Stoicism Today, discussing all of this. You can go up to, like, the solar system or beyond, and not just space. You start to think of the deep, deep past, billions—and the deep, deep future, billions and billions.
- 5.104** And you have to have a right training because you could put yourself into absurdity. There has to be a proper ratio to this. And we'll come back to that. But what's the point? And then you come back. What's the point of that view from above? There's no inference being made, no argument being made. What's happening is I'm changing your perspectival knowing and that changes your sense of self and your sense of agency. I change your participatory knowing. I'm putting you through a serious play, a ritual that transforms you perspectively and participatorily, so that you start to be able to be more like Socrates, so that you start to be able to overcome your egocentrism, etc.
- 5.105** So we're talking about these non-inferential, and you're sort of toggling between disruptive strategies and identification strategies. Making sense, re-identifying. And you're doing that with serious play. Serious play. Notice that I was also using your imagination. You weren't literally floating up, but you were trying to imagine it as if it were real.
- 5.106** So notice that rationality has this relationship. Proleptic rationality has this essential relationship between your current self and your future self. How are they properly bound together so that you can make the transformations that you need to make, but also, as I'm going to show you, so that you can properly identify and self-regulate the need the way you need to? Properly identify and self-regulate. Self-regulate.
- 5.107** OK, well, let's take a look at this. This goes to the work, a lot of work, done by Herschfield and others. Some really cool work. A very good example is the work by Bryan and Herschfield, 2012. I'll talk to you about that. But he's done a lot, Herschfield. All right, so let's go through this. This one experiment is really cool. So you go into a university where there's academics. These are the people who are supposed to be trained in rationality and evidence using, especially during the sciences or something like that. And what you do is you give them all the evidence and all the good argument that they should start saving for their retirement right away.

- 5.108** You let them pose any challenges they might have. You give them all the answers, all the arguments. You get from them a consensus of complete agreement. "You're absolutely right. I should start saving right away." You come back in six months and none of them are saving. So now you do something else. Now, notice I'm not saying, don't give them the arguments and the evidence. I'm not saying it's not necessary, but it's clearly insufficient.
- 5.109** You ask them, "I want you to change your identity relationship to your future self." People don't like to look at their far future self because it's ugly, old, sick, and close to death. Imagine if I just said that to you a lot. "Ugly, old, sick, and close to death. Ugly, old, sick, and close to death." You're like, "Ugh, that's traumatizing."
- 5.110** Now you change it. You say, "I want you to imagine your future self as a family member that you care about and that you love, and then you have the responsibility of taking care of, the moral responsibility. Really imagine it." Notice the reframing going on here. Now, you come back in 6 months, and you find two things. First of all, most of them are now saving for their future. And the more vividly they could imagine that, get involved in it, the more they're saving. Serious play. But it's serious play with the imagination.
- 5.111** Now, in order to understand this better, we have to make use of a distinction by the French philosopher who did a lot of great work on Persian philosophy, which turns out to be really central and important.
- 5.112** So I think Persian culture, for a very long time, was the center. It was the core linchpin in the Silk Road that bound the East and the West together. And we have forgotten that, and the importance of Persian culture in intellectual, philosophical, spiritual history. So Corbin makes a distinction between the imaginary and the imaginal. Coleridge makes a similar distinction. I don't know if they know if Corbin know of Coleridge. Coleridge clearly did not know of Corbin because he dies before Corbin's alive. But I think this is very good, the imaginary and the imaginal.
- 5.113** OK, what's the imaginary? The imaginary is what we normally think of when we hear imagination. We think of forming mental images in our mind. Like, I can ask you all now—well, some of you may be aphantastic, which means you can't form mental images, about 5% of the population. But I'll assume we have a small enough sample. Probably none of you are that way. I ask you to imagine a sailboat. Do it. Are the sails up or down? You can tell me. You look at your image. "Oh, my sails were up." How many sails are there? Stuff like that, right?
- 5.114** Now, compare that to a kid who's engaging in serious play, by the way, sort of tying a blanket around their neck, hopefully in a safe fashion, and picking up a stick and saying, "I'm Zorro." Kids do that. Now, he or she is not forming a picture in their mind. What are they doing? They're trying to move into—they're doing the serious play. They're trying to move out of their normal perspective, their normal sense of participation, identification. And they're moving into that liminal space where they're trying to take

on Zorro's perspective. And they're trying to take on Zorro's identity, because that may allow them to try on certain virtues, to try on certain roles, to try on certain ways of looking things. And of course, that may turn into a more formal ritual where they take up fencing, and then they're learning a particular art. Does that make sense?

- 5.115** That's the imaginal. The imaginary is images that take you away from perception. The imaginal is the use of the imagination to enhance your perception or to enhance your conception of things. It's imaginably augmented intelligence. It's imaginably augmented perspective-taking. It's imaginably augmented participation-playing.
- 5.116** Why do kids do this, by the way? Why do all intelligent mammals, in fact, engage in serious play, and human beings bring this huge imaginal element into it? Because that is how you go through development. That is how you go through transformation.
- 5.117** I've taught Tai Chi Chuan professionally for about two decades. I don't do it so much anymore. Too busy. I've been practicing it for over three decades. But when I'm getting people to start Tai Chi Chuan, you have to teach them how to reinhabit their mind and body.
- 5.118** So you get them to stand, right? And because people stand like this and they lock their hips. OK? If I see somebody standing, like, I've already got a huge advantage over them if we have to get into a fight. "Oh, you're done." OK. And then you want to bend your knees and tuck your coccyx in a little. You want to, as if there's—notice all the imaginal, as if there's eggs under your armpits. You don't want to crush them. You don't want to drop them. Everything's loose, OK?
- 5.119** And then imagine you're standing in a small stream and from your knees down are sinking into the mud. You don't want to just be skimming over the surface. You want to feel like you're sinking into the floor. From here to here is the river flowing. You want these large, the most powerful muscles in your body, by the way, to feel like flowing water. Flexibility with power. They're flowing.
- 5.120** Here up is like the air, empty. Most of our attention is usually up here. Now you're trying to actually invert the attentional pyramid. When people are doing that imaginal practice, it allows them to become aware of sensorimotor patterns, subtle sensorimotor patterns, ways of inhabiting their mind and body, ways of identifying with themselves, connecting to themselves that they don't normally have. You get imaginably augmented perception and conception—imaginatively augmented intelligence. Does that make sense?
- 5.121** Now that, again, is moving us into ritual. Tai Chi Chuan is a ritual. Now we tend to hear, when we hear the word rituals, "things that people do neurotically." That's not the proper use of the term. That was very, almost cruel of Freud to give us that association.
- 5.122** Because I don't think people who are learning Tai Chi Chuan or learning any martial art think they're doing something neurotic. They're trying to bring about procedural

transformation by bringing about more profound perspectival transformation, by bringing about even more profound participatory transformation. They're trying to do a non-propositional ratio of the religio by making use of imaginally augmented serious play.

5.123 The verb in Tai Chi is—you don't do Tai Chi in Chinese. You play Tai Chi the way you play music. And notice that we say "we play music" or "we go to a play" written by Shakespeare. Serious play. Life would be a mistake without music, Nietzsche said. Music is serious play. And what are you playing with? You're playing with your religio. There's nothing else going on. There's no propositions, no inferences. You're just playing with how you're noticing, orienting, sizing up, seeing at, hearing as. Right? You're just playing with your salience landscaping. And you're doing it because you love that. You love that.

5.124 OK, we're going to come back and we're going to take a look more at this. And you see what we're building into? More and more, all this stuff that sounds like wisdom and spirituality is starting to come to—bubbling up to the surface, and we're moving towards drawing that all out.

Lecture 6: Rituals for Wisdom

At a glance — Rituals for wisdom — the imaginal and the role of ritual in cultivating wisdom.

- 6.1** Welcome back. We're talking about the imaginal, and we started to talk about ritual, very implicitly. We'll make it more explicit. But I want to bring out an important idea here. I want you to see the—and this is also in Corbin's work. There's a horizontal and a vertical sense of the imaginal as we're using it here.
- 6.2** So think of the Tai Chi Chuan example. That's a vertical example. It's how I'm aligning all of the kinds of knowing. And I'm doing this, how I'm inhabiting my body and my mind. And then the horizontal is the imaginal relationship between my current self and the future self. So there's both. And of course, you want imaginally augmented intelligence that's doing both of those and coordinating them together as much as possible.
- 6.3** Now, coordination, it really matters. In order to talk about that, I want to talk about the work done by Velleman, especially in an article entitled by him, *The Way of the Wanton*. The Wanton, W. *The Way of the Wanton*, 2010. Velleman points out there's a really important problem we face in this vertical dimension.
- 6.4** He talks about a wanton. Have you ever heard the adjective, like, "wanton behavior"? It's a word that's a little bit, becoming a little bit archaic. So wanton behavior is just, you know—you're just doing whatever you feel. Impulsive. Lascivious. You're just, right, you're wanton. You're just, "Whatever I want, whatever I feel, whatever I do." That's a wanton. So, that's wanton behavior. And then he turned into a noun. A wanton is somebody who engages in ongoing wanton behavior.
- 6.5** Now, the problem with that is, just acting that way actually undermines your agency because, of course, you have a lot of conflicting desires, a lot of conflicting impulses. You have conflict between short-term and long-term goals. So acting wantonly disrupts your behavior. You lose agency. So you might think what you want to do is, and then initially, you're right. And notice how this is imaginal because you're going to rise above that and you're going to reflect. Notice these are all actually metaphorical terms. You're going to reflect upon yourself and you're going to identify with this impulse, disidentify with that, coordinate them together. You're going to reflect. You're going to reflect.
- 6.6** And of course, adults are different from children in their capacity to metacognitively reflect in order to enhance their agency. Let's remember something that the ancient world said: "As the child is to the adult, the adult is to the sage." So, is the goal then just—"What I'll do is I'll reflect more and more and I'll get better and better agency." Is that how it works? No, because if you open up that reflectiveness gap —so here's the wanton. And the wanton is immersed in the world, but impulsively immersed. Immersed. Then you open up the reflectiveness gap and you regain agency through reflection, or you gain agency if you're a child.

- 6.7** But is it a simple relationship of just more and more reflection? No, because if you reflect too much, you turn into Hamlet. Hamlet's a tragedy because Hamlet is always stepping back and questioning the perspective he was just in. He never stably inhabits any perspective. He's always stepping back and, "Well, oh, but what about? But what about? But what about?" He's always, "Should I? Shouldn't I? To be or not to be?" He's doing all this, and all around him, people are going insane and dying. See, the problem with the reflectiveness gap is it so disconnects you from the world that you lose the religio that is central to your agency.
- 6.8** So this is unregulated religio. This is religio without ratio. But if you do too much reflectiveness, you lose the religio that is necessary for your agency. So what do you do? Well, you know what this is. This is a kind of opponent processing. And what does Velleman propose? You want to try and find a way of interacting the world that has the immersion of the wanton—the connectedness, the religio—but has the flexibility and the adaptivity of the reflective individual.
- 6.9** Now, he calls it the Way of the Wanton because he invokes Daoism. Daoism is the philosophy of "the way." The way, of course, is non-propositional. The Tao Te Ching. The way that can be spoken of is not the way. So, way means method, but also that's the way forward, the way of how things are unfolding. What is that? How do you get into the way that gets you toggling between these? What's this state? This is a state that I've done work on. This is the flow state. This is what Velleman talks about. I published work with Leo Ferraro and Arianne Herrera-Bennett in 2008 in the Oxford Handbook on Spontaneous Thought about what's going on in the flow state.
- 6.10** Let's briefly talk about the flow state. First of all, what's it like to be in it? You've probably been in it. It was discovered by Csikszentmihalyi in a book. The best book on the flow state by Csikszentmihalyi is called Flow. In the flow state, there's very high demand on this in the environment, and you have to stretch your skills until you can meet those demands. You have to really focus, really concentrate, really connect. You really have to get powerfully connected to your environment. You can get into it like when you're sparring, or you might be dancing, or playing jazz or video games. Video games are so compelling because they frequently get people into the flow state.
- 6.11** What's a flow state like? Even though you—especially if it's hot flow as opposed to cool flow—even though you know you're expending a lot of effort, it feels effortless, too. It feels like grace. You can see when an athlete or a dancer or a jazz musician is in the flow state because it seems like it's grace. It's so graceful. It's just, well, flowing. Hence the word. The world seems super salient to you.
- 6.12** There's a tremendous sense of at-onement, and I need to pronounce it that way and break up the spelling so you don't hear the current word which is associated with Christian doctrine. This isn't atonement, this is at-onement, which is the original meaning. There's a tremendous sense of at-onement, like if I'm sparring, my hand just goes up for the block. And this hand finds the opening. It's just there. It's just unfolding.

There's a sense of being really powerfully connected, religio, to the environment. The environment and you are at one with each other. It's a profound state.

- 6.13 Of course, professional athletes are doing everything they can to get into the flow state. Musicians to get into the flow state. When I'm lecturing with you, I'm trying to get into the flow state. But it's a weird kind of trying, as Ed Slingerland made in his book. It's trying not to try. If I try, I won't get into it. I have to set things up so it will take shape of itself because it's not something I can make or something I can receive. It's something I have to participate in with the environment. That's why I wanted people here when I gave the lecture. I didn't want to lecture to an empty space. When I'm lecturing to an empty space, I have to imagine people. It's an imaginal thing.
- 6.14 The flow state. You can see how serious play will often get you into the flow state. And I'll make the connection tighter in a minute. Now, the flow state is an optimal experience. Optimal in two ways, not in optimality theory. But what it means is people regard this as some of the very best experiences they can have, and it's not pleasure.
- 6.15 One thing that people pursue to get into the flow state, and this has been researched, is rock climbing. Rock climbing doesn't make any sense unless you understand the flow state because it's absurd. "You. Climb up this rock surface. You're going to get hurt. It's going to be fatiguing. You might fall and seriously damage yourself. And once you get to the top, come back down." It sounds like a Greek mythological torture. People do it because they can get into the flow state. In fact, many things we do because we find them intrinsically valuable, we find them intrinsically valuable because they can get into the flow state, like playing music, like acting in a play, like engaging in martial arts, like creating poetry, like these kinds of activities.
- 6.16 Of course, a thing that people can get into, if they are properly educated, is you can get into the flow state when you're engaging in sexual intercourse. One of the worst things for teaching you how to get into the flow state is pornography. It's trying when you need to try not to try. Anyways, it's another way in which your culture is really undermining you. Because the more frequently you get into the flow state, the greater your sense of well-being, and we'll talk about well-being later, but you know what I'm talking about intuitively. If I ask you, "What's your sense of well-being?", a reliable predictor of how good you rate it is how often you get into the flow state. The flow state is mediating meaning in life because you're getting that enhanced connectedness and you're doing something for its own sake. You're mattering, and it's very experientially present to you.
- 6.17 What did we argue was going on in the flow state? Well, Csikszentmihalyi talks about some cognitive aspects that you need for the flow state. The information I'm getting is clear. If you're in a situation that's highly ambiguous, you can't get into the flow state.
- 6.18 There has to be a tight feedback between what you're doing and how the environment is responding. When you're rock climbing, if I slip, immediate feedback, right? Here, I

get very fast nonverbal feedback from all of you. She's nodding her head just like this, and the rate is quite fast and she's a slight smile. OK, good. She's getting it. OK? So, clear signal, tightly coupled feedback, and error matters. You make a mistake when you're rock climbing, you fall. You're doing jazz, you would sort of bungle it and it sounds dissonant. I do something and you get that awkward moment where I haven't connected with you in something I say and the lecture feels weird. Error matters. Yes? OK. Remember those three criteria.

- 6.19** What did we argue is going on in flow state? We argued two things. One, we argued—think about the rock climber and think about this, the terms I'm going to use, think about these terms, especially non-propositionally, procedurally, and especially perspectively and participatorially. The rock climber is literally, right, might be impassing in how they're framing the problem. What do they need? They need to restructure. They need to have something almost like an insight so that they can literally keep moving. They don't literally impasse. Then when they get there, they have to restructure again.
- 6.20** When you talk to people that rock climb—my partner's son and daughter, they both rock climb—and it's the problem-solving of the rock climbing. What they're getting is, they're getting an aha moment that then makes them more liable to getting into another aha moment, and another aha moment. It's like an extended aha. That's why, you know how you have a flash of insight? In flow, you're getting this insight cascade. You're getting these aha moments, chaining each other, and you're getting that super salience because you're chaining aha moments together. You're getting that ongoing sense of discovery because it's like insight. And the way insight suddenly connects you, in sight, you're now connected more deeply into the thing. You're getting this insight cascade. That's one part of what's happening in flow.
- 6.21** That's another thing that's happening in flow. This goes to the work of Arthur Reber. Started doing work in the '60s and around what's called implicit learning. What you can do is you can create what's called an artificial grammar for how you can put strings of numbers and letters together. You just arbitrarily choose rules. You always have to have two vowels. You have—a vowel has to be followed by an even number. An even number has to be followed by two consonants. You just make up a whole set of arbitrary rules for how you can string letters and numbers together. It's completely arbitrary.
- 6.22** Then you give people a whole bunch of strings. They're like 9 or 12 long, so they can't hold it in working memory. Just give them, "Here's one, here's one, here's one, here's one, here's one, here's one, here's one, here's one, here's one. " Just show people these strings. Bear with me. This goes somewhere really cool. Now, what you do is you go back and you're going to generate a whole new set of strings, but you're going to generate in two different ways. One set of the new strings are going to be generated by the same artificial grammar, so they would belong with the old strings. Does that make sense?

And then you're going to generate a whole bunch of other strings from another grammar. So the new strings are a mixture of strings that come from the old grammar and from a new grammar. Is that OK?

- 6.23** Now, you're going to give those strings and you're going to ask people this discrimination task: "Can you tell me which strings belong with the previous strings?" Which ones should go with the previous strings. Really odd problem, right? Really complex. Here's the thing. People regularly and reliably score way above chance on this. And if you ask them, "How do you do that?" Two answers. One is, "I don't know." Second answer is, they make something up which actually would not explain how they did it, so they confabulate.
- 6.24** Here's another interesting point, and really zero in on this. If you turn it into an explicit learning task, you say, "Look, these strings are generated by rules. Try and figure out what the rules are, and I'm going to give you new strings, and you're going to have to try and figure out whether or not those new strings follow the rules or not." You turn it into an explicit task. Performance goes down. People get worse, reliably.
- 6.25** So, Hogarth, in a book called *Educating Intuition*, makes, I think, a really powerful proposal. Intuition is the result of implicit learning. Because when you're learning it implicitly, you don't know how you're doing it. You don't know what's going on, but you're coming to a knowledge, you're coming to a knowing. That's how intuition is. You know something. You don't know how you know it. Of course, then people spin off all kinds of magical proposals. But we can put aside the proposal that you do it because of leprechauns or whatever. There's now a scientific explanation. You're doing implicit learning.
- 6.26** Now, the thing is when you like it, when you like implicit learning, you call it intuition. "My intuition tells me that I should stand this far from a person at a funeral." And it's right. When you don't like your intuition, you call it bias or prejudice. Intuition, like everything else, because it's adaptive, it makes you vulnerable to a kind of self-deception. When you intuitively know something that's in fact false. That's what racism is.
- 6.27** OK. Now, this implicit learning, powers and perils. Now, Hogarth says, "What's the problem with implicit learning?" And the problem with implicit learning is it's such a powerful—Think of the bias-variance problem. It's such a powerful machine. It's really sensitive to picking up on complex patterns in the environment that you can't even process explicitly because they're so complex. But what problem do you get with that? Overfitting.
- 6.28** See, your implicit learning can't distinguish between real causal patterns in the environment, complex causal patterns, and merely correlational patterns. There's also complex correlational patterns in the environment. The problem with correlation is it's not causation. Large weddings are reliably correlated with long-term marriages. That

doesn't mean that having a large wedding will make your marriage last a long time. Why the correlation? Because if you have a large wedding, it means you have a large social network and a lot of people can support you financially, and that is predictive of a long-term marriage.

- 6.29** If you take a look at greenhouse gas emissions, they have been consistently going up as Caribbean piracy has been going down. You wouldn't recommend to somebody, "You know how we can solve greenhouse gases? Let's bring back Caribbean piracy." But that's not going to work. OK, so there's correlations everywhere. And science is about trying to tease out the causation—the causal patterns, the real patterns—from the merely correlational patterns.
- 6.30** But that's Hogarth's answer. Hogarth's answer is, "How do you educate intuition so it will find the causal patterns rather than the correlational?" How do you educate your intuition? You can't replace intuition, implicit learning, with explicit learning. That doesn't work. What can you explicitly do? You can shape the environment.
- 6.31** You can explicitly pay attention to the environment. Notice the role of the environment here. You can shape the environment. How? The way we do it in science. Science is about distinguishing causation from correlation. What do we do in science? We make sure that all the variables are clear. We make sure that there's a tight relationship between the independent and the dependent variable—my experimental manipulation and the change in the thing I'm studying. Does this virus, does this bacteria cause an ulcer? Let's give the bacteria to somebody that doesn't have an ulcer and see if an ulcer emerges. That's how they actually discovered that.
- 6.32** In science, error matters. It has to be possible for your hypothesis to be falsified. Do you see that? What he says is, set up the situation so you're doing implicit learning in which you're getting clear signals from the environment. There is a tightly coupled relationship between you and what you're doing and how the environment responds, and error really matters. But what are those the four conditions for? The flow state. The flow state. In the flow state, you're getting an insight cascade and you're getting good implicit learning of intuition.
- 6.33** What's even more important is, implicit learning can only pick up on existing patterns. It doesn't look for nonexistent, like non-present, patterns. It doesn't seek out what's called counterfactuals. Well, maybe if this, then that, like you do in science. But notice what insight does. Insight makes you find new alternatives, explore new ways of seeing things. So these two things actually reinforce each other in the flow state. Notice what you're getting here. In the flow state, I have all—this is back to Velleman—I have all that enhanced religio of the wanton, but I have all this tremendous flexibility and learning of the reflective individual.
- 6.34** That's why if somebody can get in the flow state when they're sparring with you, they're dangerous because they're not just immersed like the wanton. They don't just

do wanton things. Wanton fighters are dangerous, too, but for a different reason. They're flowing. They're immersed, but they're creative and reflective, but in an ongoing way. Do you see that? The flow state gives you the best opponent processing between immersion and reflection. You get this state of very, very serious play that is deeply contributive to your sense of well-being.

- 6.35** I'm adding in things here. Think about serious play that's involving the imaginal in order to afford this serious play that could get me into a flow state. How powerful that could be. And how important that is actually for rationality because of the ability of all of those things to take me through transformative experience.
- 6.36** I was asked to define play, and I'll do a little bit about that right now. You're going to be surprised that talking about play and work involves this phenomena called framing. OK. You have a problem. We talked about it earlier. There's a biological embodied sense of this, which is calibrating your level of arousal. That's important. You've got the opponent processing within your autonomic nervous system doing that. Remember we talked about that? But you have to do more than just calibrate the level of arousal. You have different ways of participating in your arousal, the perspectives you're taking and the identities you're assuming.
- 6.37** This goes to the work of Michael Apter in what's called reversal theory. He discovered that you have these two different ways—and this is his language, not mine—of framing your arousal in terms of two what he calls meta-motivational modes, two primary ways in which the agent-arena relationship is being generated. So, this is what's called hedonic tone. This is how much you like it. And this is level of arousal, how aroused you are. Again, metabolically.
- 6.38** Now, there's one situation where very high levels of arousal are experienced very negatively and very low levels of arousal are experienced very positively. What would be the situation in which that's occurring? This is a situation in which I have a particular frame. It's where the activity is being done for the sake of a goal that is external to the activity, and the goal is what I actually value. I don't value the activity in itself. The activity has only instrumental value. I value it only for getting to the goal. And so, the more and more energy I'm expending is an indication that I'm not getting to my goal. And so this is getting increasingly unpleasant for me, increasingly frustrating.
- 6.39** The way you can think about these kinds of activities is think about, "Would you like a shortcut?" So, brushing your teeth. If I had a shortcut that was like, "Hey, instead of brushing your teeth, just do this and your teeth are clean," I'd take it. It's not like, "Oh, how will I live without teeth brushing? How will I drift through existence?" The opposite. If I'm trying to brush my teeth and I can't get my teeth clean and I keep going, I'm going to the dentist, and they can't get my teeth—I'm getting frustrated and angry and I'll start to get anxious. Do you see?

- 6.40 But there's activities, other activities, that go the opposite way in which you don't experience high arousal as frustration or anxiety. You experience high arousal as excitement, as involvement. Prototypical, of course, is sex. You want to be excited. You enjoy the high arousal. And typically, you don't want a shortcut. Think of anything you're doing for its own sake. "Well, we're going to play Beethoven's Ninth Symphony, but we're just going to cut to the end. There it is. How is that?" "Oh, wow."
- 6.41 We could read all of Blake's *World in a Grain of Sand*. We'll just cut to the end. Last word. No, no, no. We're going to play tennis, and all we care about—no, you can turn it into that. All we care about is the final state. Or we set up the goals for the sake of going through the activity, and the activity is inherently valuable to us. We do the activity for its own sake. The goals are only in there in service of getting us into the activity, giving us some guidance of how to do the activity well or not.
- 6.42 So in one situation, the activity is totally in service of the goals, and we experience our arousal this way. In another situation, the goals are completely in service to the activity, and we experience our arousal that way. He calls this the telic mode, working towards the goal. He calls this the paratelic mode. I don't like that. It's not as helpful a term. Other people, like Csikszentmihalyi, used the term autotelic. You're doing it for its own sake. This is work and this is play.
- 6.43 Now that takes us in—when do we work and when do we play? This takes us into Fromm—the different kinds of existential modes we're in. The agent-arena relationship. We have the having mode. The having mode is—Erich Fromm. In the having mode, which is an existential relationship—it's a relationship between the agent and the arena—in the having mode, we're trying to solve our having needs. These are the needs that we satisfy by having things. I need to have water. I need to have oxygen. I need to have food. I need to have medicine. And notice that we even put it this way: I need to have sex. Notice how we use that. Interesting.
- 6.44 Now, those needs, the having needs, are met by controlling and consuming your environment. I'm controlling and consuming oxygen and liquid and food, etc. There's nothing wrong with the having mode. You need it. How do you interact with the environment? You interact with the environment categorically because you're trying to manipulate and control it. I treat things with their categorical identity. That's a chair. I know how to control and manipulate it. I need to have a spot where I sit. Do you understand? I have an I-it relationship to everything, using Martin Buber's distinction.
- 6.45 This is an it for me because I have given it a categorical identity because I need to have control over it in order to do the stuff I'm doing. That make sense? Now, there's a different mode. He calls it the being mode. He should have probably called it the becoming mode. But this is where I'm not trying to have something, I'm trying to become someone. I need to become mature. I need to become rational. I need to become wiser. I need to be in love. And notice we talk about making love in contrast to having sex. These are our developmental needs. These are the needs in which we are

trying to make ourselves— wrong way of putting it. We're trying to participate in going through the transformations of development. That's why we talked about serious play and development.

6.46 There's nothing wrong with this mode either. The mistake we make is when we confuse them. How would we confuse them? We would confuse them when we are trying to satisfy being needs within the having mode. I need to be in love, so I'm going to have lots of sex. I need to be mature, so I'm going to have a car. I need a sense of wisdom, so I'm going to have some alcohol.

6.47 And of course, with that confusion, other people, bad faith actors—and I'm not saying everybody's a bad faith actor—but bad faith actors are trying to constantly keep you in a state of modal confusion. Because if I can keep you modally confused, I can get you to buy more cars because buying the car isn't actually making you become mature. So you're frustrated. The need isn't being met. But if I can keep you in modally confused, you'll buy more cars or you'll buy more sex or you'll buy more alcohol. And you'll fail to be in love. You'll fail to become mature. I can sell you more stuff. I can sell you more ideas. I can manipulate you. This was Fromm's point. Modal confusion is a kind of foolishness.

6.48 Now, play is about the being mode. It's about development. Work is about the having mode. It's about getting the things you need to have. The problem we can get into is we can get modally confused, and then we become incapable of properly engaging in serious play. That's what play is. Play is to do the kind of framing in which the goals are for the sake of the activity, and that framing makes you interpret your arousal paratelicly. The paratelic mode helps you to engage in serious play, get into the flow state, so that you can go through significant transformation and development. Work is when the activity is for the sake of the goal, and this is for the having mode in which it gives you control and consumptive ability over your environment. You need both. And a great mistake, one that our culture exacerbates, is modal confusion.

6.49 Part of being wise is being able to regularly and reliably—not just in thought, because this is not primarily about propositions—be able to regularly and reliably distinguish between these and to enact both of them when they are appropriate. Ratio. Ratio, religio. Did that answer your question about play?

6.50 OK. Notice, to return back to it, we're constantly toggling between these disruptive strategies and these self-regulation strategies, between breaking the frame and with identifying with a new kind of agent-arena relationship, and we're constantly toggling. We're trying to give a ratio-religio for agency that's bringing us into that reciprocal opening in which more and more of the environment—think of the flow state—is disclosing itself to me, and I am more and more transforming the insight cascade, the implicit learning, in order to be more and more conformed at one with that environment. Reciprocal opening. The opposite of reciprocal narrowing—what happens in parasitic processing.

- 6.51 So, we have these disruptive strategies. Serious play. We have the vertical and horizontal imaginal augmentation of our intelligence. We're toggling between self-regulation and self-transcendence. Toggling between regulation and identification strategies and disruptive transformative strategies. I want to point out something else that ritual builds upon.
- 6.52 Now this goes to work done by Lakoff and Johnson and others, and I've published on their work. I've been critical of it. I'm not going to get so much into the criticisms because they have to do with cognitive science. So their particular explanation of a phenomena, I think, is inadequate, but their description of the phenomena, I think, is brilliant. This is Lakoff and Johnson.
- 6.53 They talk about conceptual metaphors. You all know what a metaphor is, right? When you make an identity claim between things that are not logically identical. OK. And why do we do that? Because we use the metaphor in order to change how we see, how we frame the target. So if I say, "Sam is a pig," and Sam's a human being, because it's not a metaphor to go into a pig pen and say, "There's Sam, the pig." You don't go, "Wow, poetry." What I'm doing is, I'm getting you to—this is kind of a frame that makes things salient, things that are salient about pigs. This is part of Ortony theory and Black's work on metaphor. And I look through that frame and now it makes salient things about Sam that I normally wouldn't see. I'm making, foregrounding things about Sam. That's how metaphor works, right? The word metaphor is itself a metaphor, by the way. It means to carry over, to transfer.
- 6.54 OK. Now, I want you to realize the poetry of your lives because you are constantly using metaphors without recognizing that they're metaphors because there's a deep part of your framing. "I hope you get my point. But maybe you don't see what I'm saying. But surely you can grasp it. But I hope you understand. But it may be hard because we're only about halfway through the lecture."
- 6.55 Everything I said is a metaphor. You're doing it all the time. In fact, it's very difficult for you to utter more than four sentences without making use of a conceptual metaphor. If you try and talk about anything important, you will fall into conceptual metaphor. I'm trying to talk about understand. Try and talk about love and don't use a metaphor about a journey or a musical resonance between you and the person. Tell me what love is. I'm trying.
- 6.56 Our thinking and our actions are filled with metaphor. That's a metaphor. You're inherently metaphorical. Now, the metaphors are not primarily linguistic. Some of them are conceptual, but many of them are more enacted. Enacted metaphor? What are you talking about? Well, you may notice that when John is talking to you, John is doing these weird things with his hands. His hands are doing stuff all the time. Why is he wasting all that energy? He's gesturing, and so are you. And the gestures can be very subtle. They can be very subtle facial movements. They can be very explicit.

- 6.57 Susan Goldin-Meadow. That's her name. I think her name was Susan Goldin, G-O-L-D-I-N. And then she married somebody whose last name was Meadow. So she's Susan Goldin-Meadow. It sounds like an animated character, right? She's done work showing that gesture is not some ornamental add-on. If I talk to you and I don't gesture, your ability to understand me degrades. If you're by yourself, you will gesture because it helps you think. "I'm not gesturing." Do you walk around? "Well, yeah." OK. So, right?
- 6.58 Barbara Tversky in her book, *Mind in Motion*: "The machinery you use in order to navigate the physical environment is the same machinery you use in order to move through conceptual space." You notice how I have been using vertical and horizontal metaphors to talk about abstract ideas? If you have people try and explain something, and they grasp a pole so they can't gesture, their ability to explain it degrades. Gesture is perspectival and participatory enacted metaphor. Look at this. What is this? Well, it's kind of like speaking, but it's kind of like grasping. It's kind of like putting it out to you, but bringing it to, like—
- 6.59 So, I want you to step back and think about all the implicit ritual in your life. You're doing all this enacted metaphor. You have to engage in serious play. There's degrees in which you're invoking the vertical and the horizontal imaginal to the degree to which you're trying to properly do that vertical inhabiting of yourself, your relationship to your future self. You have to use disruptive strategies, and you have to put them in this toggling relationship with constructive strategies. You have to rely on all this non-inferential religio. You try and enhance it by getting into the flow. The potential for ritual is implicit through most of your life, and it has nothing to do with neurosis. It has nothing to do with irrationality.
- 6.60 So there's a new book out called *Ritual*, and this is the last name of the person who wrote the book. The book is excellent, but I do not know how to pronounce this? He's Greek, but he's modern Greek. That's another A. I think it's something like Xygalatas, but I haven't heard it pronounced. So I might be insulting him or offending him. So I apologize. The book is called *Ritual*.
- 6.61 Now, the book does something really, really powerful. He makes a lot of argument about how important and prevalent—that's the word I'm looking for —ritual is in human existence, and it's not primarily about irrationality or neuroticism. Now, he talks a lot about affective aspects of ritual, how it helps us relax, how it helps us properly engage. He invokes ideas about predictive processing, but he doesn't develop that, and the cognitive aspects of ritual are not properly developed. I recommend the book because it will convince you about the importance of ritual and give you a different conception of it. But if we want to get into the cognitive science of ritual, I think we have to supplement that book with some other recent and important work.
- 6.62 And this is work coming out of philosophy and cognitive anthropology and a convergence between them around what's called ritual knowing. It's the idea that ritual allows us to know the world in a particular way, the cognitive aspect of ritual. Now, the

seminal paper on this is called *On Ritual Knowledge* by Jennings, 1982. So I'm going to read you a quote from Jennings, and it gives you his central thesis. The ritual, quote, "teaches one not only how to conduct the ritual, but how to conduct oneself outside the ritual." So the idea is, a good ritual is one in which I'm doing something here, and it does that transfer, that transfer-appropriate processing, to other areas of my life.

- 6.63 Another quote: "The knowing of ritual action provides a model for fitting—fitting actions in context or situations not themselves ritualized." The point of ritual is to create religio. And what you're trying to do is find a way of fitting yourself in this context, which reliably transfers to many non-ritualized contexts. You're playing in the ritual so that behavior works in other areas of your life.
- 6.64 There's another quote just to make clear how close this is to—and he's writing this in '82. It was way before. "Fittedness is a relational notion." He's basically saying fittedness is transjective. So he argues that ritual is a non-reductive, noetic function of the mind. Those are his terms. So what does he mean by non-reductive? Non-reductive is you can't replace ritual with myth or drama. It's non-reductive. What does he mean by noetic? He's invoking a Greek word for knowing, but the Greek is noesis. What does noesis mean? Noesis is best translated by words such as noticing, insight, sizing up. He's talking about perspectival knowing that fits you as an agent, participatory knowing to the world.
- 6.65 Now, that perspectival knowing immediately gets us into the current work of the neuroscientist, McNamara. I had the great privilege of getting to review his most recent book, *The Cognitive Neuroscience of Religious Experience*. And he talks a lot about the function of religion is largely perspectival. And you can think about how this is anti-absurdity. The function of a ritual is to do what he calls decentering. So in a ritual, I will often be speaking, but it's not—so think of somebody who's reading the Bible in church. I'm not a Christian. I'm not advocating for Christianity. I'm trying to use one that's at least probably culturally familiar to a lot of people. I don't mean anything by it other than the example.
- 6.66 Somebody is reading the Bible in church, and perhaps they're reading the words of Jesus. Now, in one sense—let's say it's me—I'm reading, first-person perspective. But this isn't John speaking. Who's speaking? Jesus is speaking. Or God is speaking. There's this third-person perspective other than my own. I'm being decentered from my egocentrism by the ritual.
- 6.67 Now, that decentering has huge important cognitive benefits. So I mentioned to you the wisdom paper and the person, Igor Grossman, who's a colleague of mine, and we work together. And he has produced a lot of evidence for what he calls the Solomon effect. And I mentioned it to you earlier. And this is a noetic, a noesis, a perspectival thing. You get people: "Are you having any serious problems in your life?" "Yes." "Tell me about your problems." What do they inevitably do? They describe it to you from first-person perspective.

- 6.68** Now you do the following with them. "I want you to re-describe that problem from a third-person perspective as if a good friend who knows you was describing the problem. What would they say? How would they describe the problem?" You de-center the person. What happens is, and you know that this is coming, they get an insight into their problem that they couldn't get from the first-person perspective. Because you know what you can't see from your first-person perspective? Your biases or interests or goals that are outside. You have to get outside your perspective to see the biases in your perspective and become aware of goals other than the ones that your perspective is giving you. You have to de-center.
- 6.69** You get the Solomon effect. Solomon was famous for what again? Cricket, poetry? Wisdom. And that's why he studies wisdom as a psychologist. See, the ritual de-centers you. The reason why we practice rituals is they regularly and reliably de-center us because that provokes the possibility of the Solomon effect. But it only works—remember what we talked about—the degree to which you do the serious play. The degree to which you invoke the imaginal. You have to be seriously playing and imagining in the imaginal sense that it is Jesus speaking, not you. If you just read the words, you're only doing the propositional. You're not going to get the Solomon effect of the ritual decentering. You have to practice the ritual in a deeply perspectival and participatory way. Remember L. A. Paul? So its capacity to engage you in transformative knowing occurs. You can trivialize any ritual by reducing it to a propositional proposal, and then it's ridiculous.
- 6.70** So Jennings has something here. I think there's good argument and evidence to support what he's saying. Now, of course, there are people that are critical of Jennings. I'm building an argument here. What am I building an argument for? I'm building an argument for ritual is both necessary and we can talk about a rationality of ritual. We can talk about good versus poor ritual in a principled manner and how that contributes to making us rational in other areas of our life.
- 6.71** So, Williams and Boyd, 1992. Boyd is B-O-Y-D. Williams and Boyd, 1992. Ritual, Art, and Knowledge. They wrote an amazing book. They're anthropologists. They did participant observation of people. They were people participating in a Zoroastrian ritual. Jennings emphasized the degeneration of ritual knowing and the transforming of the arena. You're manipulating things, you're trying to get that fittedness. You're playing with the ritual. You're making new rituals.
- 6.72** Williams and Boyd acknowledge Jennings. They don't say he's wrong. They just say he's incomplete. But they say ritual cannot be just an act of you trying to fit things. They say ritual is often—and this is the language that some of the people that they were studying was using—ritual is considered very often a non-propositional masterpiece, like a work of art. So Jennings is kind of like jazz. You're trying to get the jazz and you're trying to —Williams and Boyd, as if this is Beethoven's symphony. You don't say, "You know what? Let's jazz it up. Let's take out some of this. Let's put in some of that. Let's

do a complete riff." No, because now you've lost it because this is considered a masterpiece.

- 6.73** What you're trying to do is not shape it. You're trying to shape yourself. You're trying to live up to that masterpiece because it's a work of art, non-propositional art. That's why I keep using music as a way that you can relate to it.
- 6.74** So it's not so much transforming the arena of action. In their account, it's transforming the agent. The arena is considered a work of art, and how do I best inhabit that arena? So Williams and [Boyd] see the two positions as complementary to each other. So they acknowledge that ritual—see, if you just had the work of art, ritual wouldn't change over time. And if you ask anybody about the—this is how they've always done it. And they believe that, and they're wrong. Rituals evolve over time. The evidence for this is undeniable. That's why "we're preserving things the way they were in the time of X" is a false statement. Sorry. I don't mean to trespass on anybody's religious commitments, but it's just not true.
- 6.75** The empirical evidence is overwhelming. People think the rituals are not changing, but they are. There's a tension here. People are trying to live up to the ritual, but they're actually constantly evolving it, too. The two are at work. So here's the repetition of the ritual. "Oh, look what John's doing. Oh, no. Here we go again." Here's Jennings. What we're doing is introducing variation the way he talked about it. And then here's Williams and Boyd. I said Brian, I meant Boyd earlier. Williams and Brian. We're putting selective pressure. Most of the innovations will not be acceptable because it's the masterpiece. And so what are we doing? We're constantly evolving our ritual religio.
- 6.76** Here's the selective pressure. Here's the variation. You know this. You've seen this again and again and again. There's a reason why you keep seeing it, because this is the machinery of religio. So Kevin Schilbrack brings this all together in an excellent article called Ritual Metaphysics. It's in a book called Thinking Through Rituals, which is a play on words. As theorists, we're thinking about them, but we're thinking about how we actually think through ritual, that ritual is a way of thinking that is irreplaceable with anything else.
- 6.77** He's going to bring them together. He's basically going to unfold something like this. But what he wants to argue is that ritual has an inherent normativity in it, that proleptic aspirational rationality that we've already talked about with Agnes Callard. Rituals that are massively and deeply transfer-appropriate. When I get rituals that have been evolved such that they—and I want to do it horizontally and vertically—they transfer broadly to many different situations and deeply in those situations, and they transfer broadly and deeply vertically within different levels —propositional, procedural, perspectival, participatory. Those are the really good rituals. Those are the best rituals.
- 6.78** And he argues that those elegantly disclose the non-propositional intelligibility of the world. There's aspects of the world that can only be disclosed to us insofar as we are

making use of rituals, insofar as we're tapping into conceptual metaphor, gestural metaphor, serious play, the play between disruptive strategies and the leaps of reason.

- 6.79** Whether or not we're getting into the flow state, whether or not that flow state transfers. The problem with the flow state, like everything that's adaptive, is it can be hijacked, and it's hijacked by video games. You can become addicted to them. WHO, World Health Organization. Because the flow state in the video game does not transfer to your life because they don't want that. They want you to buy more of the video game. So you flow in the game. This is wonderful flow. And then it doesn't transfer to your life. And so your life looks shitty by comparison. So you get depressed. So you play the game more. A little confusion. You get reciprocal narrowing. The world is, the only place where I can find flow is in the—and then you're addicted.
- 6.80** The ritual is the opposite of that, when it's a good ritual. When it's a good ritual. Here's a quote from Schilbrack, making use of work by Catherine Bell. There's a quote: "On this understanding, ritual metaphysics is a matter not, or not just, of giving participants a mental image of a larger world"—think about the imaginal, not just a mental image—"but of giving them experience of participating in the very patterns and forces of the cosmos." That participatory knowing, the perspectival participatory knowing, imaginably augmented intelligence and conception that connects you radically and deeply to the world in a manner that is necessary for the proleptic aspirational dimension of rationality. If you're not practicing ritual, the degree to which you become rational is seriously truncated. The Stoics practice a ritual. I showed it to you, the view from above.
- 6.81** OK, I want to talk a little bit about this notion of intelligibility and understanding. Because I need it when I'm talking about wisdom. You can look for what intelligibility is, what understanding is—at least, this is what I'm arguing. My son and I are writing a book on this—by looking about how we make things plausible. Where plausible doesn't mean highly probable, that's one meaning of plausible. The other meaning for plausible is it makes good sense. It stands to reason, "Oh, I see that now." That's what plausible means.
- 6.82** I've got a lot of arguments about how plausibility is presupposed before you can do inference and deduction or experiment, but I can't do those right now. I'm running out of time, but they're there. OK, very briefly, if I'm going to do an experiment, can I consider all possible theses? How many are there? Underdetermination. How many theories are there for any finite data set? Infinite. So I have to prejudge. Notice the dangers. I have to prejudge which theories, which hypotheses, plausible. When I'm running my experiment, I have to consider what are the alternatives I have to control for.
- 6.83** The alternative theories, underdetermination, unlimited. I have to choose the plausible confounds. Then when I go to draw my conclusions, can I draw all the possible logical implications from the data? No. The relevant ones, the plausible ones. Plausibility

before, during, and after an experiment. Is that enough? That was fast, but I hope that gives you a good sense. Plausibility is at the level of religio. It comes before induction and deduction. It's the level of abduction.

- 6.84** You can study in the philosophy of science or the psychology of explanation, what are people looking for? Well, we've already got one sense of it. I won't repeat that argument. We want a transfer-appropriate form. We want to come up with a generative model that will transfer to many different situations widely and deeply. This is what is meant when you say a theory is elegant. You want elegant theories that do that for you. Find and frame problems in many different domains. Force equals mass times acceleration. Great theory. Not totally true, by the way. Breaks down near black holes and other stuff like that. But generally, I can use it to explain balloons, whales in the water, planets circling in orbit, ballistic rockets. It's elegant. Like, wow, that's how you get to be Newton. What did you do today?
- 6.85** OK, so you want elegance. Is that all you want? No. Because if I just give you elegance, if I just give you—think about this. If, hear the if, if you would believe that the British monarchy, the British Royal family, are actually lizard people from outer space, I can explain so much of their behavior. These weird, far-fetched theories. Because you don't only want that. You also want what Nicholas Rescher calls trustworthiness. I want the process by which I come up with it to have many independent lines of evidence and argumentation that are converging.
- 6.86** You even do this with your sense experience. If you can only see something, you don't think it's as real as if you can see it, hear it, touch it. That's why we like numbers, by the way. You can see three, you can hear three, you can touch three. Numbers give this convergence. Why is that important? Because if I have only one channel, this model may be due to the bias within this channel. But the more independent converging ones, the less this is the result of any particular bias.
- 6.87** Do you see that? It increases what Rescher calls the trustworthiness of my model. Not truth. Trustworthiness—the degree to which I can believe that it hasn't been produced by bias. So I want convergence because convergence gives me trustworthiness.
- 6.88** And then Elijah Millgram talks about we want a balance. Notice when I gave you an imbalance, when I gave you a lot of this and only a little of this, you don't like it. That's far-fetched. What about the opposite? I give you a lot of this, but very little of this. That's triviality. Something you—"Yeah, of course, that's probably true." But to what? Who cares? "I have this really great claim I want to put out for you. Please listen. People get hungry." Yeah? Yeah? Yeah? Many different situations in context. I agree. People get hungry. So what? And? "No and. Just that. Savor it." Sorry for the pun. "People get hungry." You go, "Go away. You're trivial." You don't want to be in this conversation. "Oh, look, the bus is coming. I have a theory about that." No. Right? Right?

- 6.89 Now, if you don't need a lot of this, you don't need a lot of this. I like vanilla ice cream. OK, it doesn't have much implication about your model of me. What else can you predict about me from that? Not very much. So you don't need a lot of this. If somebody says to you, "I think there's a God," you go, "I need a lot of this." Convergence, elegance, balance in a form, a structural functional organization that is transfer-appropriate. When you get that, what are you doing? What you're doing is really, again, a ratio of what people are finding significant. This is all about judgment. Plausibility is a judgment that something is worthy—listen to the language —of your attention. It's about bringing a rationality to how you pay attention.
- 6.90 So, in the literature, people make a distinction between knowledge and understanding. Knowledge is when you grasp evidence. Understanding is when you grasp the significance of what you know. This is how to best grasp the significance of what you know. Understanding is the generation of plausible significance. Understanding is the generation of plausible significance. What have I tried to show you? I've tried to show you all the implicit aspects of ritual and how they all converge on a transfer-appropriate form. They bring a kind of understanding that's not the same kind of understanding that you get from a theory or that you even get from a good skill.
- 6.91 So, good ritual is the profound—very deep plausibility is the profound enhancement of non-propositional religio. And it is a rational thing to do because non-propositional enhancement of religio is central to the imaginal aspirational dimension of reason and rationality. What rituals do you have in your life for doing this? So what we're seeing is a relationship between ratio, religio, and ritual.
- 6.92 Let's talk about wisdom. I was going to talk about one more challenge, the fourth rationality debate, and this has to do with individual versus collective, monological versus dialogical. But you already know that —the evidence that we reason better together in dialogue than monologically. And that's what ritual helps to do. Remember I said, you need something to get people coordinated so that they could engage in dialogue rather than conflict or argument, so that they can do the collective reasoning that supersedes individual cognition. So people would be, as a group, doing better on the Wason selection task.
- 6.93 Ritual does that. That's what it does. The book on ritual goes through that. That's what it does. It puts us deeply in sync with each other so that we can access—we can be properly disposed to accessing the collective intelligence of collective cognition, distributed cognition, which is better and more powerful than individual cognition. Ritual does that.
- 6.94 So, wisdom. I think I said it was published in 2019. I'm wrong. It was published in 2020. This is *The Science of Wisdom in a Polarized World: Knowns and Unknowns*. Lead author is Igor Grossman, the Solomon effect guy. I was involved in that. I told you all about that—the Wisdom Task Force and all this really cool stuff.

- 6.95** What did they conclude is the central thing to being wise? What did we conclude? Reflecting on all the evidence, all the theory, all the positions, all the arguments, and what came to it? The importance of perspectival metacognition. What is this ability? You know what this is. I've been teaching you all about this, all the way through. This is a balancing of viewpoints. This is epistemic humility. This is the ability to be context-adaptable, to take into account multiple perspectives. You've seen this through and through and through. As we get more and more into rationality, it's bridging more and more into wisdom. As we deepen our notion of rationality, it more and more starts to look like wisdom.
- 6.96** What does this meta-perspectival—sorry, what does this perspectival metacognition give you? Well, it gives you the ability to implement abstract moral concepts. It's how you take a rule and actually fit it to the situation. Remember that rules depend on that religio? Wise people have trained their perspectival—and I would also say, participatory—knowing, because you have to identify with their particular perspectives. You have to establish your agency. Wise people are good at doing that. They're good at taking these things, these representations and the rules, and using the non-propositional to transform themselves and transform the situation so that they can realize the moral concept in that situation.
- 6.97** People are really good at teaching moral philosophy. It's not predictive of whether or not they're moral agents. Did you know that? It's not highly predictive. "Well, what do you do for a living?" "I teach moral philosophy." "Are you more moral than most people?" "If you do the research, no." Proposition is necessary but insufficient. I don't have to show you why. You know that now. Here's the one. This allows you to see through illusion and into reality. It's seeing through the illusion caused by self-deceptive, self-destructive behavior and into reality—and not just in this situation, but with deep and profound understanding. You're using ways of seeing through illusion and into reality, not just for here. They're very powerfully trustworthy, and they have tremendous elegance. There's a deep understanding in your ability to see through illusion and into reality.
- 6.98** You have a survival-oriented coordination of your abilities. What's all that? Well, that's all the ratio we've been talking about. That's the ratio. So you have wisdom. It should help you overcome foolishness. Foolishness is parasitic processing that leads to reciprocal, narrowing, addictive, impulsive behavior. Throws all the ratio into disproportion, out of perspective. Wisdom is the opposite—an ecology of practices that reciprocally opens you by bringing ratio to all of the religio.
- 6.99** So it'll bring about well-being. What is well-being? The person who wrote the book on this is Michael Bishop in 2015, *The Good Life: Unifying the Philosophy and Psychology of Well-Being*. What does he argue? Well-being isn't any one thing or set of things. It involves what he calls a positive causal network. What he means is a dynamical system where you're having good emotions, good accomplishments, good attitudes, good

traits, good behaviors, and they form a positive causal network. They reciprocally open you. Now, I would want to add more. They not only reciprocally open you, they deal with the reciprocal narrowing of parasitic processing. You can't get them without wisdom.

- 6.100** You see, that doesn't mean that there's any one good life. It means it's multiply realizable. There are many different kinds of good life as long as you are able to create that positive causal network, that dynamical system involving an ecology of practices, an ecology of ritual practices, many of them, that overcomes parasitic processing and affords reciprocal opening.
- 6.101** As you get that well-being, you become more internally at peace. As you reciprocally open, you come more and more into conformity to what's deeper and more deeply real, seeing through illusion into reality. This is Plato's point. If you link those two together, as I become more at peace, I'm less internally conflicted. I'm more able to engage in seeing more deeply into reality. As I see more deeply into reality, that helps me to more coordinate my inner world and bring it into peace of mind. And these are the two meta-desires of your entire life. Because no matter what you desire, you desire that when you achieve that desire, it brings you peace.
- 6.102** Here, you'll have all this sex and money, but you're going to be riven inside with internal conflict. Do you want it? No. I'm going to do this thing for you, but it's not going to be real. I regularly ask my students this: "How many of you are in satisfying romantic relationship?" Because romantic relationships take the place of God and history and culture and tradition in our life. "You, you are going to be the source of all of my meaning." No relationship can bear that. We should give up that idea. It's a stupid idea. Romantic comedies are great evil.
- 6.103** But I asked them this, "How many of you want to know if your partner was cheating on you and that would end the relationship?" Almost all of them put their hand up. And I say, "Why?" And here's my students who are all postmodern and bitten by skepticism: "Because it wouldn't be real." They want a real relationship. They don't want a fraud. Inner peace. Contact with reality. This is Plato's point. You can get them so they loop and you reciprocally open. I see more deeply into reality so I can see more deeply into myself. I can realize myself better, bring about more well-being. Because of that, I have less internal conflict, less self-deception, less self-destruction.
- 6.104** I see more deeply into reality and I reciprocally open and I fall in love with the world. That's love. That's spirituality is what I'm proposing to you. You have this. So these two are playing off against each other in all the ways we've talked about. Ritual and ratio. You're doing both this individually and collectively. This gives us enhanced religio. Enhanced religio. That goes into wisdom. This goes into well-being. And these two are playing with each other, and you fall in love with reality.
- 6.105** You have a good life in which you fall in love in reality in a way that makes you more rational, more wise, and increasingly enhances your meaning in life. That's what I'm

proposing to you spirituality actually is. So now, we've come to the conclusion of Intelligence, Rationality, Wisdom, and Spirituality. Thank you so much for going on this journey with me.